

Coherent and ultra-low-power EDSR with a flopping-mode spin qubit in germanium

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Hole spin qubits in semiconductor quantum dots (QDs) enable high-fidelity all-electric control, but conventional electric dipole spin resonance (EDSR) can require substantial rf drive power at the low magnetic fields that are favorable for qubit coherence and readout. In planar Ge hole spin qubits, this can reach -27 dBm at the device, posing challenges for scalable architectures due to heating and crosstalk. Here, we demonstrate a flopping-mode (FM) qubit in Ge, where a single spin is delocalized in a double QD, combining first-order protection against charge noise with exceptionally efficient electric driving. By mapping out coherence sweet-spots as a function of magnetic field orientation we achieve $T_2^* = 1.4$ μ s, $T_2^{\text{Hahn}} = 11.5$ μ s, $T_2^{\phi, \text{CPMG}32} = 130$ μ s, and $T_1 = 226$ μ s, and a single-qubit gate fidelity of up to 99.76 % for a gate time $t_{X\pi} = 88$ ns. Importantly, these results are obtained at a nearly in-plane magnetic field of 5 mT using only -52 dBm drive power at the device. We further find that qubit relaxation in this regime is consistent with a two-photon Orbach process, providing a route for further optimization. Our results demonstrate that FM-EDSR supports ultra-low-power, high-fidelity single-qubit operations, improvements that could benefit scalable hole-spin-based architectures and hybrid spin-photon interfaces.

INTRODUCTION

Hole spin qubits in planar strained Ge/SiGe heterostructures have emerged as a promising platform for quantum computing and simulation [1–4]. Their highly anisotropic and voltage-dependent g -tensor enables fully electrical spin control via g -tensor magnetic resonance (g -TMR), yielding single-qubit gate fidelities in excess of 99.9% [5–9]. Recent progress in coherence optimization [6, 10], high-fidelity two-qubit gates [9], robust spin readout [11] and coherent shuttling [8] has relied on operation in low in-plane magnetic fields below 100 mT, where charge noise sensitivity and the spin relaxation rate are reduced. The same low-field regime, however, decreases the Rabi driving efficiency and typically requires drive powers around $P_{\text{drive}} \approx -27$ dBm (or equivalently 10 mV drive voltage) at the device for practical gate speeds [7, 8, 12]. In semiconductor spin qubit platforms, a large driving power has been associated with heating [13], readout fidelity degradation [14, 15] and cross-talk [12, 16], making power-efficient control increasingly important as qubit arrays grow [9, 12, 17–20].

Several approaches can be employed to improve the power-efficiency of single-qubit gates. For instance, it has been observed that Rabi driving is moderately improved for certain hole occupations [12]. Alternatively, a large g -tensor variability in neighboring quan-

tum dots can be utilized to perform high-fidelity hopping gates via baseband pulsing, which reduces dielectric losses [9, 21]. Furthermore, both EDSR and baseband gates can be extended to shuttling links where the qubit probes larger magnetic field or g -tensor gradients [8, 18, 22]. Another approach is to operate in the flopping-mode regime, where a single charge carrier is delocalized across a double quantum dot (DQD) [23–25]. At the charge symmetry point, the increased electric dipole can enhance spin–electric coupling by orders of magnitude. This has enabled strong spin–photon coupling for electrons in Si/SiGe [26–28], and for holes in Si nanowires [29, 30], with theory predicting fast, low-power and high-fidelity single-qubit gates [31–34]. The enhanced dipole, however, presents a trade-off: the same charge hybridization that strengthens electrical control can also increase charge-noise sensitivity and relaxation. Whether strong transverse spin–electric coupling can coexist with weak longitudinal susceptibility and long relaxation times therefore remains a key question for high-fidelity operation.

In this work, we demonstrate that these requirements can be simultaneously satisfied in a flopping-mode (FM) hole-spin qubit in planar Ge. Pauli spin blockade (PSB)-based readout using an ancillary quantum dot (QD) spin enables operation at low magnetic fields, where the spin coherence is improved. We study the coherence anisotropy with respect to the magnetic field orientation and identify dephasing sweet-spots where the Zeeman energies in the two QDs are degenerate [6, 35]. At these sweet-spots, we observe that charge noise is suppressed to first order, and that second order charge

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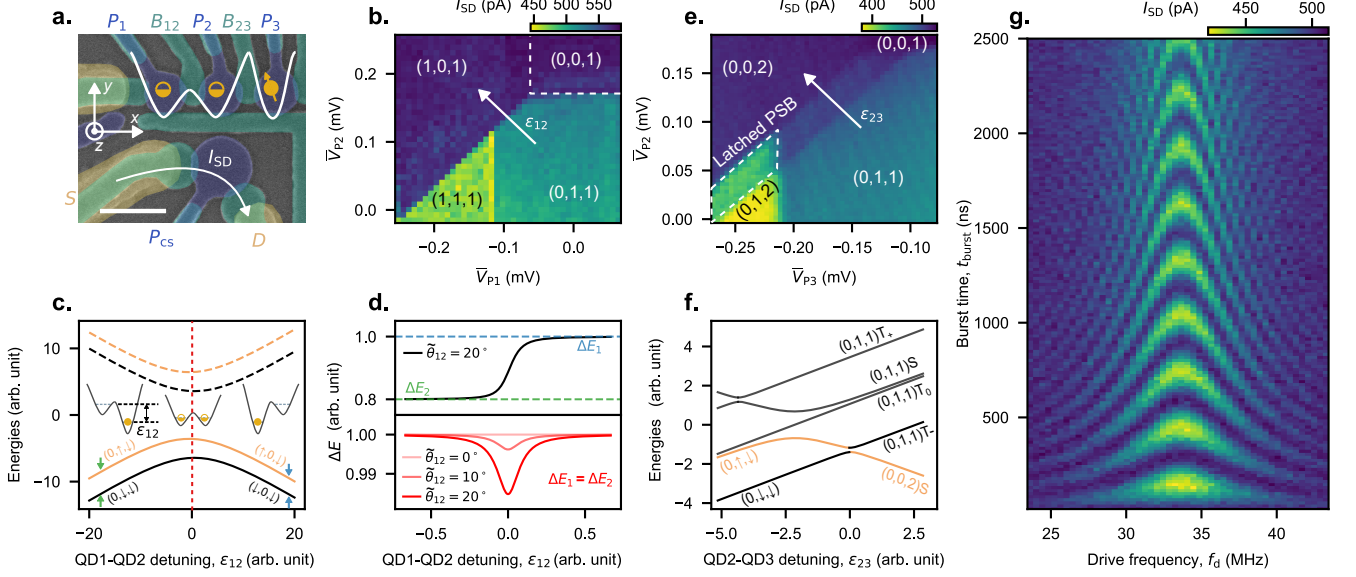


FIG. 1. **a.** False colored scanning electron micrograph of a QD device identical to the one utilized in this work. Plungers P_i are used to tune the chemical potential of QD_i , and barriers B_{ij} are used to control tunnel coupling between QD_i and QD_j . S and D indicate the source and drain ohmic contacts of the SHT charge sensor. The dc current I_{SD} through these contacts is used to charge-sense the triple QD. **b.** Pulsed DQD charge stability diagram measured as a function of the virtualized gate voltages $\bar{V}_{P1}$ and $\bar{V}_{P2}$, near (1,0,1)-(0,1,1) charge transition. The DQD energy detuning ϵ_{12} is tuned along the direction illustrated by the white arrow. **c.** Eigenenergies as a function of energy detuning ϵ_{12} near the (1,0,1)-(0,1,1) charge transition calculated using the Hamiltonian in Supplementary section S5 while assuming a finite magnetic field. Solid black and orange lines denote spin- $\downarrow$ and spin- $\uparrow$ charge ground states respectively. Dashed black and orange lines indicate spin- $\downarrow$ and spin- $\uparrow$ charge excited states. Blue (green) arrows represent the Zeeman splitting of the hole in $QD_{1(2)}$. The FM spin qubit, denoted by the red dashed line. **d.** Energy difference between the black and orange solid branches shown in **c.**, calculated by assuming different (top panel) and equal (bottom panel) Zeeman energies in QD_1 and QD_2 for different values of Larmor vector misalignment θ_{12} . Blue and green dashed lines in the top panel illustrate the Zeeman energy of the hole in QD_1 and QD_2 . **e.** Pulsed DQD charge stability diagram (see text) measured as a function of the virtualized gate voltages $\bar{V}_{P2}$ and $\bar{V}_{P3}$, near the (0,1,1)-(0,0,2) charge transition. The DQD energy detuning ϵ_{23} is tuned along the direction illustrated by the white arrow. The region in (0,1,2) enclosed by the white dashed lines corresponds to the latched-PSB readout window. **f.** Calculated eigenenergies as a function of energy detuning ϵ_{23} near the (0,1,1)-(0,0,2) charge transition. S, T_0 , T_+ and T_- indicate spin singlet and triplet states. **g.** Rabi chevron pattern of the FM spin qubit measured at a detuning sweet-spot for $\mathbf{B}$ orientation $\hat{\mathbf{b}}_3$ (see section Anisotropic Coherence) with rf drive frequency f_d , drive amplitude 550 μV on $\bar{V}_{P2}$ and drive duration t_{burst} . Here, $|\mathbf{B}| = 5 \text{ mT}$ and $t_c/h = 13 \text{ GHz}$.

noise coupling, together with the Overhauser field fluctuation, is limiting the longitudinal coherence time of the qubit. Furthermore, when operating at the sweet-spots, we observe a Rabi driving efficiency normalized by the qubit frequency, exceeding 0.15 mV^{-1} , a two to three orders of magnitude improvement over g -TMR drive of single-QD Loss-DiVincenzo (LD) spin qubits in Ge [7, 8, 12, 36]. Dynamical decoupling extends the coherence time to $T_2^{\phi, \text{CPMG}32} = 130 \mu\text{s}$, and randomized benchmarking yields average physical-gate fidelities of up to 99.76(1)% with drive powers below -51 dBm (0.6 mV drive voltage) at the device. We further characterize spin relaxation as a function of magnetic-field orientation, detuning and tunnel coupling, finding behaviour consistent with a second-order Orbach process induced by lattice phonons or thermal photons [37]. Finally, a comparison with both an LD qubit in the same device and recent

spin qubit experiments in Ge places the FM qubit in a favourable power-fidelity regime.

Flopping-mode qubit

Fig. 1a displays a false-colored scanning electron micrograph of a device identical to the one used in this work, defined on a planar Ge/SiGe heterostructure grown by reverse-grading on a Si substrate [38]. Holes are accumulated in a 20 nm strained Ge quantum well residing 47 nm below the surface and contacted with annealed Pt-GeSi ohmic contacts [39]. We make use of two gate layers, with barriers and screening gates in the first layer, and plungers in the second layer. The three uppermost plunger gates P_i , with $i = 1, 2, 3$ are primarily used to define a triple-QD, while the barriers B_{12} and B_{23} are used

to control the inter-dot tunnel couplings. The larger QD under P_{cs} is used as a single-hole transistor. It is probed in transport by monitoring the dc-current I_{SD} through the source (S) and drain (D) ohmic contacts (Supplementary section S1), allowing for sensitive detection of the triple-QD charge occupation down to the single-hole level. An external magnetic field ($\mathbf{B}$) is applied using a three-axis vector magnet, with the coordinate frame shown in Fig. 1a.

The dc voltage setpoint is set to the middle of the (0,1,1) charge region, where (n_1, n_2, n_3) denotes the charge configuration in the triple-QD, and n_i designates the number of holes in QD $_i$ under plunger P_i . All gate voltages are virtualized to compensate for cross-capacitance, with $\bar{V}_i$ denoting the pulsed virtual gate voltage of gate i (see Supplementary section S14 for more details). Charge stability diagrams are measured by pulsing the virtual plunger gates to each point and reading out I_{SD} after a settling time. The pulsed-gate charge stability diagram of the DQD, defined under P_1 and P_2 , is shown in Fig. 1b. It reveals the inter-dot transition between (0,1,1) and (1,0,1), which is the operating point for the FM qubit. We define the energy detuning $\varepsilon_{12} = \mu_2 - \mu_1$, where μ_i is the chemical potential of QD $_i$. The corresponding detuning direction in gate-voltage space is shown as a white arrow in Fig. 1b.

The energy diagram of the leftmost DQD in the presence of a finite $\mathbf{B}$ is presented in Fig. 1c, as a function of ε_{12} . Black and orange solid lines represent the charge ground states having spin-down ($|\downarrow\rangle$) and spin-up ($|\uparrow\rangle$) respectively, where the splitting between the two states is primarily given by the Zeeman energy (E_z) of the hole. Similarly, black and orange dashed lines correspond to the charge-excited $|\downarrow\rangle$ and $|\uparrow\rangle$ states. For large detunings $|\varepsilon_{12}| \gg t_c$, where t_c is the tunnel coupling, the eigenstates correspond to the localized charge states $|L\rangle = |(1, 0, 1)\rangle$ and $|R\rangle = |(0, 1, 1)\rangle$. At the charge symmetry point ($\varepsilon_{12} = 0$), the charge eigenstates are the symmetric (bonding, $|+\rangle = (|L\rangle + |R\rangle)/\sqrt{2}$) and anti-symmetric (antibonding, $|-\rangle = (|L\rangle - |R\rangle)/\sqrt{2}$) states, separated by an energy $2t_c$ [40, 41]. The FM qubit is encoded in the Zeeman-split bonding states, $|0\rangle = |+, \downarrow\rangle$ and $|1\rangle = |+, \uparrow\rangle$ [24]. At $\varepsilon_{12} = 0$, the charge density is delocalized across the DQD, implying an enhanced charge dipole moment, and therefore faster electric dipole spin resonance (EDSR) drive compared to that of an LD qubit [23, 24].

Importantly, the Larmor vector, and therefore E_z can be site-dependent due to g -tensor variability between QDs [6, 42]. The energy splitting between the Zeeman-split charge ground states ΔE at $\varepsilon_{12} = 0$ is given by [9, 43]:

$$\Delta E(\varepsilon_{12} = 0) = \frac{1}{2} \sqrt{(E_{z,1} + E_{z,2})^2 - 4E_{z,1}E_{z,2} \sin^2(\tilde{\theta}_{12}/2)},$$

where $E_{z,i} \ll 2t_c$ is the Zeeman energy in QD $_i$ and $\tilde{\theta}_{12}$ is a rotation angle taking into account site-dependent Larmor vector directions and spin-flip tunneling induced

by spin-orbit interaction (SOI) (see Supplementary section S5) [42, 44]. The top panel of Fig. 1d illustrates $\Delta E(\varepsilon_{12})$, for the case where the two QDs exhibit very different E_z . Here, ΔE approaches $E_{z,1(2)}$ for $\varepsilon_{12} \gg 0$ ($\varepsilon_{12} \ll 0$) denoted by a blue (green) dashed line. In contrast, the bottom panel presents $\Delta E(\varepsilon_{12})$ for the case when the Larmor vectors of the two QDs have comparable magnitudes ($E_{z,1} \approx E_{z,2}$) and $\tilde{\theta}_{12}$ is finite. In this regime, $\Delta E(\varepsilon_{12})$ exhibits a local minimum at $\varepsilon_{12} = 0$, corresponding to a coherence sweet-spot where the spin is first-order insensitive to detuning charge noise (i.e., $\partial \Delta E / \partial \varepsilon_{12} = 0$). We emphasize that the depth of the ΔE minimum relative to $E_{z,1(2)}$ is independent of t_c , in contrast to the regime $E_z \approx 2t_c$ often considered in the context of strong spin-photon coupling to electron spins [26, 27]. Consequently, the operating point considered here is simultaneously a detuning and tunnel-coupling sweet-spot. Since the g -tensors of Ge QDs are inherently anisotropic, electrically tunable, and site dependent [2, 42, 44–47], the conditions $E_{z,1} \approx E_{z,2}$ together with a finite $\tilde{\theta}_{12}$ can be systematically realized by choosing an appropriate magnetic-field orientation, as discussed later.

Readout of the FM qubit is performed using latched Pauli spin blockade (PSB) [11, 48], employing the spin ancilla in QD $_3$. Fig. 1e presents the pulsed charge stability diagram of the DQD formed under P_2 and P_3 , near the (0,0,2)–(0,1,1) charge transition, where the latched PSB window is revealed by a pulsed-gate measurement (Supplementary section S2). The relevant energy diagram is shown in Fig. 1f, as a function of the DQD energy detuning $\varepsilon_{23} = \mu_2 - \mu_3$. To initialize the FM qubit, a (0,0,2)S singlet is first prepared by ramping into the (0,0,2) charge region and waiting for the spin to relax, followed by adiabatic mapping to (0, $\downarrow$, $\downarrow$) [11]. The exchange between QD $_2$ and QD $_3$ is then turned off by applying a positive voltage pulse on barrier $\bar{V}_{B23}$. This is followed by a negative pulse on $\bar{V}_{B12}$ to increase t_c . We then ramp to the charge symmetry point between (1,0,1) and (0,1,1) ($\varepsilon_{12} = 0$), which is the operating point of the qubit (for more details see Supplementary section S2).

An example chevron pattern of the FM qubit operated at the first-order detuning sweet-spot is shown in Fig. 1g, obtained by radio-frequency (rf) drive of the virtual voltage $\bar{V}_{P2}$ with the frequency f_d , drive duration t_{burst} , and drive amplitude $A_{d,P2} = 550 \mu V$ on $\bar{V}_{P2}$. We extract a Rabi frequency $f_{Rabi} = 3.5$ MHz and qubit frequency $f_{FM} = 33.7$ MHz. The detailed configuration of this measurement is discussed next.

Anisotropic coherence

To accurately describe the dynamics of a single spin confined in a DQD, we consider a model Hamiltonian that incorporates the g -tensors of the spin in QD $_1$ (g_1) and QD $_2$ (g_2), their dependence on gate voltage, and contributions from spin-flip tunneling due to SOI. To this end, we characterize the g_1 and g_2 respectively by mea-

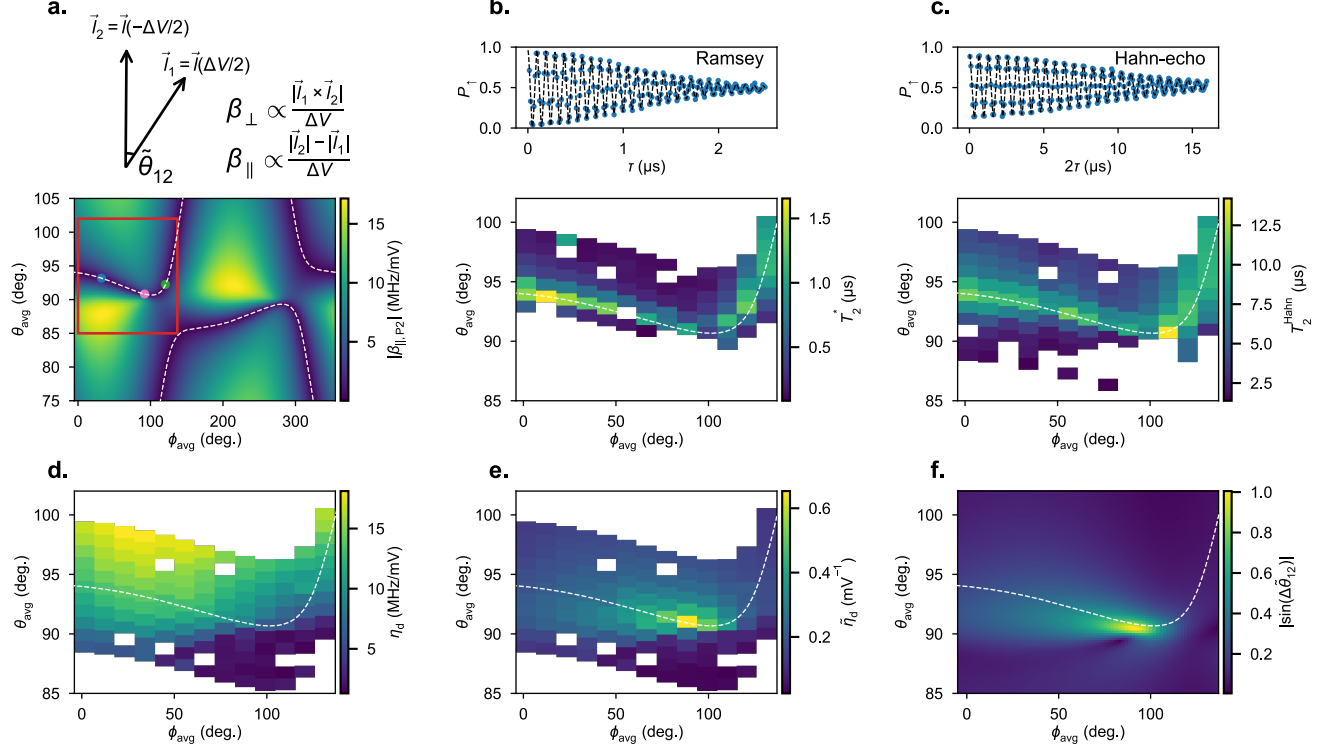


FIG. 2. **a.** Top panel: Schematic showing the definition of longitudinal ($\beta_{||}$) and transverse ($\beta_{\perp}$) spin-electric susceptibility (LSES and TSES), with FM qubit Larmor vector $\mathbf{l}$ evaluated at two (incrementally small) voltage detunings of $\Delta V = \Delta \bar{V}_{Pi}$ from the charge symmetry point. Bottom panel: Absolute value of the LSES, $|\beta_{||,P2}|$, with respect to $\bar{V}_{P2}$. Blue, pink and green dots define the field directions, $\hat{\mathbf{b}}_1$, $\hat{\mathbf{b}}_2$ and $\hat{\mathbf{b}}_3$ respectively, corresponding to experimentally measured zero-detuning sweet-spots. **b.** Top panel: Example Ramsey decay measurement at $\mathbf{B}$ orientation $\hat{\mathbf{b}}_1$, yielding $T_2^* = 1.4 \mu\text{s}$. Bottom panel: T_2^* measured as a function of $\mathbf{B}$, with spherical angles defined in the average g -tensor frame. **c.** Top panel: Example Hahn echo measurement at $\mathbf{B}$ orientation $\hat{\mathbf{b}}_1$, showing $T_2^{\text{Hahn}} = 11.2 \mu\text{s}$. Bottom panel: T_2^{Hahn} measured as a function of $\mathbf{B}$ -field spherical angles. **d.** **e.** Rabi driving efficiency (η_d) and normalized Rabi driving efficiency ($\tilde{\eta}_d$) as a function of $\mathbf{B}$ direction. **f.** Calculated $|\sin(\Delta\tilde{\theta}_{12})|$ with $\Delta\tilde{\theta}_{12}$ denoting the change in angle of the effective FM qubit Larmor vector for $\Delta V = \Delta \bar{V}_{P2} = 250 \mu\text{V}$. All measurements taken at $\varepsilon_{12} = 0$, with fixed $|\mathbf{B}| = 5 \text{ mT}$ and $t_c/h = 13 \text{ GHz}$.

asuring the Zeeman splitting of the spin in each QD as a function of the $\mathbf{B}$ direction (see Supplementary section S6) [6, 49]. We also characterize the gate-voltage dependence of each g -tensor using a modified Hahn echo sequence [6], and measure the energy splitting at $\varepsilon_{12} = 0$ as a function of $\mathbf{B}$ direction, from which we estimate a spin-flip tunneling angle $\theta_{\text{SO}} = 14.7(6)^\circ$, similar to values extracted from the exchange interaction in other works [42, 44] (see Supplementary section S6). Based on this model, we calculate the effective Larmor vector of the FM qubit $\mathbf{l}(\bar{V}_{P1}, \bar{V}_{P2})$, near $\varepsilon_{12} = 0$, assuming linear gate-voltage dependence of the g -tensor [9]. This is used to characterize the longitudinal spin-electric susceptibility [6, 35] (LSES) of the FM qubit at $\varepsilon_{12} = 0$ with respect to $\bar{V}_{Pi}$, given by $\beta_{||,Pi} = \frac{d\mathbf{l}}{dV_{Pi}} \cdot \mathbf{l}/(h|\mathbf{l}|)$, as illustrated in the top panel of Fig. 2a. The LSES measures the sensitivity of the qubit frequency to gate voltage, which implies that minimizing $|\beta_{||,Pi}|$ is beneficial for suppressing

charge-noise-induced dephasing.

The bottom panel in Fig. 2a presents the calculated $|\beta_{||,P2}|$, as a function of the magnetic field orientation (see Supplementary S7 for $\beta_{||,P1}$). Here, ϕ_{avg} and θ_{avg} are spherical coordinates, used to denote the $\mathbf{B}$ direction in the average g -tensor frame of g_1 and g_2 , with $\theta_{\text{avg}} = 90^\circ$ corresponding to the xy (sample) plane of the average frame (see Supplementary Fig. S3). Importantly, the white dashed curve in Fig. 2a marks the contours $\beta_{||,P2} = 0$, or LSES sweet-lines [35], where the Zeeman energies of the two QDs are degenerate at $\varepsilon_{12} = 0$, resulting in a first-order detuning sweet-spot. We note that the geometry of the FM qubit sweet-line resembles that of the LD qubits in QD₁ and QD₂, with a sweet-spot generally occurring at small out-of-plane $\mathbf{B}$ tilts (see Supplementary Fig. S5).

To confirm the noise suppression at the sweet-line, we measure the longitudinal qubit coherence times T_2^* and

T_2^{Hahn} as a function of ϕ_{avg} and θ_{avg} at $\varepsilon_{12} = 0$, around the region denoted by the red square in Fig. 2a, for fixed $|\mathbf{B}| = 5$ mT and $t_c/h = 13$ GHz (see Supplementary section S13). At each field direction, the control pulses and $\varepsilon_{12} = 0$ are calibrated as detailed in Supplementary section S3. Representative coherence decay traces are shown in the top panels of Fig. 2b,c for the $\mathbf{B}$ orientation $\hat{\mathbf{b}}_1$ (blue point on the sweet-line in Fig. 2a), yielding $T_2^* = 1.4$ μs and $T_2^{\text{Hahn}} = 11.2$ μs from a fit to a generalized exponential decay model (see Supplementary Equation S10). The full maps in Fig. 2b,c show that the coherence maxima track the calculated LSES sweet-line over magnetic field orientation, directly demonstrating first-order suppression of charge-noise sensitivity. Along the sweet-line, T_2^* reaches up to 1.5 μs and T_2^{Hahn} about 15 μs . T_2^* reduces rapidly away from the sweet-line, whereas Hahn echo coherence is more robust, especially at larger out-of-plane angles.

Importantly, suppressing the longitudinal susceptibility does not compromise electrical control. We define the Rabi driving efficiency $\eta_d = f_{\text{Rabi}}/A_{d,P2}$ and the normalized Rabi driving efficiency $\tilde{\eta}_d = f_{\text{Rabi}}/(A_{d,P2}f_{\text{FM}})$. From the same chevrons used to obtain the coherence maps (Fig. 2b,c), we extract $\eta_d > 5$ MHz/mV and $\tilde{\eta}_d > 0.15$ Hz/(Hz mV) along the sweet-line (Fig. 2d, e). $\tilde{\eta}_d$ is two to three orders of magnitude larger than values reported for g -TMR-driven LD qubits [7, 12, 36]. Noting that the power of the qubit drive tone approximately scales as V_{drive}^2 [9, 13], the significantly enhanced η_d and $\tilde{\eta}_d$ of the FM qubit facilitate minimally dissipative qubit operations, as will be clear in the comparison with LD spin qubits later in Fig. 5.

The origin of the large transverse spin-electric susceptibility (TSES), defined as $\beta_{\perp, P_i} = |\frac{d\mathbf{l}}{dV_{P_i}} \times \mathbf{l}|/(h|\mathbf{l}|)$, is revealed by the relative orientation of the local Larmor vectors. At the sweet-line their magnitudes are nearly matched, whereas their directions remain misaligned. Resonant gate modulation therefore changes primarily the direction of the effective Larmor vector, while leaving its magnitude comparatively insensitive, analogous to iso-Zeeman EDSR [49–51]. In this regime, $\beta_{\perp, P_i}$ scales approximately as $f_{\text{FM}}|\sin(\Delta\hat{\theta}_{12})|$, where $\Delta\hat{\theta}_{12}$ is the rotation of the effective Larmor vector with respect to the change of a gate-voltage by ΔV_{P2} (see top panel in Fig. 2a). The calculated $|\sin(\Delta\hat{\theta}_{12})|$ map in Fig. 2f closely follows the measured angular dependence of $\tilde{\eta}_d$ in Fig. 2e. Thus, the same inter-dot Larmor-vector misalignment that provides strong transverse electrical control can coexist with longitudinal noise protection. The key feature of the FM operating point is therefore that the longitudinal and transverse electrical susceptibilities can be engineered separately: matching the Larmor-vector magnitudes suppresses dephasing, while their finite relative angle preserves strong electrical driving.

For the remainder of the study, we focus on three representative $\mathbf{B}$ directions on the sweet-line (see Fig. 2a) denoted by unit vectors $\hat{\mathbf{b}}_1$ (blue), $\hat{\mathbf{b}}_2$ (pink) and $\hat{\mathbf{b}}_3$ (green).

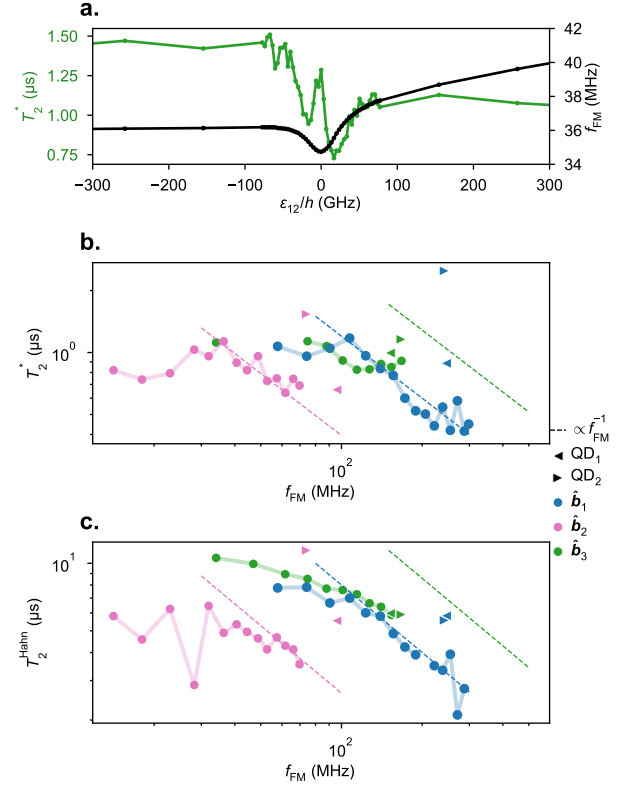


FIG. 3. **a.** FM qubit frequency f_{FM} (black) and T_2^* (green) as a function of ε_{12} , extracted from the same Ramsey measurement, with $|\mathbf{B}| = 5$ mT and $t_c/h = 13$ GHz. **b.** **(c.)** T_2^* (T_2^{Hahn}) as a function of f_{FM} for the $\mathbf{B}$ orientations $\hat{\mathbf{b}}_1$ (blue), $\hat{\mathbf{b}}_2$ (pink), and $\hat{\mathbf{b}}_3$ (green), and $t_c/h = 13$ GHz. Each dashed line is proportional to $1/f_{\text{FM}}$ with prefactors related by the extracted second derivatives of the qubit spectra (see text). The LD qubit coherence times in QD₁₍₂₎ are measured at the center of (1,0,1) and (0,1,1) represented with left (right) pointing triangles.

Within the angular range in Fig. 2b-f, $\hat{\mathbf{b}}_1 = (\phi_{\text{avg}}, \theta_{\text{avg}}) = (32.36^\circ, 93.16^\circ)$ maximizes f_{Rabi} , $\hat{\mathbf{b}}_2 = (91.80^\circ, 90.83^\circ)$ maximizes $f_{\text{Rabi}}/f_{\text{FM}}$, and $\hat{\mathbf{b}}_3 = (120.6^\circ, 92.25^\circ)$ gives the lowest Rabi frequency. We also fix $|\mathbf{B}| = 5$ mT and $t_c/h = 13$ GHz unless stated otherwise.

Coherence scaling

We next examine the mechanisms limiting qubit coherence at the detuning sweet-spot. To first explicitly demonstrate the charge detuning noise protection at the sweet-spot, we present f_{FM} and T_2^* as a function of ε_{12} for $\hat{\mathbf{b}}_3$ in Fig. 3a, using a Ramsey sequence with an interleaved detuning pulse. As expected, f_{FM} exhibits a minimum at $\varepsilon_{12} = 0$ (i.e. $\partial f_{\text{FM}}/\partial \varepsilon_{12} = 0$). Accordingly, T_2^* exhibits local minima at the flanks (where $|\partial f_{\text{FM}}/\partial \varepsilon_{12}|$ is maximal), and a maximum at $\varepsilon_{12} = 0$, clearly demonstrating the protection against detuning noise. We note

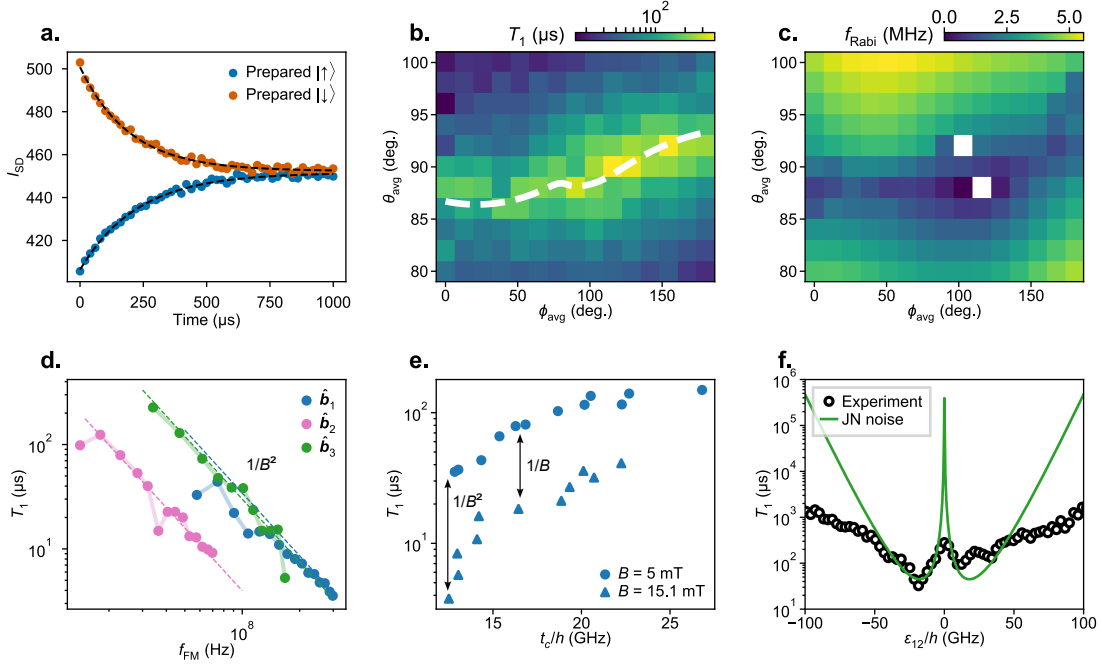


FIG. 4. **a.** Spin relaxation for $\mathbf{B}$ direction $\hat{\mathbf{b}}_3$, $|\mathbf{B}| = 5$ mT, $t_c/h = 13$ GHz. Measured for spin-up and spin-down initializations. **b.** T_1 dependence on $\mathbf{B}$ direction for spin-down initialization. Dashed line shows analytical T_1 sweet-spot (see Supplementary section S172). **c.** Rabi frequency dependence on $\mathbf{B}$ direction with $A_{d,P2} = 600$ μ V. **d.** T_1 as a function of qubit frequency at zero detuning, measured for directions $\hat{\mathbf{b}}_{1,2,3}$. Dashed lines are obtained as $T_1 = \tilde{\eta}_d^{-2} k f_{\text{FM}}^{-2}$, where k is a fixed constant for all three $\mathbf{B}$ directions. **e.** Scaling of T_1 with tunnel coupling t_c for $\hat{\mathbf{b}}_1$ and two different magnetic field magnitudes $|\mathbf{B}| = 5$ mT and 15.1 mT. At $t_c/h = 13$ GHz the $\mathbf{B}$ scaling of T_1 follows $1/|\mathbf{B}|^2$, consistent with **d.**, while for higher t_c it appears to satisfy $1/|\mathbf{B}|$. **f.** Scaling of relaxation time with energy detuning ε_{12} . Green curve corresponds to the calculated T_1 , assuming a Johnson-Nyquist (JN) noise with a characteristic impedance of 250 Ω and temperature of 300 mK.

that the T_2^* at $\varepsilon_{12} = 0$ remains comparable to that of the LD qubits in the deeply detuned single-QD regimes. This suggests that T_2^* of the qubit both in the FM and LD regimes is mainly limited by hyperfine noise for the given $\mathbf{B}$ [6, 52, 53].

In Supplementary section S9 we present an analysis of the noise power spectral density (PSD) of the FM qubit extracted using CPMG and Ramsey measurements for $\hat{\mathbf{b}}_3$ [54, 55]. Sweet-spot operation suppresses the noise by about an order of magnitude compared to flank operation at frequencies above 100 kHz. We further observe an extension of the longitudinal coherence to $T_2^{\phi, \text{CPMG}} = 130(3)$ μ s with 32 refocusing pulses.

The scaling of the coherence times with $|\mathbf{B}|$ provides additional insights about the origin of decoherence mechanisms [6]. We vary $|\mathbf{B}|$ in the range 5 mT to 25 mT for the $\mathbf{B}$ orientations $\hat{\mathbf{b}}_1$, $\hat{\mathbf{b}}_2$ and $\hat{\mathbf{b}}_3$ and present the measured T_2^* and T_2^{Hahn} as a function of f_{FM} in Fig. 3b, and c. For relatively low qubit frequencies ($f_{\text{FM}} < 100$, 40, and 200 MHz for $\hat{\mathbf{b}}_1$, $\hat{\mathbf{b}}_2$ and $\hat{\mathbf{b}}_3$ respectively) the coherence plateaus or decreases with lower f_{FM} , which we attribute to the finite broadening of the hyperfine line approaching the Larmor frequency of the ^{73}Ge isotope [6, 55]. We also note that T_1 is not limiting the coherence times in this experiment, with the corresponding values later discussed

in Fig. 4d.

At higher frequencies, the coherence times T_2^* and T_2^{Hahn} appear to follow $T_2 \propto 1/f_{\text{FM}}$ for $\hat{\mathbf{b}}_1$. This is consistent with the trend observed for charge-noise limited coherence times in planar Ge LD qubits [6, 9]. The coherence times at the FM sweet-spot are expected to follow $T_2 = 1/(\sigma_\varepsilon^2 \chi f_{\text{FM}})$, for a value $\chi = |\frac{\partial^2 f_{\text{FM}}}{\partial \varepsilon^2}|/f_{\text{FM}}$ that is assumed to be frequency-independent, and an effective detuning noise amplitude σ_ε [24, 30]. We extract χ from the measured qubit spectra at specific $|\mathbf{B}|$ values, which allows for a comparison between charge-noise limited coherence times for the respective $\mathbf{B}$ orientations. Assuming constant σ_ε^* and $\sigma_\varepsilon^{\text{Hahn}}$, same for the different $\mathbf{B}$ orientations (visually fitted), we present the calculated $1/((\sigma_\varepsilon^*)^2 \chi f_{\text{FM}})$ and $1/((\sigma_\varepsilon^{\text{Hahn}})^2 \chi f_{\text{FM}})$ in Fig. 3b and c (dashed lines). Apparently, for $\hat{\mathbf{b}}_1$ and $\hat{\mathbf{b}}_2$, the measured coherence times asymptotically approach the charge-noise-limited line for higher qubit frequencies. For the orientation $\hat{\mathbf{b}}_3$ the model predicts a later onset of charge-noise-limited decay, beyond the measured range. This suggests that the qubit coherence in this regime is instead hyperfine-limited, consistent with the previous observation in Fig. 3a.

We additionally investigate the coherence of the LD qubits in QD₁ and QD₂ by evaluating T_2^* and T_2^{Hahn} at

the nominal center of the (1,0,1) and (0,1,1) charge regions. The measurements are taken for $\hat{\mathbf{b}}_1$, $\hat{\mathbf{b}}_2$ and $\hat{\mathbf{b}}_3$ with $|\mathbf{B}| = 20$ mT, with the results shown as triangles in Fig. 3b,c. For the cases where charge noise is expected to limit the FM qubit coherence ($\hat{\mathbf{b}}_1$ and $\hat{\mathbf{b}}_2$ orientations), the LD qubit coherence time is significantly larger than that of the FM qubit. However, for $\hat{\mathbf{b}}_3$ the coherence of the FM qubit and the LD qubits are almost identical, further supporting our hypothesis that the FM qubit is limited by hyperfine interactions in this regime.

Spin relaxation

To understand the physical process limiting the spin lifetime, we measure the relaxation time T_1 as a function of $\mathbf{B}$, ε_{12} and t_c . Fig. 4a shows the measured T_1 decay at $\varepsilon_{12} = 0$ for $\hat{\mathbf{b}}_3$, $|\mathbf{B}| = 5$ mT and $t_c/h = 13$ GHz. Here, we prepare either a $|\uparrow\rangle$ or a $|\downarrow\rangle$ state, and measure its population decay as a function of time as represented by the orange (initialized in $|\uparrow\rangle$) and blue (initialized in $|\downarrow\rangle$) datapoints. They both decay into the maximally mixed state with a similar decay time of $T_1 \approx 226$ μ s.

In Supplementary section S16, we present a comprehensive treatment of T_1 relaxation of the FM qubit taking into account thermal photons (e.g. from gate lines and insufficient packaging), lattice phonons and charge noise. By absorbing a photon or a phonon close to the charge transition frequency $2t_c/h$, an FM qubit can be excited from the ground ($|+\rangle$) to the excited ($|-\rangle$) charge state manifold. Then, this excited charge state can relax back to the FM qubit subspace (i.e. the ground charge state) by emitting a phonon or a photon in a mechanism known as the Orbach process [37]. Because each of the absorption and emission events can be accompanied by a spin flip due to the effective transverse magnetic field gradient, the second-order process can result in the observed decay into the mixed state. We note that the first-order process involving direct emission or absorption of phonons at the qubit frequency is negligible in our regime $f_{\text{FM}} \sim 10 - 100$ MHz due to the reduced density of states of the environmental bath. In our setup, we expect the thermal photons, rather than the lattice phonons, to play the dominant role in the second-order process. This is because lattice phonons are expected to be well-thermalized to the base temperature of the cryostat $T_{\text{base}} \sim 10$ mK ($k_B T_{\text{base}}/h \sim 200$ MHz $\ll 2t_c/h$).

In Fig. 4b we present the measured T_1 decay as a function of the $\mathbf{B}$ field direction. Here, the qubit is prepared in the $|\downarrow\rangle$ state and $|\mathbf{B}|$ is kept constant at 5 mT. The white dashed line corresponds to the calculated contour where T_1 is expected to maximize due to a reduction of the effective transverse magnetic field gradient (see Supplementary section S16). It closely follows the maximum of the measured T_1 . Fig. 4c presents the corresponding Rabi frequency for each field orientation, which is showing an inverse dependence to the measured T_1 [24]. This confirms that the effective transverse magnetic field gra-

dient, which drives the qubit excitation, also mediates the relaxation process.

Fig. 4d shows T_1 measured for the $\mathbf{B}$ orientations $\hat{\mathbf{b}}_1$, $\hat{\mathbf{b}}_2$ and $\hat{\mathbf{b}}_3$ (see Fig. 2a), while varying f_{FM} with the magnetic field strength. Dashed lines correspond to a fit to $T_1 = \tilde{\eta}_d^{-2} k f_{\text{FM}}^{-2}$, where k is a fixed constant and $\tilde{\eta}_d$ is the normalized Rabi driving efficiency extracted separately for each orientation. We observe a quadratic decay of T_1 , which is consistent with either the second-order photon (or phonon) process, or first-order coupling to photons, and rules out $1/f$ charge noise (see Supplementary section S16) [30].

Since the Orbach process depends on a finite thermal population of the excited charge state it should be exponentially suppressed for higher values of t_c . We measure T_1 as a function of t_c/h at $\varepsilon_{12} = 0$ at the $\mathbf{B}$ orientation $\hat{\mathbf{b}}_1$ for two different magnetic field strengths $|\mathbf{B}| = 5$ and 15.1 mT. The results are shown in Fig. 4e. Indeed, we observe that T_1 increases with t_c for both magnetic field strengths. For relatively low $t_c/h \lesssim 15$ GHz, the T_1 at $|\mathbf{B}| = 5$ mT is approximately nine times larger than that at $|\mathbf{B}| = 15.1$ mT, which is consistent with the $1/f_{\text{FM}}^2$ trend shown in Fig. 4d. However, for $t_c/h \gtrsim 15$ GHz, the T_1 at $|\mathbf{B}| = 5$ mT is about three times larger than the value at $|\mathbf{B}| = 15.1$ mT, suggesting instead a $1/f_{\text{FM}}$ scaling of the T_1 . This trend points towards the Orbach process being suppressed at larger values of t_c , with relaxation instead dominated by charge noise which results in $T_1 \sim 1/f_{\text{FM}}$.

Next, the T_1 is extracted as a function of ε_{12} (Fig. 4f) for $\mathbf{B}$ orientation $\hat{\mathbf{b}}_3$ and $|\mathbf{B}| = 5$ mT. At zero detuning the T_1 is enhanced, which is a hallmark of the second-order process (see Supplementary section S16). A theory calculation superimposed on the data (green curve) qualitatively captures the trend. Here, we have assumed a Johnson-Nyquist noise generated by five 50 Ω rf gates acting as an effective 250 Ω noise source to qualitatively capture the trend. The data was then fitted by setting a photonic bath temperature of 300 mK. The asymmetry in the data may be due to a two-level fluctuator displacing the inter-dot detuning during the measurement (as discussed in Supplementary section S12). We note that the model predicts a sharply peaked T_1 exactly at $\varepsilon_{12} = 0$, where the second-order process is negligible. We attribute this discrepancy to the model not considering the low-frequency charge noise, which effectively modulates ε_{12} . Given that the second-order relaxation process rapidly increases away from $\varepsilon_{12} = 0$, a small detuning modulation can effectively reduce the measured T_1 . This suggests that reducing low-frequency charge noise can not only improve longitudinal coherence times of the qubit, but also indirectly enhance T_1 .

High-fidelity, ultra-low-power EDSR

The qubit fidelity is characterized using Clifford randomized benchmarking [56], for the $\mathbf{B}$ orientations $\hat{\mathbf{b}}_1$ and $\hat{\mathbf{b}}_3$, $|\mathbf{B}| = 5$ mT and $t_c/h = 13$ GHz. Decay traces of the average survival probability after the recovery Clifford are shown in Fig. 5a,b. Here, the drive amplitudes were fixed to 460 μ V and 550 μ V resulting in f_{Rabi} of 5.7 MHz and 3.5 MHz, at $f_{\text{FM}} = 58.8$ MHz and $f_{\text{FM}} = 33.7$ MHz respectively. The drive amplitudes were chosen to satisfy the validity of the rotating wave approximation, while maximizing gate speed. The average physical gate fidelities are evaluated to $\mathcal{F}_{1\text{qb}} = 99.76(1)\%$ and $99.74(1)\%$ as outlined in Supplementary section S15. The corresponding qubit coherence metrics are summarized in Supplementary Table S2. Our results demonstrate that flopping-mode EDSR is compatible with high-fidelity operations, while simultaneously reducing control power by several orders of magnitude compared to g -TMR operation at similar qubit frequencies [7, 8, 12, 36].

We further illustrate this advantage by explicitly comparing our results with those reported in the literature. In Fig. 5c, we plot V_{drive}^2 as a function of the gate infidelity $1 - \mathcal{F}_{1\text{qb}}$, where V_{drive} denotes the rf drive amplitude at the device. We compile results from recently reported Ge hole spin qubits operated at magnetic fields below 100 mT, with the field oriented approximately in the sample plane [7, 8, 12, 36]. Each point is color-coded according to the qubit frequency reported in each study. The stars, indicated by a red arrow, represent the FM qubit in this work for the two configurations shown in Fig. 5a,b. We observe about a two-order-of-magnitude reduction in drive power for the FM qubit compared to typical LD qubits, while maintaining comparable gate performance. We also show a comparison to an LD qubit (marked with a triangle) measured in our device in QD₂ and driven with the same gate ($\bar{V}_{\text{P2}}$) as the FM qubit, with a field of $|\mathbf{B}| = 15$ mT oriented along the in-plane hyperfine sweet-spot (see Supplementary section S12). Notably, we observe an improvement by a factor of three in the infidelity of the FM qubit, while drive power is reduced by two orders of magnitude. Assuming a linear scaling of f_{Rabi} with f_{qubit} and drive amplitude, we infer a four-order-of-magnitude reduction in driving power for the FM qubit compared to an LD qubit, when operated at the same f_{Rabi} and f_{qubit} .

CONCLUSIONS

We have demonstrated a flopping-mode hole-spin qubit in a Ge DQD, showing that it can support high-fidelity EDSR control of a single spin at low magnetic fields, while reducing the required drive power by two orders of magnitude compared to state-of-the-art single-QD hole spin qubits in Ge. We systematically map the magnetic field sweet-lines where the Zeeman energies of the two QDs are degenerate and the qubit becomes first-

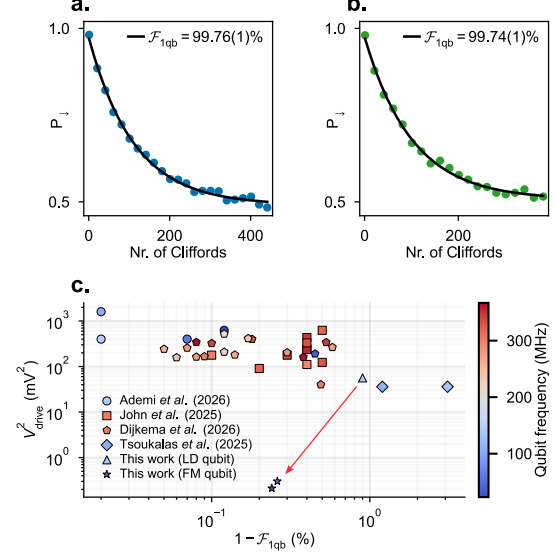


FIG. 5. **a.** **(b.)** Single-qubit Clifford randomized benchmarking experiments for $\mathbf{B}$ orientations $\hat{\mathbf{b}}_1$ ($\hat{\mathbf{b}}_3$) for $|\mathbf{B}| = 5$ mT and $t_c/h = 13$ GHz. The qubit is driven with an rf amplitude $A_{d,\text{P2}} = 460$ μ V (550 μ V) at the device. The average physical gate fidelity calculated from the exponential fits (black lines) is shown in the insets. Corresponding longitudinal and transverse coherence metrics are summarized in Supplementary Table S2. **c.** Compiled data from references [7, 8, 12, 36], showing the square of the drive amplitude at the device, V_{drive}^2 and the physical single-qubit gate infidelity. Stars denote the two measured configurations in **a.** and **b.** Triangle denotes an LD qubit measured at the hyperfine sweet-spot in the same device (Supplementary section S12). Points are color coded according to qubit frequency.

order insensitive to detuning fluctuations. The coherence measurements on the sweet-lines reveal a crossover from a low-field regime consistent with hyperfine-limited dephasing to a higher-field regime governed by second-order charge-noise coupling. The key feature of these operating points is the ability to separately engineer the longitudinal and transverse spin-electric susceptibilities: matching the Larmor vector magnitudes between the two QDs significantly reduces dephasing, while their finite relative angle preserves a large transverse spin-electric susceptibility. This allows Rabi control at MHz frequencies with sub-millivolt drive amplitudes at the device, while operating at a low magnetic field where the qubit coherence is optimized. Randomized benchmarking yields physical-gate fidelities of up to 99.76(1)%, similar to median values reported for LD qubits in state-of-the-art Ge devices, while maintaining a high gate speed of $t_{X\pi} = 88$ ns [7, 12].

The relaxation measurements expose a complementary trade-off: the transverse Larmor-vector gradient that enables efficient control also opens a relaxation channel. Our modeling indicates that thermal photons are the

dominant relaxation mechanism in the present setup. Improved control-line filtering, radiation-tight packaging, and larger tunnel couplings could enhance T_1 by reducing thermal excitations, while lower charge noise would reduce fluctuations away from the FM operating point ($\varepsilon_{12} = 0$) thereby allowing the qubit to better exploit the sharp T_1 enhancement at this point.

These results suggest a practical operating strategy for larger spin-qubit arrays: the FM regime could be activated for ultra-low-power resonant control, while idling and two-qubit operations could be performed in the LD qubit regime, where longer coherence and spin lifetime may be available. Because the power reduction is obtained per driven qubit, its absolute benefit grows with the number of simultaneously operated qubits and directly addresses the aggregate rf-power budget of large semiconductor-qubit arrays. Highly uniform neighboring g -factor magnitudes would be advantageous for realizing FM sweet spots across an array, while gate-voltage tunability of the g -tensors provides an additional degree of freedom for engineering the finite inter-dot Larmor-vector tilt required for efficient EDSR and sweet-spot operation [12, 20, 57]. Isotopically purified heterostructures may enable FM-qubit operation in out-of-plane fields [53], where more uniform Larmor vectors are expected and the SOI vector remains perpendicular to the quantization axis, enhancing the drive via spin-flip tunneling.

Beyond scalable control, the large electric dipole of the FM qubit is also attractive for coherent spin-photon interfaces at higher qubit frequencies, and the high driving efficiency opens access to spin dynamics beyond the

rotating-wave approximation [33, 58].

Acknowledgment

We acknowledge the staff of the Binnig and Rohrer Nanotechnology Center (BRNC) for their contributions to the sample fabrication, and Alberto Bordin for providing feedback towards the manuscript. P.S. acknowledges support from the Swiss State Secretariat for Education, Research and Innovation (SERI) under contract number MB22.00081. This research was funded in part by NCCR SPIN, a National Centre of Competence in Research, funded by the Swiss National Science Foundation (grants 51NF40-180604 and 51NF40-225153) and by the Swiss National Science Foundation (grant 200021-188752).

Author contributions

AO and WJ conceptualized the experiment. AO performed the measurements with the support of WJ and KT. AO and WJ analyzed the data. PZ and SB developed the relaxation theory model. FO, FDP, EA, YR, WJ and AO implemented the experimental set-up. MM and FJS fabricated the device. IS, LM, LS, KT and AO contributed to device testing and development. KT designed the device. AO and WJ wrote the manuscript with input from all authors. PHC and PS supervised the project.

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SUPPLEMENTARY MATERIALS FOR: COHERENT AND ULTRA-LOW-POWER EDSR WITH A FLOPPING-MODE SPIN QUBIT IN GERMANIUM

S1. Experimental setup

The measurements are performed in a Bluefors LD25 dilution refrigerator (bottom-loader) with a base temperature of around 10 mK, equipped with a three-axis superconducting vector magnet (American Magnetics) with maximum range $\pm(1,1,6)$ T for (z,x,y) directions (as defined in Fig. 1a) operated in direct current mode. As shown in the measurement setup schematic in Supplementary Fig. S1, the dc gate voltages are applied with a 24-channel QDevil QDAC-II digital-to-analog converter (DAC) through twisted pair cables made of phosphor bronze. This is followed by a low-pass filtering stage at the mixing chamber plate using a QDevil filter box (225 MHz cut-off) and on-PCB low-pass filtering (50 kHz cut-off). We note that the PCB in this experiment is not light-tight (there is no sample lid), which offers a possible route to the improvement of T_1 . RF pulsing and readout are performed using the Quantum Machines OPX1000, passing through a total attenuation of 26 dB and combined with the dc voltages through on-PCB bias tees (0.1 s time constant). The source and drain of the SHT are connected to two individual Basel Precision Instruments SP 983C IV converters with a gain of 1×10^6 V/A and 30 kHz bandwidth. The voltage signals are subtracted and amplified using a Stanford Research Systems SR560 differential amplifier with a gain of 10^2 and 100 kHz bandwidth. The 50 Ω output port of the amplifier is fed to a readout channel of the OPX1000 for signal integration, while the 600 Ω output port is fed into a Keysight 34465A digital multimeter (DMM) for acquiring dc current maps.

S2. Initialization and readout

The dc voltage setpoint is set to approximately the middle of the (0,1,1) charge region. Initialization into $|\downarrow, \downarrow\rangle_{\text{QD2, QD3}}$ is performed by ramping with a 3 μs ramp (adiabatic with respect to the ST^- anti-crossing) into the (0,0,2) charge region and waiting for 15 μs to ensure spin relaxation, before symmetrically ramping back to the (0,1,1) region. This is followed by a voltage pulse of approximately +8 mV on the virtual barrier $\bar{V}_{\text{B23}}$, which ensures the suppression of the exchange interaction with the ancilla spin. We then ramp down the voltage on the virtual barrier $\bar{V}_{\text{B12}}$ using a ramp of 5000 ns, by a value of approximately -15 mV, depending on the target t_c . This is followed by a ramp to the charge symmetry point between (1,0,1) and (0,1,1), the operating point of the FM qubit, using a 500 ns ramp that is adiabatic with respect to t_c . For all measurements except T_1 measurements: Before performing gates we wait for a default duration of 3.5 μs . This ensures stabilization of the voltage level. For T_1 measurements: We perform an X_π pulse immediately after ramping to the charge symmetry point. After performing the qubit operations we symmetrically ramp back to (0,1,1) and reverse the sequence of ramps on $\bar{V}_{\text{B12}}$ and $\bar{V}_{\text{B23}}$. This is followed by an adiabatic 3 μs ramp to the PSB region between (0,1,1) and (0,0,2), followed by a 16 ns ramp to the latched region in (0,1,2) for readout. We perform current-based SHT readout (see experimental setup section) with a 50 μs settling time, followed by a 50 μs integration time.

The readout fidelity is strongly dependent on magnetic field magnitude and orientation. This may be due to relaxation and mapping errors inherent to the latched PSB readout [11]. We do not observe any significant reduction in the visibility from detuning to the inter-dot unless the wait time is significant compared to the T_1 time.

S3. Detuning and gate calibration

The FM qubit operation point is sensitive to the inter-dot detuning which is subject to temporal drifts and jumps. We therefore rely on a modified spin-echo protocol to re-calibrate the detuning setpoint:

$$X_{\pi/2} - \Delta\bar{V}_{\text{P2}}(\tau) - Y_\pi - \tau - Z(\varphi) - X_{\pi/2}. \quad (\text{S1})$$

Here, the rf-driven gates are performed at the FM qubit operating point ($\varepsilon_{12} = 0$). $\Delta\bar{V}_{\text{P2}}$ denotes the detuning from the sweet-spot, given by an adiabatic pulse on $\bar{V}_{\text{P2}}$ of duration τ . A virtual phase update $\varphi = 2\pi f_{\text{virt}}\tau$ is applied before the final spin projection for easier fitting. When sweeping $\Delta\bar{V}_{\text{P2}}$, a phase is picked up when the qubit is displaced from its initial detuning position. This results in a change in the frequency of the probability oscillations as demonstrated in a representative calibration measurement in Fig. S2a. The oscillations are fitted for each detuning

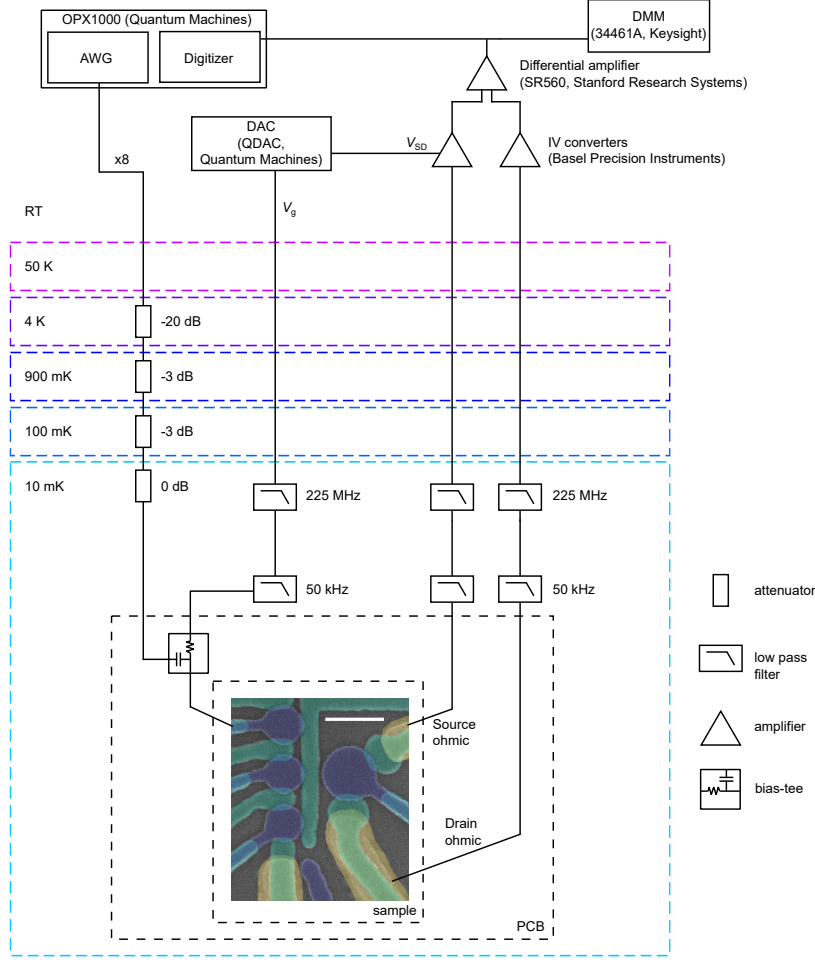


FIG. S1. Measurement setup schematic.

value, with the corresponding deviation from f_{virt} plotted in Fig. S2b. We fit the data to Eq. (S21) and use this to find the qubit frequency minimum for achieving sweet-spot operation. The zero-detuning point ($\varepsilon_{12} = 0$) and t_c are calibrated using the same protocol, but taking a wider range of detunings ($|\varepsilon_{12}| \gg t_c$), with an example shown in Fig. S9c.

S4. Magnetic field offset calibration

The magnetic field shows hysteresis, which results in sub-mT offsets that change approximately linearly with applied magnetic field strength [6, 42]. Due to the large out-of-plane g -tensor tilts (1.9° and 3.7°) in our sample it is important to initially calibrate the offsets in all three axes of the vector magnet. The magnetic-field offsets are determined recursively for the axes $i \in \{x, y, z\}$. The nominal field B_i is swept while the remaining field components are held fixed, and the field that minimizes the qubit frequency, $B_{\text{min},i}^+$, is extracted from spectroscopy measurements. The signs of the two orthogonal field components are then reversed and the measurement is repeated to obtain $B_{\text{min},i}^-$. The offset along axis i is estimated as $B_{\text{offset},i} = (B_{\text{min},i}^+ + B_{\text{min},i}^-)/2$. The procedure is performed sequentially for z , y , and x , and the entire sequence is repeated twice to refine the offset estimates. We find the offsets $B_{\text{offset},z} = -297 \mu\text{T}$, $B_{\text{offset},y} = -473 \mu\text{T}$ and $B_{\text{offset},x} = 402 \mu\text{T}$. Subsequent corrections are done using only the z component as in other works.

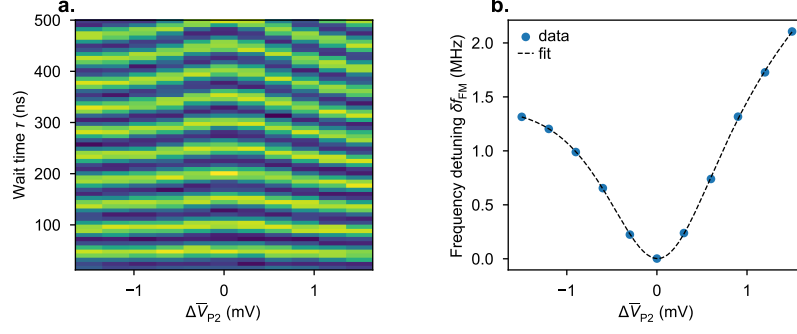


FIG. S2. **a** Representative example of a Hahn echo measurement used to calibrate the detuning sweet-spot condition, with $\Delta\bar{V}_{P2}$ denoting the deviation from a previous calibration of the sweet-spot. **b** Qubit frequency detuning extracted from the data in **a** using a cosine fit. Black dashed line corresponds to a fit to the FM model Hamiltonian. The point of zero derivative of the model is used to extract the new sweet-spot condition (in this case closely coinciding with the previous calibration).

S5. Flopping-mode qubit Hamiltonian

In this section we describe the comprehensive model that takes into account the g -tensors of QD₁ and QD₂, their gate-voltage dependence, and the spin-orbit interaction [6, 42, 44, 45], to describe the dynamics of the FM qubit in our system.

The Hamiltonian of an FM qubit which does not take into account the spin-flip tunneling term due to the spin-orbit interaction (SOI) is typically written as (S2) [24].

$$H_0 = \frac{\varepsilon_{12}}{2}\tau_z + t_0\tau_x + \sum_{i=1,2} \frac{\mu_B \mathbf{B} \cdot g_i(\bar{V}_{P1}, \bar{V}_{P2}) \boldsymbol{\sigma}}{2} \frac{\tau_0 + (-1)^i \tau_z}{2}, \quad (\text{S2})$$

Here, τ_i corresponds to the Pauli- i operator in the charge basis $|L\rangle = |(1, 0, 1)\rangle$ and $|R\rangle = |(0, 1, 1)\rangle$, and $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ is the Pauli spin operator. $g_{1(2)}(\bar{V}_{P1}, \bar{V}_{P2})$ is the g -tensor of QD₁₍₂₎ dependent on the gate-voltages $\bar{V}_{P1}$ and $\bar{V}_{P2}$ in the lab frame, where we assume a linear dependence of g -tensors on the gate-voltages as shown in (S3).

$$g_{1(2)}(\bar{V}_{P1}, \bar{V}_{P2}) = g_{1(2)}(\bar{V}_{P1}^{1(2),0}, \bar{V}_{P2}^{1(2),0}) + \frac{\partial g_{1(2)}}{\partial \bar{V}_{P1}}(\bar{V}_{P1} - \bar{V}_{P1}^{1(2),0}) + \frac{\partial g_{1(2)}}{\partial \bar{V}_{P2}}(\bar{V}_{P2} - \bar{V}_{P2}^{1(2),0}) \quad (\text{S3})$$

Here, $\bar{V}_{P1}^{1(2),0}, \bar{V}_{P2}^{1(2),0}$ are the dc-voltages at which $g_{1(2)}$ were characterized by sweeping the magnetic field orientation [6]. We show the extraction of $g_{1(2)}(\bar{V}_{P1}^{1(2),0}, \bar{V}_{P2}^{1(2),0})$, $\partial g_{1(2)}/\partial \bar{V}_{P1}$ and $\partial g_{1(2)}/\partial \bar{V}_{P2}$ in Supplementary section S6.

SOI results in a spin-flip tunneling term (S4) [59, 60].

$$H_{\text{SOI}} = t_0 \tan(\theta_{\text{so}}) \tau_y \mathbf{n}_{\text{so}} \cdot \boldsymbol{\sigma} \quad (\text{S4})$$

where $\mathbf{n}_{\text{so}}$ is a spin-orbit vector whose norm is unity, θ_{so} is the spin-orbit angle which describes the amount of the spin rotation around $\mathbf{n}_{\text{so}}$ in case of a tunneling event. The effective tunnel coupling t_c is equivalent to $t_c = \sqrt{t_0^2 + t_0^2 \tan^2(\theta_{\text{so}})} = t_0 / \cos(\theta_{\text{so}})$.

The total Hamiltonian of the FM qubit is written as (S5).

$$H_{\text{FM}} = H_0 + H_{\text{SOI}} \quad (\text{S5})$$

In the spin-orbit frame, where the g -tensors are effectively transformed by $g_1^{\text{so}} = g_1 R_{\text{so}}(\theta_{\text{so}})$ and $g_2^{\text{so}} = g_2 R_{\text{so}}(-\theta_{\text{so}})$, the Hamiltonian reduces to Eq. (S6) [60]. $R_{\text{so}}(\theta)$ is a matrix representing the rotation around $\mathbf{n}_{\text{so}}$ by an angle θ .

$$H_{\text{FM}} = \frac{\varepsilon_{12}}{2}\tau_z + t_c\tau_x + \sum_{i=1,2} \frac{\mu_B \mathbf{B} \cdot g_i^{\text{so}}(\bar{V}_{P1}, \bar{V}_{P2}) \boldsymbol{\sigma}}{2} \frac{\tau_0 + (-1)^i \tau_z}{2}, \quad (\text{S6})$$

We note that we have neglected contributions from magnetotunneling due to the low qubit frequencies considered here [61].

S6. g -tensor characterization and spin-orbit interaction measurement

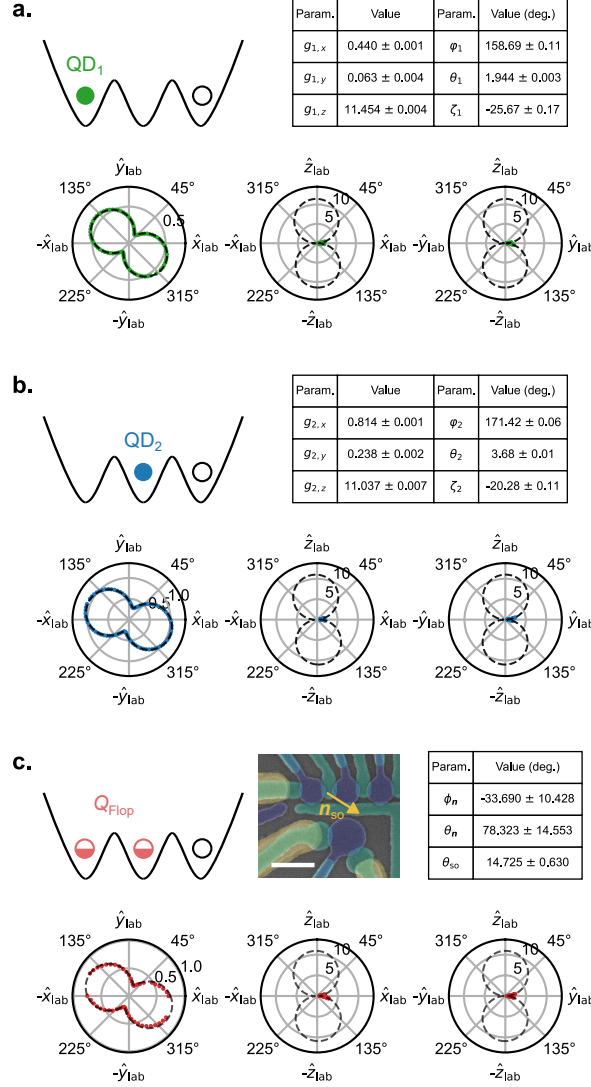


FIG. S3. **a.** **(b.)** g -tensor characterization of the hole in QD₁(2). Bottom three panels correspond to the measured effective g -factor $g^* = E_z/(\mu_B |\mathbf{B}|)$ for different magnetic orientations in the xy -, xz - and yz -plane in the lab frame respectively. The black dashed line in each panel represents the fit to the g -tensor model. The corresponding set of parameters ($g_{1(2),x}, g_{1(2),y}, g_{1(2),z}, \varphi_{1(2)}, \theta_{1(2)}, \zeta_{1(2)}$) extracted from the fit are shown in the table in the top right panel. **c.** Effective g -factor of the FM qubit for different magnetic field orientations in the xy -, xz - and yz -plane in the lab frame (bottom three panels). The black dashed line in each panel is the fit to the FM qubit model (Supplementary section S5) which also takes into account the spin-orbit interaction. Yellow arrow in the device SEM (top middle panel) illustrates the direction of the fitted spin-orbit vector $\mathbf{n}_{so}$. The fitted values representing the spin-orbit vector ($\phi_n, \theta_n, \theta_{so}$) are presented in the table (top right panel, see text).

The Zeeman splitting E_z of a heavy-hole (HH) state is governed by an anisotropic 3×3 g -tensor $\mathbf{g}$, resulting in a field-direction-dependent splitting $E_z = \mu_B |\mathbf{B} \cdot \mathbf{g}|$. In practice, this anisotropy is site-dependent, for instance due to strain gradient present in the material, and electrostatic confinement anisotropy [2, 42, 45–47]. We therefore characterize the g -tensor of QD₁ and QD₂ respectively, to systematically map the FM qubit operational sweet-line as a function of magnetic field direction shown in Fig. 2.

To characterize the g -tensor of QD₁₍₂₎, we prepare the $|L(R), \downarrow\rangle$ state deep in (1,0,1) ((0,1,1)) charge configuration, by first initializing in (0,0,2)S, then adiabatically pulsing to a point in the charge stability diagram, $(\Delta \bar{V}_{P1}, \Delta \bar{V}_{P2}) = (-5, +5)$ mV $(+5, -5)$ mV away from the charge symmetry point. The qubit frequency is

measured as a function of $\mathbf{B}$ direction using adiabatic rapid passage (ARP) spectroscopy with a chirped rf tone [62], and fitted to $E_z(\mathbf{B}) = \mu_B |\mathbf{B} \cdot \mathbf{g}_{1(2)}|$ to obtain $g_{1(2)}$ [6, 49]. Representative measurements of the effective g -factor $g^* = E_z/(\mu_B |\mathbf{B}|)$ of QD₁₍₂₎ as a function of the azimuthal (ϕ_{lab}) and the zenith (θ_{lab}) angles in the lab frame are shown in Fig. S3a(b). The g -tensor, $g_{1(2)}$, in its diagonalized form is given by $\text{diag}(g_{1(2),x}, g_{1(2),y}, g_{1(2),z})$, with the principal values $g_{1(2),y} < g_{1(2),x} < g_{1(2),z}$ defining the g -tensor frame. The laboratory frame g -tensor is obtained by applying the Euler rotations $R(\varphi_{1(2)})R(\theta_{1(2)})R(\zeta_{1(2)})$ to $\text{diag}(g_{1(2),x}, g_{1(2),y}, g_{1(2),z})$ with zyz Euler rotation convention [6]. The extracted parameters of $g_{1(2)}$ are summarized in the tables in Fig. S3a(b), while the corresponding fits are shown as black dashed lines. As can be inferred from the measured E_z along ϕ_{lab} for QD₁ and QD₂ (left panel in Fig. S3a and b), the g -tensor is not uniform across the QDs despite their similar lithographic sizes and shapes (see Fig. 1a). Especially, the g -factors along the principal axes in QD₂ are larger by about a factor of two compared to QD₁. We attribute the inhomogeneity in both the overall magnitude and tilt to strain induced by the ohmic contact (brown structure on the left side of QD₁ in Fig. 1a) [47], and by lattice mismatch in the material stack of the reverse-graded heterostructure employed in this work [39].

To accurately capture the system Hamiltonian it is also necessary to take into account the finite gate-voltage tunability of the g -tensors. We therefore examine the dependence of each g -tensor on the plunger gate voltages $\bar{V}_{P1}$ and $\bar{V}_{P2}$, reconstructing the full g -tensor derivatives $\partial g_{1(2)}/\partial \bar{V}_{P1}$, and $\partial g_{1(2)}/\partial \bar{V}_{P2}$ [6]. In a typical Hahn-echo sequence $X_{\pi/2} - \tau - Y_{\pi} - \tau - X_{\pi/2}$, the qubit frequency is kept fixed throughout both free evolution segments represented by τ . In order to evaluate $\partial \Delta E_z / \partial \bar{V}_{Pi}$ at an arbitrary point in the charge stability diagram, one can instead detune the qubit by $(\Delta \bar{V}_{P1}, \Delta \bar{V}_{P2})$ relative to the charge symmetry point during both τ segments. The qubit phase pick-up due to an incremental change in $\bar{V}_{Pi}$ close to the point $(\Delta \bar{V}_{P1}, \Delta \bar{V}_{P2})$ can be measured by slightly changing the amplitude of the first and second τ segment, by a value $\pm \delta V$ on $\bar{V}_{Pi}$, as shown in Supplementary Fig. (S4)b. Moreover, to get a more accurate estimate, we sweep the value of δV in a range ± 1.75 mV and fit the result to a linear model, which gives the qubit frequency derivative. To find $\partial \Delta E_z^{1(2)} / \partial \bar{V}_{P1}$, we pulse to $(\Delta \bar{V}_{P1}, \Delta \bar{V}_{P2}) = (6 \text{ mV}, 0)$ $((-6 \text{ mV}, 0))$. To evaluate $\partial \Delta E_z^{1(2)} / \partial \bar{V}_{P2}$, we pulse to $(\Delta \bar{V}_{P1}, \Delta \bar{V}_{P2}) = (0, +6 \text{ mV})$ $((0, -6 \text{ mV}))$.

In this way, we measure $\partial \Delta E_z^{1(2)} / \partial \bar{V}_{P1}$ and $\partial \Delta E_z^{1(2)} / \partial \bar{V}_{P2}$ as a function of magnetic field direction, and fit to Eq. S7 to extract $\partial g_{1(2)} / \partial \bar{V}_{P1}$. The measured $\Delta g_{1(2)}^* = (\mu_B |\mathbf{B}|)^{-1} \partial \Delta E_z^{1(2)} / \partial \bar{V}_{P1}$ are represented by green (blue) points in Supplementary Fig. S4, with the dashed lines representing $\Delta g_{1(2)}^*$ calculated using the fitted parameters (see Table S1 for the extracted parameters).

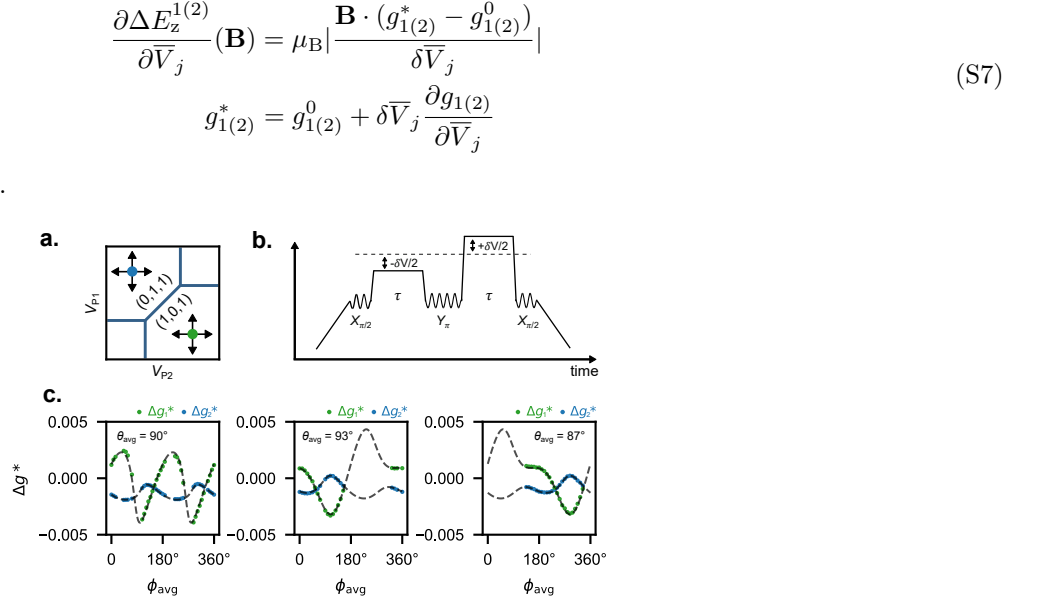


FIG. S4. g -tensor derivative measurements. **a.** Schematic of the g -tensor derivative measurement. The change of the Zeeman energies of each QD is measured by varying the gate-voltages in the direction denoted by the arrows at a fixed point in the charge stability diagram. **b.** Schematic of the Hahn-echo like pulse sequence utilized for measuring the g -tensor derivative. **c.** $\Delta g_{1(2)}^*$ measured by varying $\bar{V}_{P1}$ at different magnetic field directions represented by ϕ_{avg} for different values of θ_{avg} .

Each of the g -tensor derivatives is represented by $(g'_{1(2),x}, g'_{1(2),y}, g'_{1(2),z}, \phi'_{1(2)}, \theta'_{1(2)}, \zeta'_{1(2)})$ where q' represents the derivative of the variable q with respect to the gate voltage. Based on the extracted $\partial g_{1(2)}/\partial \bar{V}_{P1}$ and $\partial g_{1(2)}/\partial \bar{V}_{P2}$,

TABLE S1. Parameters representing the g -tensor derivatives $\frac{\partial g_1}{\partial \bar{V}_{P1}}$, $\frac{\partial g_1}{\partial \bar{V}_{P2}}$, $\frac{\partial g_2}{\partial \bar{V}_{P1}}$, and $\frac{\partial g_2}{\partial \bar{V}_{P2}}$. Each parameter ($p'_i = \partial p_i / \partial \bar{V}_j$) is extracted by fitting the measured Zeeman splitting variation with respect to the modulation of $\bar{V}_j$, to the model Eq. S7.

	$\frac{\partial g_1}{\partial \bar{V}_{P1}}$	$\frac{\partial g_1}{\partial \bar{V}_{P2}}$	$\frac{\partial g_2}{\partial \bar{V}_{P1}}$	$\frac{\partial g_2}{\partial \bar{V}_{P2}}$
$g'_{i,x}$ (mV $^{-1}$)	$(7.11 \pm 0.21) \times 10^{-4}$	$(-1.05 \pm 0.02) \times 10^{-3}$	$(-1.56 \pm 0.01) \times 10^{-3}$	$(9.21 \pm 0.03) \times 10^{-4}$
$g'_{i,y}$ (mV $^{-1}$)	$(6.19 \pm 0.79) \times 10^{-4}$	$(-8.27 \pm 0.87) \times 10^{-4}$	$(-9.78 \pm 0.11) \times 10^{-4}$	$(2.28 \pm 0.07) \times 10^{-4}$
$g'_{i,z}$ (mV $^{-1}$)	$(6.51 \pm 0.07) \times 10^{-3}$	$(-4.03 \pm 0.07) \times 10^{-3}$	$(-1.36 \pm 0.03) \times 10^{-3}$	$(7.28 \pm 0.02) \times 10^{-3}$
φ'_i (deg./mV)	$(4.01 \pm 0.02) \times 10^{-1}$	$(-4.06 \pm 0.02) \times 10^{-1}$	$(-4.83 \pm 0.04) \times 10^{-2}$	$(-2.24 \pm 0.02) \times 10^{-2}$
θ'_i (deg./mV)	$(5.11 \pm 0.08) \times 10^{-3}$	$(-9.25 \pm 0.09) \times 10^{-3}$	$(-3.05 \pm 0.04) \times 10^{-3}$	$(-4.69 \pm 0.02) \times 10^{-3}$
ζ'_i (deg./mV)	$(1.80 \pm 0.04) \times 10^{-1}$	$(-2.28 \pm 0.05) \times 10^{-1}$	$(-6.54 \pm 0.66) \times 10^{-3}$	$(7.45 \pm 0.04) \times 10^{-2}$

we reconstruct $g_{1(2)}(\bar{V}_{P1}, \bar{V}_{P2})$ for arbitrary $(\bar{V}_{P1}, \bar{V}_{P2})$ assuming its linear dependence on the gate-voltages, as anticipated in Eq. (S3).

Having quantified the g -tensors and their dependence on gate voltages, we now turn to estimate the spin-orbit interaction (SOI) in the system. Finite SOI induces spin-flip tunneling between the two QDs (Eq. (S4)) which also affects the FM qubit energy (Eq. (S5)). We measure the FM qubit g -factor at the charge symmetry point ($\varepsilon_{12} = 0$) for various $\mathbf{B}$ directions, shown as red datapoints in the bottom three panels of Supplementary Fig. S3c. We estimate $g_{1(2)}(\bar{V}_{P1}, \bar{V}_{P2})$ at the charge symmetry point by linear extrapolation of the g -tensor values in QD $_{1(2)}$, using the measured g -tensor derivatives. We then fit the measured f_{FM} to the FM qubit Hamiltonian Eq. (S5), while putting the spin-orbit vector $\mathbf{n}_{\text{so}}$ and the spin-orbit angle θ_{so} as the fitting parameters. Here, $\mathbf{n}_{\text{so}}$ is parameterized using the spherical coordinates $(\varphi_{\mathbf{n}}, \theta_{\mathbf{n}})$ as $\mathbf{n}_{\text{so}} = (\cos(\varphi_{\mathbf{n}}) \sin(\theta_{\mathbf{n}}), \sin(\varphi_{\mathbf{n}}) \sin(\theta_{\mathbf{n}}), \cos(\theta_{\mathbf{n}}))$. The black dashed lines in the bottom three panels of Supplementary Fig. S3c correspond to the model fit, with the fit parameters shown in the table in the upper right panel of Supplementary Fig. S3c. The spin-orbit vector $\mathbf{n}_{\text{so}}$ is visualized as a yellow-arrow on the SEM image in the top middle panel of Supplementary Fig. S3. We note that $\mathbf{n}_{\text{so}}$ is expected to point in-plane (i.e. $\theta_{\mathbf{n}} = 90^\circ$) for a Rashba-type SOI, with the electric field along the out-of-plane direction. The fitted $\mathbf{n}_{\text{so}}$ is tilting by about 12° in the out-of-plane direction ($\theta_{\mathbf{n}} \sim 78.3^\circ$). This could possibly be explained by a small stray electric field in the in-plane direction.

S7. LSES and TSES calculation

Based on the Hamiltonian described in Supplementary S5, and the parameters extracted in Supplementary S6, the effective Zeeman splitting $E_z(\bar{V}_{P1}, \bar{V}_{P2}; \mathbf{B})$ at an arbitrary point in the DQD stability diagram $(\bar{V}_{P1}, \bar{V}_{P2})$ can be calculated for a given $\mathbf{B}$. By evaluating $E_z(\bar{V}_{P1}, \bar{V}_{P2}; \mathbf{B})$ for different magnetic field directions, effective g -tensor $g(\bar{V}_{P1}, \bar{V}_{P2})$ is obtained, and utilized to build the FM qubit Larmor vector $\mathbf{l}(\bar{V}_{P1}, \bar{V}_{P2}; \mathbf{B})$, as reported in the main text.

The LSES of the FM qubit as a function of ϕ_{avg} and θ_{avg} with respect to $\bar{V}_{P1}$ is presented in Supplementary Fig. S5a. It is numerically calculated, by first evaluating $g(\Delta \bar{V}_{P1} = +\delta \bar{V}_{P1}, \Delta \bar{V}_{P2} = 0)$, $g(\Delta \bar{V}_{P1} = -\delta \bar{V}_{P1}, \Delta \bar{V}_{P2} = 0)$ and $g_0 = g(\Delta \bar{V}_{P1} = 0, \Delta \bar{V}_{P2} = 0)$. Then, we obtain $g'_{0,P1} = \partial g_0 / \partial \bar{V}_{P1} = (g(\Delta \bar{V}_{P1} = +\delta \bar{V}_{P1}, \Delta \bar{V}_{P2} = 0) - g(\Delta \bar{V}_{P1} = -\delta \bar{V}_{P1}, \Delta \bar{V}_{P2} = 0)) / 2\delta \bar{V}_{P1}$. Following the definition of LSES (Eq. S8), we obtain the LSES $\beta_{\parallel, P1(2)}$ for arbitrary $\mathbf{B}$.

$$\beta_{\parallel, P1(2)} = \frac{\mu_B}{h|\mathbf{B} \cdot \mathbf{g}_0|} (\mathbf{B} \cdot \mathbf{g}'_{0, P1(2)}) \cdot (\mathbf{B} \cdot \mathbf{g}_0) \quad (\text{S8})$$

The LSES with respect to $\bar{V}_{P2}$, $\beta_{\parallel, P2}$ is presented in Supplementary Fig. S5d. We note that $\mu_B \mathbf{B} \cdot \mathbf{g}'_{0, P1(2)}$ corresponds to the derivative of the Larmor vector $\frac{d\mathbf{l}}{d\bar{V}_{P1(2)}}$ shown in Fig. 2a, with $\mu_B \mathbf{B} \cdot \mathbf{g}_0$ being $\mathbf{l}$ at the given configuration.

The TSES can be calculated using (Eq. S9), where the TSES with respect to $\bar{V}_{P1}$ ($\beta_{\perp, P1}$) and $\bar{V}_{P2}$ ($\beta_{\perp, P2}$) are shown in Supplementary Fig. S5b and e respectively.

$$\beta_{\perp, P1(2)} = \frac{\mu_B}{h|\mathbf{B} \cdot \mathbf{g}_0|} |(\mathbf{B} \cdot \mathbf{g}'_{0, P1(2)}) \times (\mathbf{B} \cdot \mathbf{g}_0)| \quad (\text{S9})$$

The TSES normalized by the FM qubit frequency, $\beta_{\perp, P1(2)} / f_{\text{FM}}$ is illustrated in Supplementary Fig. S5c and f respectively. We further show the calculated LSES, TSES and the normalized TSES of the Loss-DiVincenzo (LD)

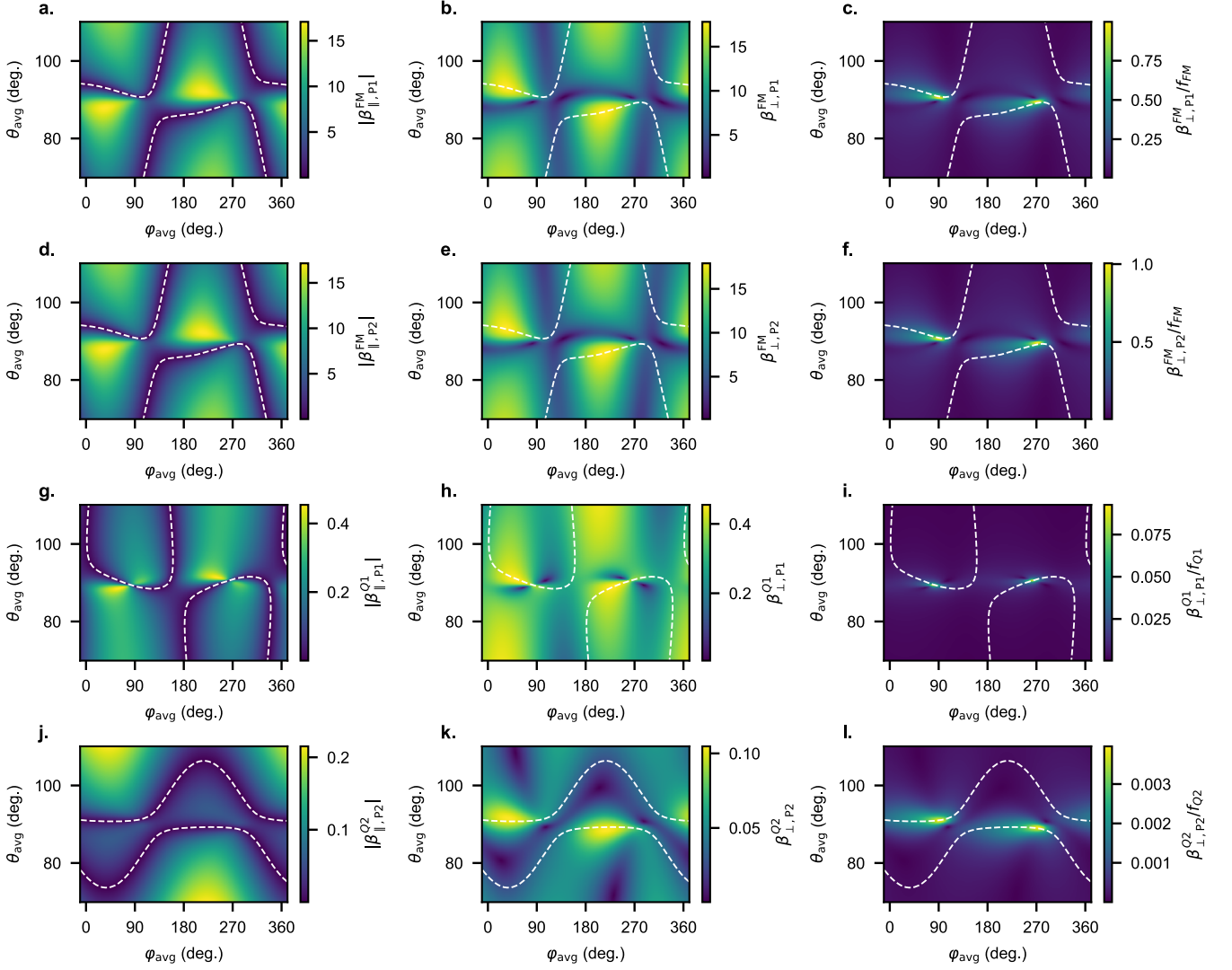


FIG. S5. **a (d).** Calculated LSES of the FM qubit, $\beta_{\parallel, P1(2)}^{\text{FM}}$, as a function of the magnetic field angles φ_{avg} and θ_{avg} with respect to $\bar{V}_{P1}$ ($\bar{V}_{P2}$) modulation. **b (e).** Calculated TSES of the FM qubit, $\beta_{\perp, P1(2)}^{\text{FM}}$ as a function of the magnetic field angle φ_{avg} and θ_{avg} with respect to $\bar{V}_{P1}$ ($\bar{V}_{P2}$) modulation. **c (f).** Calculated TSES normalized by FM qubit frequency $\beta_{\perp, P1(2)}^{\text{FM}}/f_{\text{FM}}$ as a function of the magnetic field angle φ_{avg} and θ_{avg} with respect to $\bar{V}_{P1}$ ($\bar{V}_{P2}$) modulation. **g (j).** Calculated LSES of the Loss-DiVincenzo (LD) qubit in $\text{QD}_{1(2)}$, $\beta_{\parallel, P1(2)}^{\text{QD}}$, as a function of the magnetic field angle φ_{avg} and θ_{avg} with respect to $\bar{V}_{P1}$ ($\bar{V}_{P2}$) modulation. **h (k).** Calculated TSES of the LD qubit in $\text{QD}_{1(2)}$ with respect to the plunger gate voltage $P1(2)$, $\beta_{\perp, P1(2)}^{\text{QD}}$ as a function of the magnetic field angle φ_{avg} and θ_{avg} with respect to $\bar{V}_{P1}$ ($\bar{V}_{P2}$) modulation. **i (l).** Calculated TSES normalized by LD qubit frequency $\beta_{\perp, P1(2)}^{\text{QD}}/f_{\text{QD}}$ as a function of the magnetic field angle φ_{avg} and θ_{avg} with respect to $\bar{V}_{P1}$ ($\bar{V}_{P2}$) modulation. White dashed curves in each plot denote the sweet-line where the corresponding LSES becomes 0.

qubits in $\text{QD}_{1(2)}$ with respect to the corresponding virtual plunger gate voltages $\bar{V}_{P1(2)}$ (Supplementary Fig. S5g-i(j-l)). These results clearly demonstrate that the TSES of the FM qubit is two-orders-of-magnitude larger than that of the LD qubits in the same device, which allows ultra-low-power, and fast Rabi oscillation of the FM qubits.

TABLE S2. Qubit coherence metrics measured for the same configurations as the RB results in Fig. 5.

	$\hat{\mathbf{b}}_1$	$\hat{\mathbf{b}}_3$
$ \mathbf{B} $	5 mT	5 mT
t_c/h	13 GHz	13 GHz
f_{FM}	58.8 MHz	33.7 MHz
$A_{\text{d,P2}}$	460 μV	550 μV
f_{Rabi}	5.7 MHz	3.5 MHz
T_2^*	1.44(1) μs , $\beta = 1.83(5)$	1.40(2) μs , $\beta = 1.64(4)$
T_2^{Hahn}	11.28(7) μs , $\beta = 2.10(5)$	11.50(6) μs , $\beta = 2.42(5)$
$T_2^{\phi, \text{CPMG}}$	82(6) μs (24 decoupling pulses), $\beta = 1.1(1)$	130(3) μs (32 decoupling pulses), $\beta = 3.2(2)$
T_1	53(2) μs	226(6) μs
$T_{2\rho}^{\text{rot}}$	11.8(2) μs	21.1(5) μs
$\mathcal{F}_{1\text{qb}}$	99.76(1)%	99.74(1)%

S8. FM qubit coherence time extraction

Qubit dephasing times T_2^* and T_2^{Hahn} are obtained by fitting the probability decay to the model

$$P(t) = A \exp \left[- \left(\frac{t}{T_2} \right)^\beta \right] \cos(2\pi ft + \varphi) + C \quad (\text{S10})$$

where β is left as a free parameter. Additional corrections to this formula can apply due to the quadratic coupling to the noise at sweet-spots, beyond the scope of this work [63].

Presumably due to the quasi-static fluctuations of the Rabi frequency, we find that the Rabi quality factor is not a representative number for gate fidelity. We report an example Rabi decay trace in Fig. S6c for $\mathbf{B}$ direction $\hat{\mathbf{b}}_3$ and the same parameters as for the RB experiment in Fig. 5b of the main text. The quasi-static nature of the Rabi decay can be seen by performing a rotary echo experiment, where the phase of the resonant drive is flipped by π at half time ($t_{\text{burst}}/2$). This significantly prolongs the driven coherence time $T_{2\rho}^{\text{rot}}$ (Fig. S6d). The rotary echo data is fitted using exponential decay (with $\beta = 1$). In Table S2 we report the coherence times and gate parameters for the two configurations in Fig. 5 of the main text.

S9. Flopping-mode qubit noise spectrum

We characterize the FM qubit performance at the detuning sweet-spot for the field direction $\hat{\mathbf{b}}_3$, fixing $B = 5$ mT and $t_c/h = 13$ GHz. The corresponding chevron plot is shown in Fig. 1g of the main text. To examine the noise affecting the qubit energy splitting, we perform CPMG spectroscopy [54], varying the number of Y_π pulses from $N_\pi = 1$ to $N_\pi = 32$, and sweeping the inter-pulse separation τ as defined in the schematic in Fig. S6b. The coherence decay traces, shown in Fig. S6a, are fitted with a decay model to find the CPMG dephasing time T_{2,N_π} and normalized as described in Supp. Inf. section S10. The extracted values of T_{2,N_π} as a function of N_π are shown in the top right inset of Fig. S6b, with a linear log-log fit giving the slope $\gamma = 0.72$ (Fig. S6b inset). If we assume a colored noise model $S(f) = S_0/f^\alpha$ for the PSD, we find $\alpha = \frac{\gamma}{1-\gamma} = 2.58$, which is not consistent with a simple charge-noise picture.

To further study this trend we plot the noise power spectral density (PSD) extracted from the CPMG data (see Supplementary section S10), shown on the bottom right in Fig. S6b with the same color coding. Moreover, we also extract the PSD on the flank of the spectrum (magenta datapoints, bottom right in Fig. S6b), at the detuning $\varepsilon_{12}/h = -15$ GHz $\approx -t_c/h$, where the LSES is measured to be the steepest. This condition is explicitly marked with a magenta line in Fig. S6e, where the qubit frequency is shown as a function of ε_{12} (zoom-in of Fig. 3 main text). The CPMG data displayed in Fig. S6b only includes datapoints with $N_\pi \geq 8$. In Fig. S6f a zoom-in on the data is shown, where we additionally include points obtained from $N_\pi < 8$, marked with crosses. This allows for a comparison between the datasets over a wider frequency range, although we note that the reconstruction is less accurate for lower number of decoupling pulses [54]. The sweet-spot data reveals a localized feature in the PSD which could possibly be explained by a two-level fluctuator (TLS). We note that beatings likely induced by a TLS were observed in the Ramsey decay of the LD qubit in QD₂ (see Supplementary section S12). The ^{73}Ge isotope

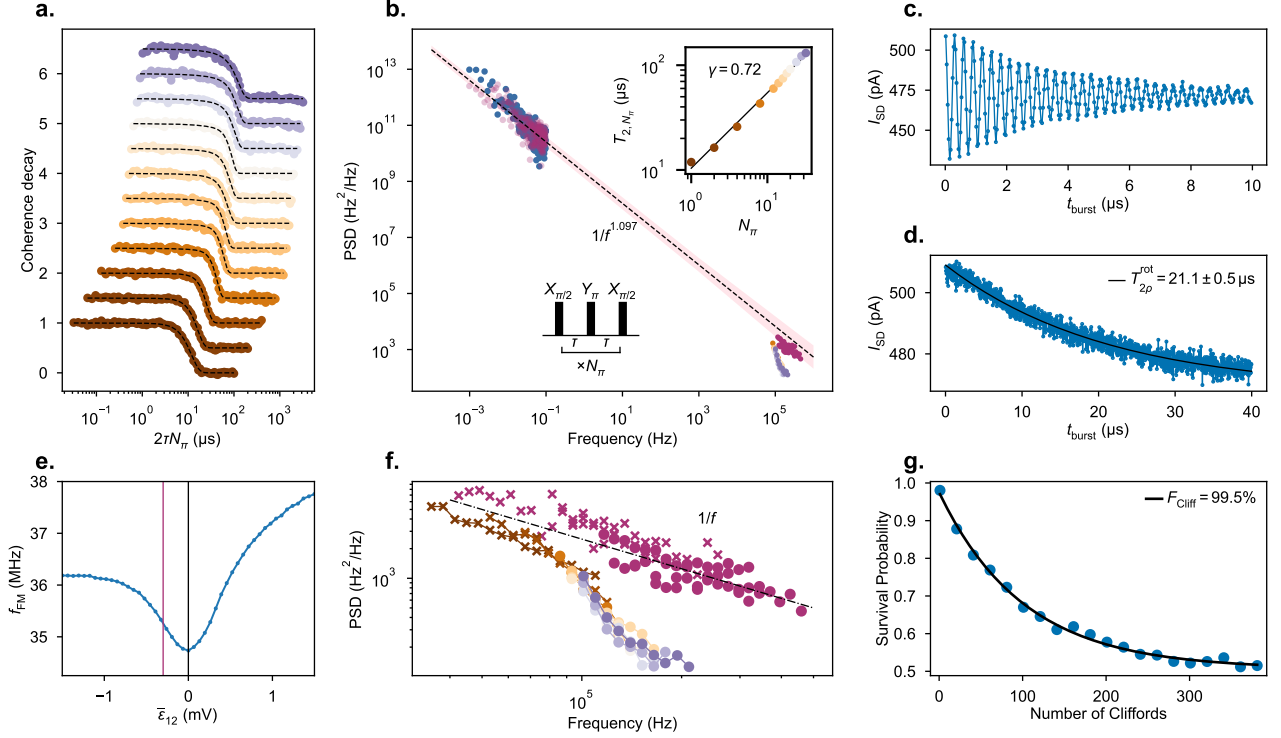


FIG. S6. Qubit characterization for the $\mathbf{B}$ orientation $\hat{b}_3$, $|\mathbf{B}| = 5$ mT, $t_c/h = 13$ GHz. **a.** CPMG decay traces for different number of Y_π pulses: $N_\pi = 1, 2, 4, 8, 12, 14, 16, 18, 20, 24, 28, 32$. The x-axis corresponds to the total duration of the sequence as defined in the inset in **b.**, where τ is the varied inter-pulse separation. An offset is added for easier visual separation. Dashed lines correspond to model fits (see text). **b.** Frequency dependence of the power spectral density. PSD extracted from the CPMG traces in **a.** shown on the bottom right, excluding points with $N_\pi < 8$. Magenta points on the bottom right are extracted from a CPMG measurement at the off-detuned point shown in **e.** Blue datapoints on the top left are obtained by averaging three PSDs extracted from time-series measurements of the FM qubit frequency at $\varepsilon_{12} = 0$, measured using Ramsey oscillations. Black dashed line corresponds to a linear fit to this data in log-log space, with the shaded blue region denoting the 1σ uncertainty of the fit. Magenta datapoints on the top left are the analogous measurement at the off-detuned point. **c.** Rabi oscillations at $\varepsilon_{12} = 0$ with a drive amplitude of $550 \mu\text{V}$. **d.** Rotary Echo measurement for the same parameters as **c.** fitted with an exponential decay model. **e.** FM qubit frequency as a function of detuning ε_{12} , extracted from Ramsey oscillations. Black solid line denotes the parking position for sweet-spot operation ($\varepsilon_{12} = 0$). Magenta line denotes the value of ε_{12} where the off-detuning PSD is extracted. **f.** Zoom-in on CPMG PSD data shown in **b.** Additionally the PSD extracted from $N_\pi < 8$ is shown as crosses. **g.** Single-qubit randomized benchmarking decay, fitted to obtain an average Clifford gate fidelity of 99.5% and average physical gate fidelity of 99.7%.

is expected to give a peak in the PSD at a much lower frequency of 7.4 kHz for the chosen value of $|\mathbf{B}|$ [6, 52]. However, we cannot rule out other effects of hyperfine interactions [52, 53]. A comparison of the data on and off the sweet-spot reveals that detuning sweet-spot operation effectively suppresses noise above 100 kHz, lowering the PSD by more than an order of magnitude. Finally, we fit the CPMG PSD on the flank with a linear function in log-log space and fixing a noise exponent of $\alpha = 1$ (dash-dotted line in Fig. S6f). With this we extrapolate a charge noise level of $S_{1/f}(1 \text{ Hz}) = 2.0 \times 10^{-12} \text{ eV}^2/\text{Hz}$, similar to values reported on the same material [6], where we have used the slope of the red line in Fig. S6e for converting Hz to eV. We note that this value shows a discrepancy with the Ramsey measurement discussed later, potentially indicating another dominating noise source for lower frequencies, or a deviation from the $1/f$ model.

The low frequency range of the PSD on the sweet-spot and on the flank is probed using Ramsey spectroscopy (see Supplementary section S11) [54, 55], where the deviations from the virtual frequency are tracked for subsequent Ramsey measurements. This allows us to track qubit frequency fluctuations at a sampling rate of $f_s = 0.2$ Hz. The PSD is obtained from the Fourier transform of the data. The blue and magenta points shown on the top left in Fig. S6b are obtained by averaging three PSDs obtained from the data shown in Supplementary Fig. S7. The PSD on the flank is fitted to obtain a noise exponent $\alpha = 1.097$. The extrapolation to high frequency values is shown

with dashed line with an error bar shown as the shaded pink region in Fig. S6b. If we try to extract a charge noise amplitude from this data, we obtain $S_{1/f^{1.097}}(1 \text{ Hz}) = 1.4 \times 10^{-11} \text{ eV}^2/\text{Hz}$, which is about an order of magnitude larger than previously reported values for charge noise in the same material [6]. We therefore expect other mechanisms to dominate the PSD in the low-frequency regime, such as magnetic noise, which would be consistent with the weak frequency dependence of the T_2^* (Fig. 3b) in this magnetic field configuration.

Another source of noise in the FM qubit is the fluctuation of the Rabi frequency. This effect is expected to be more pronounced in this type of qubit, since the Rabi frequency is strongly dependent on detuning [24]. The decay of the Rabi oscillation is shown in Fig. S6c. To verify that the noise is quasistatic we perform a rotary echo experiment (Fig. S6d), where the phase of the drive is flipped by π in the middle of the sequence. The resulting coherence is improved by about a factor of five. We thus conclude that Rabi decay is mostly quasi-static.

S10. CPMG noise spectroscopy

The CPMG sequence follows the pulse protocol:

$$X_{\pi/2} - (\tau - Y_{\pi} - \tau)^{N_{\pi}} - X_{\pi/2}. \quad (\text{S11})$$

The measurement takes between 5 and 16 minutes to complete for $N_{\pi} = 1$, and $N_{\pi} = 32$ respectively. Before every measurement, we calibrate the amplitude of the detuning pulse to find the position of the detuning sweet-spot. We perform single-shot readout, averaging the outcomes of 4000 shots. The CPMG data at finite detunings $|\varepsilon_{12}| > 0$ (magenta points Fig. S6b,f) is obtained by inserting additional detuning pulses during each τ segment:

$$X_{\pi/2} - (\Delta\bar{V}_{P2}(\tau) - Y_{\pi} - \Delta\bar{V}_{P2}(\tau))^{N_{\pi}} - X_{\pi/2}. \quad (\text{S12})$$

and using an odd number of Y_{π} pulses $N_{\pi} = 1, 3, 5, 9, 17, 33$. The PSD is extracted from the CPMG traces following the method described in Ref. [54], with the difference that here we also consider the effect of the T_1 relaxation as outlined in Ref. [64]. Each coherence decay trace is fitted using the model

$$f(T) = A \exp \left[- \left(\frac{T}{T_{2,N_{\pi}}} \right)^{\beta} \right] \exp \left[- \frac{1}{2} \frac{T}{T_1} \right] + B, \quad (\text{S13})$$

where $T = 2N_{\pi}\tau$ is the total duration of the sequence, $T_{2,N_{\pi}}$ is the coherence time, β is a free parameter capturing the noise color, T_1 relaxation is separately extracted and fixed, and A and B are fitted constants. Next, the datapoints are normalized between 0 and 1 and the effect of T_1 is divided out:

$$\tilde{f}(T) = \frac{f(T) - B}{A \exp(-\frac{1}{2} \frac{T}{T_1})}. \quad (\text{S14})$$

We retain only the data points on the steep part of the decay curve, lying between 0.15 and 0.85. Finally, the PSD in units of Hz^2/Hz is obtained using the formula

$$PSD \left(\frac{N_{\pi}}{2T} \right) = \frac{-\ln \tilde{f}(T)}{2\pi^2 T}. \quad (\text{S15})$$

S11. Ramsey noise spectroscopy

The low frequency PSD on the sweet-spot is probed using Ramsey spectroscopy with the sequence

$$X_{\pi/2} - \tau - Z(2\pi f_{\text{virt}}\tau) - X_{\pi/2}, \quad (\text{S16})$$

where τ is a varied wait time, $Z(\varphi)$ is a virtual gate and $f_{\text{virt}} = 15.625 \text{ MHz}$ is the virtual frequency. The Ramsey spectroscopy on the flank is obtained with the pulse sequence:

$$X_{\pi/2} - \Delta\bar{V}_{P2}(\tau) - Z(2\pi f_{\text{virt}}\tau) - X_{\pi/2}, \quad (\text{S17})$$

where $\Delta\bar{V}_{P2}(\tau)$ denotes a detuning pulse, adiabatic with respect to t_c , with a duration τ . To acquire a single Ramsey trace we execute a single shot for each duration τ before repeating the cycle for a total of 50 shots. The outcomes

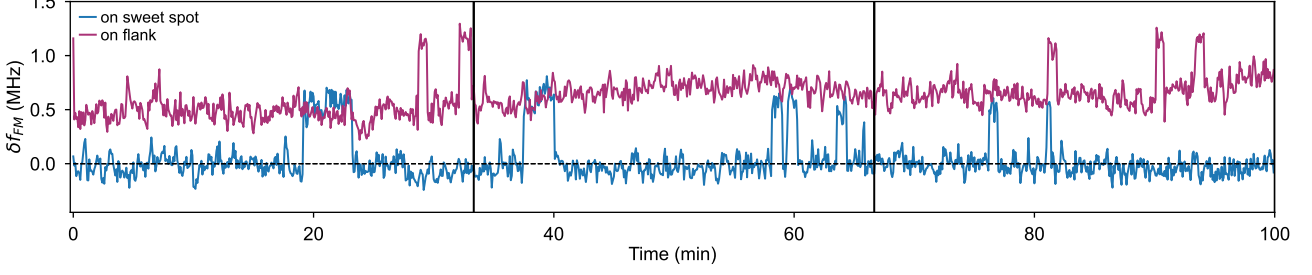


FIG. S7. Time series of the extracted FM qubit frequency fluctuation (δf_{FM}) on the flank (magenta) and on the sweet-spot of the qubit frequency spectrum shown in Fig. S6e. Black vertical lines indicate pauses in the acquisition, followed by a recalibration of the sweet-spot detuning. Dashed line at $\delta f_{FM} = 0$ indicates zero deviation from the virtual frequency. The three time traces on the sweet-spot are taken consecutively, followed by the three traces on the flank consecutively. We overlay the time series for the two detuning setpoints for easier comparison of the noise amplitudes.

are then averaged and saved in a buffer. Another Ramsey measurement is then directly repeated. All of the data acquisition is done on the OPX1000 with no software loops. We therefore rely on the deterministic timing of the hardware to ensure equal spacing between the measurements. The full duration of the experiment is approximately 33 minutes.

The decay of each Ramsey oscillation is fitted using Eq. S10, obtaining a time series of the qubit frequency detuning $\delta f_{FM}(\tau) = f_{FM} - f_{virt}$ (Fig. S7). The experiment is repeated multiple times with an interleaved calibration of the sweet-spot position, denoted with vertical black lines in Fig. S7. The PSD is calculated individually for each experiment and the results are then averaged together to obtain the PSD shown in Fig. S6. The PSD is obtained from the real fast Fourier transform (rFFT) of the data as $PSD = \frac{2|rFFT|^2}{Nfs}$. We overlay the results for on and off the flank for easier comparison of the noise amplitude. We note the presence of a two-level fluctuator which leads to frequency jumps of $\sim 15\%$ of the Rabi frequency present in both datasets. Another notable feature is the background drift which is present only in the on-flank dataset. We attribute this to the suppression of low-frequency noise in the tunnel coupling at the sweet-spot.

S12. LD qubit performance

We compare the performance of the FM qubit to an LD qubit defined in QD₂. The qubit is operated at the center of the (0,1,1) charge region with a closed inter-dot tunnel coupling between (1,0,1) and (0,1,1). We orient $\mathbf{B}$ into the equatorial plane of g_2 by performing state spectroscopy as a function of the out-of plane $\mathbf{B}$ component $\mathbf{B}_z$ (Fig. S8a) for an in-plane angle $\phi_{QD2} = 41.4^\circ$ in the g_2 reference frame. Operating close to the equatorial plane is expected to yield improved fidelities for g -TMR driven single-qubit gates [6, 12]. At the frequency minimum, with $|\mathbf{B}| = 15$ mT and a drive amplitude $A_{d,P2} = 7.5$ mV, we obtain the chevron plot in Fig. S8b. We find a Rabi frequency of 0.95 MHz and a qubit frequency of 141.9 MHz, resulting in $\eta_d = 0.13$ MHz/mV and $\tilde{\eta}_d = 0.0009$ mV⁻¹. The corresponding decay traces for Ramsey and Hahn echo experiments are shown in Fig. S8c,d. We observe a beating in the Ramsey decay but not in the echo decay, suggesting the presence of quasi-static random telegraph noise from a two-level fluctuator. We fit the Ramsey decay to the model $Ae^{-(t/T_2^*)^\alpha} \cos(2\pi ft) \cos(\pi \delta f_{TLS} t) + B$, finding $T_2^* = 2.8 \pm 0.05$ μ s. For the echo decay, we use the same model as in the main text, finding $T_2^{Hahn} = 23 \pm 0.5$ μ s. We then perform randomized benchmarking (Fig. S8e), obtaining a Clifford gate fidelity $98.5 \pm 0.1\%$ and an average primitive gate fidelity of $99.1 \pm 0.1\%$. This gives a factor 3 higher infidelity compared to the FM qubit.

S13. Lever arm and t_c estimation

We first extract the virtual gate lever arm of $\bar{V}_{P3}$ using temperature-dependent broadening of the reservoir transition [65] between the charge sectors (0,1,1) and (0,1,2), probed by varying $\bar{V}_{P3}$ in the range ± 3 mV (Fig. S9a). We set the temperature using a heater on the mixing chamber plate and monitor the temperature with a sensor also placed on the mixing chamber plate. For each temperature, the SHT current as a function of $\bar{V}_{P3}$ is fitted to the Fermi-Dirac

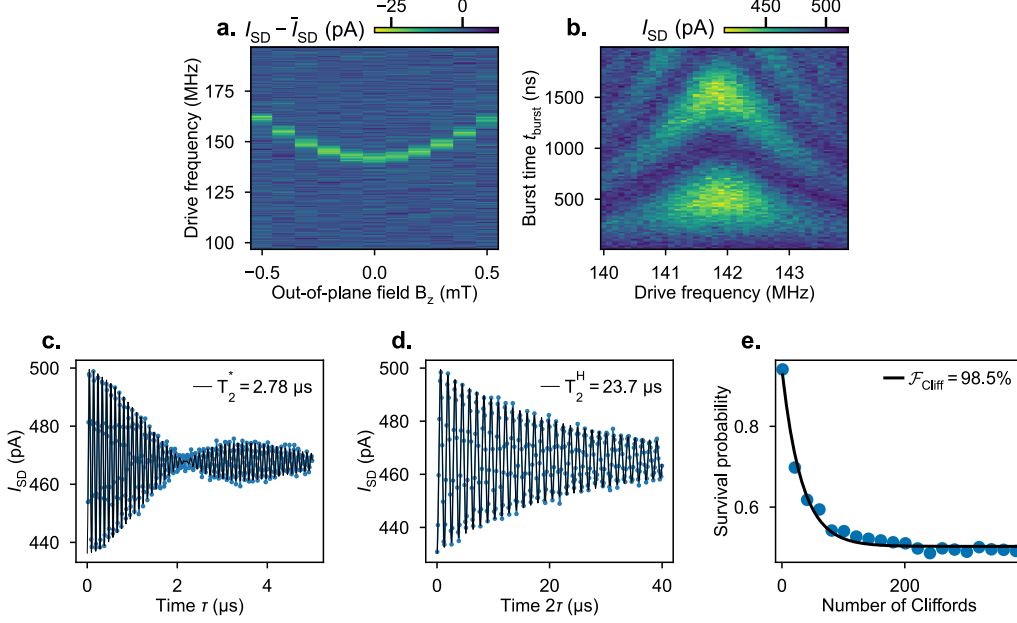


FIG. S8. **a.** ARP spectroscopy as a function of out-of-plane magnetic field strength. **b.** Chevron oscillation for the qubit under P_2 , driven with an amplitude of 7.5 mV. **c.**(**d.**), Ramsey (Hahn echo) measurement at the hyperfine sweet-spot with $B_z = 0$ and $|\mathbf{B}| = 15$ mT. **e.** Randomized benchmarking performed on the LD qubit at the hyperfine sweet-spot.

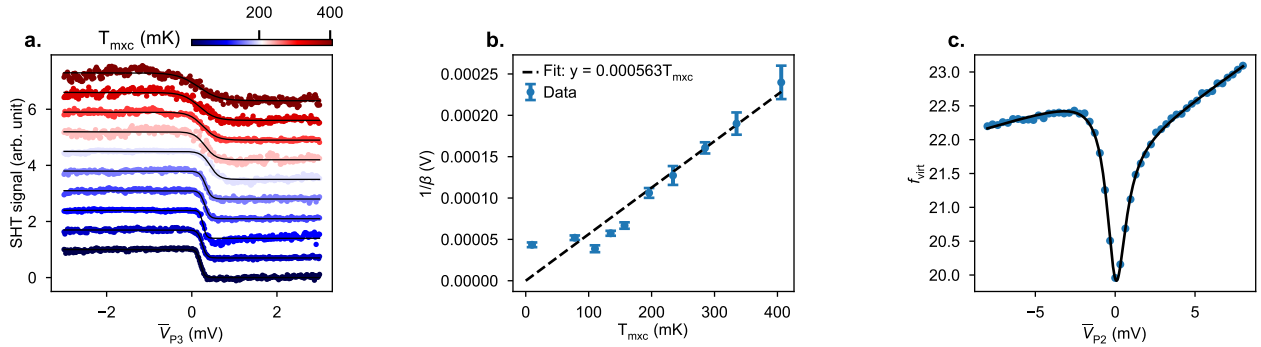


FIG. S9. **a.** Normalized sensor current as a function of virtual plunger voltage $\bar{V}_{P3}$ revealing the (0,1,2)-(0,1,1) reservoir transition. The measurement is repeated for different mixing chamber temperatures T_{mixc} and fitted with a Fermi-Dirac distribution (black lines). An offset is added for visual separation of the data. **b.** Extracted values of $\beta^{-1} = k_B T_{mixc} / \alpha_{33}$ as a function of T_{mixc} . Dashed line corresponds to a linear fit which is constrained to pass through zero. **c.** Typical measurement used for calibrating t_c and the condition $\varepsilon_{12} = 0$.

distribution:

$$f(V) = \frac{A}{e^{\beta(V-V_0)} + 1},$$

with $\beta = \frac{\alpha}{k_B T_{mixc}}$, where T_{mixc} is used as an estimate for the electron temperature and α is the lever arm. In Fig. S9b we show the extracted value of β^{-1} as a function of T_{mixc} . We overlay a least squares linear fit to the last five points, with an intercept constrained to zero. From the slope of the fit we find a lever arm of $\bar{V}_{P3}$ of $\alpha_{33} = 0.153$ eV/V, where we use the notation α_{ij} for gate $\bar{V}_i$ acting on QD_{*j*}. The slope of the inter-dot transition between (0,0,2) and (0,1,1)

in virtual gate space (Fig. 1e) is given by the expression:

$$s_{23} = \frac{\alpha_{33}}{\alpha_{22}} \quad (\text{S18})$$

We extract $s_{23} = 1.04$ from the measured charge stability diagram and use this to find $\alpha_{22} = 0.147 \text{ eV/V}$. Next, we estimate the lever arm α_{11} . As a consequence of the gate $\bar{V}_{B12}$ being pulsed to open the tunnel coupling, the lever arm of gate P_2 with respect to QD_1 changes from the uncoupled value, resulting in a relative lever arm ratio $\alpha_{21}/\alpha_{11} = 0.4$, while $\alpha_{12} = 0$ still holds. We extract the lever arm of virtual gate $\bar{V}_{P1}$ as:

$$\alpha_{11} = \frac{\alpha_{22}}{s_{12} + \alpha_{21}/\alpha_{11}} = 0.10 \text{ eV/V} \quad (\text{S19})$$

Next, we extract the detuning lever arm of the combined gate $\bar{\varepsilon}_{12} = \bar{V}_{P2} - \bar{V}_{P1}$ as:

$$\alpha_{\varepsilon} = \alpha_{22} + \alpha_{11} \left(1 - \frac{\alpha_{21}}{\alpha_{11}} \right) = 0.21 \text{ eV/V} \quad (\text{S20})$$

The tunnel coupling t_c is extracted by sweeping across the inter-dot transition between (0,1,1) and (1,0,1) with the gate $\bar{V}_{P2}$ using the echo sequence described in the detuning calibration section S3 and fitting the resulting qubit spectrum to the model:

$$f_{\text{FM}} = \frac{\mu_B B}{h} \sqrt{\frac{\varepsilon_{12}^2 + 4t_c^2}{2} (g_1^2 + g_2^2) + \frac{\varepsilon_{12}}{2} \sqrt{\varepsilon_{12}^2 + 4t_c^2} (g_2^2 - g_1^2) + 2g_1 g_2 t_c^2 \cos(\tilde{\theta}_{12})} \quad (\text{S21})$$

Here, ε_{12} is the energy detuning, $\tilde{\theta}_{12}$ is a free parameter capturing quantization axis misalignment due to g -tensor variation and spin-flip tunneling coming from SOI [42], and g_1 and g_2 are free parameters capturing the g -factors in QD_1 and QD_2 . We also allow for linear variation of the g_1 and g_2 with ε_{12} [9]. We use the standard convention where $2t_c$ is the separation between the bonding and anti-bonding states. An example of the model fit resulting in $t_c/h = 13 \text{ GHz}$ is shown in Fig. S9c.

S14. Virtual gate matrix

We note that the following virtual gate matrix leads to independent chemical potentials of QD_1 and QD_2 only in the case before $\bar{V}_{B12}$ is pulsed (see Supplementary section S2). Pulsing of $\bar{V}_{B12}$ leads to modified relative lever arms, which we consider in the lever arm extraction (section S13).

$$\begin{pmatrix} V_{P4} \\ V_{P3} \\ V_{P2} \\ V_{P1} \\ V_{B45} \\ V_{B34} \\ V_{B23} \\ V_{B12} \\ V_{RB} \\ V_{PS} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -0.1584 & -0.0075 & 0 & -0.948 & -0.4812 & 0.0522 & 0 & 0 \\ 0 & -0.2146 & 1 & -0.1892 & 0 & 0.2034 & -0.8567 & -0.4171 & 0 & 0 \\ 0 & -0.0029 & -0.327 & 1 & 0 & 0.0028 & 0.1211 & -1.5553 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ -0.0143 & -0.0258 & -0.0345 & -0.02 & -0.017 & -0.0122 & 0.0106 & 0.0269 & -0.0054 & 1 \end{pmatrix} \begin{pmatrix} \bar{V}_{P4} \\ \bar{V}_{P3} \\ \bar{V}_{P2} \\ \bar{V}_{P1} \\ \bar{V}_{B45} \\ \bar{V}_{B34} \\ \bar{V}_{B23} \\ \bar{V}_{B12} \\ \bar{V}_{RB} \\ \bar{V}_{PS} \end{pmatrix}$$

S15. Randomized benchmarking

We perform single qubit RB using the Clifford gate decomposition shown in Table S3. For each of the 101 random Clifford sequences we vary the number of Cliffords in steps of 20 as $\{1, 21, 41, 61, \dots, 381\}$ (20 lengths), followed by a recovery Clifford. The averaging is done with 500 shots in the inner loop, followed by varying the length of the sequence, and finally sweeping over 101 random Clifford sequences in the outer loop. For each of the 101 Clifford sequences we prepend an idling pulse and an X_π pulse, which are then used to normalize the probability scale. The

experiment is performed using the OPX1000 from Quantum Machines, which enables on-the-fly generation of random sequences and determination of their inversion gate using a stored Cayley lookup table. The total duration of the experiment is approximately 20 minutes. We fit the averaged decay curve using the model $Ap^n + B$ where n is the number of Cliffords and p is the decay rate. The Clifford gate fidelity is obtained as $\mathcal{F}_{\text{Cliff}} = (1 + p)/2$. The error in the fidelity is computed from the covariance of the fit. The average single qubit gate fidelity is obtained as $1 - (1 - \mathcal{F}_{\text{Cliff}})/r$, where $r = 1.875$ is the average number of gates per Clifford.

Clifford	Pulse sequence
C_0	I
C_1	X_{180}
C_2	Y_{180}
C_3	$Y_{180} X_{180}$
C_4	$X_{90} Y_{90}$
C_5	$X_{90} Y_{-90}$
C_6	$X_{-90} Y_{90}$
C_7	$X_{-90} Y_{-90}$
C_8	$Y_{90} X_{90}$
C_9	$Y_{90} X_{-90}$
C_{10}	$Y_{-90} X_{90}$
C_{11}	$Y_{-90} X_{-90}$
C_{12}	X_{90}
C_{13}	X_{-90}
C_{14}	Y_{90}
C_{15}	Y_{-90}
C_{16}	$X_{-90} Y_{90} X_{90}$
C_{17}	$X_{-90} Y_{-90} X_{90}$
C_{18}	$X_{180} Y_{90}$
C_{19}	$X_{180} Y_{-90}$
C_{20}	$Y_{180} X_{90}$
C_{21}	$Y_{180} X_{-90}$
C_{22}	$X_{90} Y_{90} X_{90}$
C_{23}	$X_{-90} Y_{90} X_{-90}$

TABLE S3. The 24 single-qubit Clifford operations and their corresponding pulse decompositions.

S16. T_1 relaxation theory

1. Flopping mode

The flopping mode Hamiltonian is

$$H = \frac{\varepsilon_{12}}{2} \tau_z + t_c \tau_x + \sum_{i=1,2} \frac{\mu_B \mathbf{B} \cdot g_i \boldsymbol{\sigma}}{2} \frac{\tau_0 + (-1)^i \tau_z}{2}, \quad (\text{S22})$$

in the position basis of each dot and with the gate-dependent g -tensors $g_i(\bar{V}_{P1}, \bar{V}_{P2})$ defined in Eq. (S3), written in shorthand.

$$g_i = R_{avg}^T \cdot R_i \cdot g_{\text{diag}}^i \cdot R_i^T \cdot R_{avg} \cdot R_{so,i} \quad , \quad g_{\text{diag}}^i = \text{diag}(g_1^i, g_2^i, g_3^i) \quad . \quad (\text{S23})$$

where the principal axis g -tensor g_{diag}^i and its zyz -rotation in the lab frame R_i can be found in Fig.S3. A further rotation R_{avg} is applied to rotate into the averaged lab frame. Finally, an antisymmetric part given by SOI rotation $g_i = g_s^i R_{so,i}$ is applied to rotate the averaged lab frame into SO frame.

For convenience we introduce the Larmor vector of the i th qubit

$$\mathbf{l}_i = \mu_B \mathbf{B} \cdot g_i \quad , \quad (\text{S24})$$

whose amplitude corresponds to the measured Larmor frequency of the i th dot $E_{z,i} = |\mathbf{l}_i|$. We remark that $E_{z,i}$ is independent of the spin-flip tunneling contribution $R_{so,i}$, but depends on the principal axes (R_i) and principal values ($g_{1,2,3}^i$) of g_s^i . For this reason, the individual g -tensor characterization only allows us to extract information on g_s^i .

2. Effective Theory

To develop an effective FM qubit model we first diagonalize the $B = 0$ part of the Hamiltonian by a unitary transformation

$$\tilde{U} = e^{-i\theta\tau_y}, \quad \tan(2\theta) = 2t_c/\varepsilon_{12}, \quad (\text{S25})$$

with identity $e^{-i\theta(\hat{n}\cdot\sigma)} = \sigma_0 \cos \theta - i(\hat{n} \cdot \sigma) \sin \theta$, leading to

$$\tilde{H} = \tilde{U}^\dagger H \tilde{U} = \frac{\Omega}{2} \tau_z + \begin{pmatrix} \sin^2(\theta) & \sin(2\theta)/2 \\ \sin(2\theta)/2 & \cos^2(\theta) \end{pmatrix} \frac{\mathbf{l}_1 \cdot \boldsymbol{\sigma}}{2} + \begin{pmatrix} \cos^2(\theta) & -\sin(2\theta)/2 \\ -\sin(2\theta)/2 & \sin^2(\theta) \end{pmatrix} \frac{\mathbf{l}_2 \cdot \boldsymbol{\sigma}}{2}, \quad (\text{S26})$$

with orbital energy

$$\hbar\Omega = \sqrt{4t_c^2 + \varepsilon_{12}^2}. \quad (\text{S27})$$

The FM qubit is encoded in the ground state projection of $\tilde{H}$, ignoring the bonding states energy offset $\frac{\hbar\Omega}{2}$:

$$H_{\text{FM}} = \frac{\mathbf{l} \cdot \boldsymbol{\sigma}}{2}, \quad (\text{S28})$$

where we defined the FM Zeeman frequency vector $\mathbf{l}$ that interpolates between the Larmor vectors $\mathbf{l}_{1,2} = \mu_B \mathbf{B} \cdot g_{1,2}$ of left and right QDs according to the detuning-dependent hybridization parameter $\eta \in [0, 1]$:

$$\mathbf{l} = \mu_B \mathbf{B} \cdot [\cos^2(\theta)g_1 + \sin^2(\theta)g_2] = \mu_B \mathbf{B} \cdot [(1-\eta)g_1 + \eta g_2] = (1-\eta)\mathbf{l}_1 + \eta\mathbf{l}_2, \quad (\text{S29})$$

$$\eta = \sin^2 \theta = \frac{1}{2} - \frac{\varepsilon_{12}}{2\sqrt{4t_c^2 + \varepsilon_{12}^2}}. \quad (\text{S30})$$

The parameter $\eta = 0$ when the wavefunction is fully localized in QD₁ ($\varepsilon_{12} \gg t_c$) and $\eta = 1$ when the wavefunction is localized in QD₂ ($\varepsilon_{12} \ll -t_c$); in FM regime $\varepsilon_{12} = 0$, $\eta = 1/2$. We emphasize that in our experiment the tunneling t_c is much larger than all Zeeman energies, such that higher order corrections to H_{FM} that lead to the emergence of sweet-spots in FM qubits are negligible here. The lowest order corrections are estimated below.

Importantly, even if the vector $\mathbf{l}$ is a linear interpolation of the two Larmor vectors $\mathbf{l}_1$ and $\mathbf{l}_2$, the measured Zeeman energy of the FM qubit, i.e. the absolute value of $\mathbf{l}$, can present a non monotonic behavior against ε_{12} when the two Larmor vectors are misaligned. This effect can be understood by explicitly observing that

$$\Delta E = |\mathbf{l}| = \sqrt{[(1-\eta)E_{z,1} + \eta E_{z,2}]^2 - 2\eta(1-\eta)(E_{z,1}E_{z,2} - \mathbf{l}_1 \cdot \mathbf{l}_2)}, \quad (\text{S31})$$

where $E_{z,i} = |\mathbf{l}_i|$ is the Zeeman energy of the i -th dot. There are two terms determining ΔE . The first term $(1-\eta)E_{z,1} + \eta E_{z,2}$ is the detuning-dependent interpolation of the Zeeman energies of each dot, that is a monotonic function of ε_{12} . The second term $2\eta(1-\eta)(E_{z,1}E_{z,2} - \mathbf{l}_1 \cdot \mathbf{l}_2)$ appears at finite hybridization parameters ($\eta \neq 0$ and $\eta \neq 1$) and captures the tilt between Larmor vectors, vanishing when $\mathbf{l}_1 \parallel \mathbf{l}_2$. Introducing the relative angle between Larmor vectors

$$\cos(\theta_{12}) = \frac{\mathbf{l}_1 \cdot \mathbf{l}_2}{E_{z,1}E_{z,2}}, \quad (\text{S32})$$

and

$$\sin(\theta_{12}) = \frac{|\mathbf{l}_1 \times \mathbf{l}_2|}{E_{z,1}E_{z,2}} \quad (\text{S33})$$

and in the FM regime $\varepsilon_{12} = 0$ (i.e. $\eta = 1/2$)

$$\Delta E_{\text{FM}} = \sqrt{\left(\frac{E_{z,1} + E_{z,2}}{2}\right)^2 - E_{z,1}E_{z,2} \sin^2\left(\frac{\theta_{12}}{2}\right)}, \quad (\text{S34})$$

Note that the angle θ_{12} depends on both symmetric measured g tensors (g_s^i) and on the accumulated spin-flip tunneling rotation $R_{so,i}$ between dot 1 and 2

$$\cos(\theta_{12}) = \frac{\mu_B^2}{\hbar^2 E_{z,1}E_{z,2}} (\mathbf{B} g_1 g_2^T \mathbf{B}) = \frac{\mu_B^2}{\hbar^2 E_{z,1}E_{z,2}} (\mathbf{B} g_s^1 R_{so} g_s^2 \mathbf{B}), \quad (\text{S35})$$

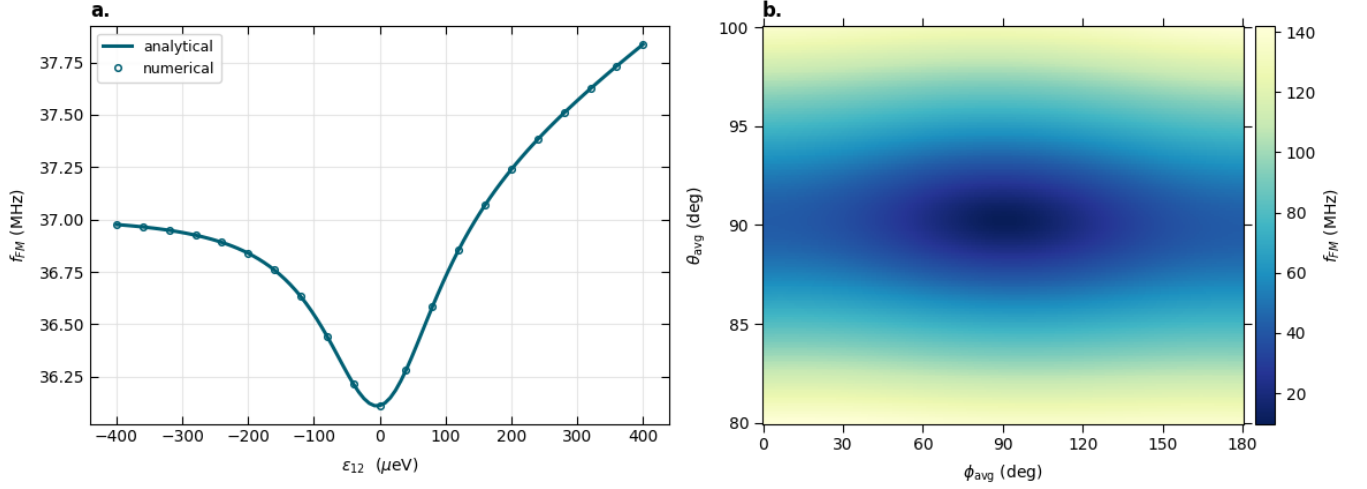


FIG. S10. **a.** FM qubit frequency as function of detuning ϵ_{12} . We use experimentally extracted tunnel coupling $t_c = 52\mu\text{eV}$, g_1 and g_2 and the ZYZ rotation angle in Fig. S3. The numerical line is obtained by numerically diagonalizing the 4 by 4 Hamiltonian in Eq. (S22) and the analytical equation is obtained by plugging in the Eq. (S28). **b.** The FM qubit frequency as function of in plane ϕ_{avg} and out of plane θ_{avg} magnetic field angles.

where $R_{so} = R_{so,1}R_{so,2}^T$.

Explicitly, the transformation that diagonalizes the H_{FM} in Eq. (S28) is

$$u = e^{-i\psi\sigma_z}e^{-i\phi\sigma_y}, \text{ with } \cos(2\psi) = l_y/l_x, \quad \tan(2\phi) = l_z/\Delta E. \quad (\text{S36})$$

3. Correction to the FM Hamiltonian

We now analyze corrections to the FM Hamiltonian H_{FM} in Eq. (S28) arising from the coupling between the bonding and antibonding orbital states and show that they scale at least as B^3/Ω^2 and can therefore be safely neglected in the low-field limit where we are operating the qubit. By defining

$$\bar{\mathbf{l}} = \eta\mathbf{l}_1 + (1-\eta)\mathbf{l}_2 = \mathbf{l}_{|\theta \rightarrow \theta + \pi/2}, \quad (\text{S37})$$

$$\Delta\mathbf{l} = \frac{\mathbf{l}_1 - \mathbf{l}_2}{2} = \frac{\mu_B \mathbf{B} \cdot [g_1 - g_2]}{2} \quad (\text{S38})$$

the DQD Hamiltonian reduces to

$$\tilde{H} = \frac{1}{2} \begin{pmatrix} \Omega\sigma_0 + \bar{\mathbf{l}} \cdot \boldsymbol{\sigma} & \sin(2\theta)\Delta\mathbf{l} \cdot \boldsymbol{\sigma} \\ \sin(2\theta)\Delta\mathbf{l} \cdot \boldsymbol{\sigma} & -\Omega\sigma_0 + \mathbf{l} \cdot \boldsymbol{\sigma} \end{pmatrix}. \quad (\text{S39})$$

The notation $\mathbf{l}_{|\theta \rightarrow \theta + \pi/2}$ indicates that $\bar{\mathbf{l}}$ is connected to the Larmor vector of the FM $\mathbf{l}$ by substituting θ with $\theta + \pi/2$, or equivalently by changing the sign of the detuning, i.e. $\epsilon_{12} \rightarrow -\epsilon_{12}$, which leads to $\eta \rightarrow (1-\eta)$ and $(1-\eta) \rightarrow \eta$.

The correction to H_{FM} can be derived by a Schrieffer-Wolff (SW) transformation. To second order and for low magnetic fields, the generator S of the SW transformation is

$$S = \frac{\sin(2\theta)}{2\Omega} \begin{pmatrix} 0 & \Delta\mathbf{l} \cdot \boldsymbol{\sigma} \\ -\Delta\mathbf{l} \cdot \boldsymbol{\sigma} & 0 \end{pmatrix}, \quad (\text{S40})$$

resulting in the effective Hamiltonian

$$H_{\text{eff}} \approx \text{diag} \left(\frac{\bar{\mathbf{l}} \cdot \boldsymbol{\sigma}}{2} + \frac{\Omega}{2} \left(1 + \frac{\sin^2(2\theta)|\Delta\mathbf{l}|^2}{2\Omega^2} \right) \sigma_0, \frac{\mathbf{l} \cdot \boldsymbol{\sigma}}{2} - \frac{\Omega}{2} \left(1 + \frac{\sin^2(2\theta)|\Delta\mathbf{l}|^2}{2\Omega^2} \right) \sigma_0 \right), \quad (\text{S41})$$

the bonding block gives the effective FM subspace

$$H_{\text{eff}} = \frac{\mathbf{l} \cdot \boldsymbol{\sigma}}{2} - \frac{\Omega}{2} \left(1 + \frac{\sin^2(2\theta)|\Delta\mathbf{l}|^2}{2\Omega^2} \right) \sigma_0 \quad (\text{S42})$$

with

$$|\Delta \mathbf{l}|^2 = \Delta l_x^2 + \Delta l_y^2 + \Delta l_z^2 = \frac{(E_{z,1} - E_{z,2})^2}{4} + E_{z,1} E_{z,2} \sin^2 \left(\frac{\theta_{12}}{2} \right) \propto B^2. \quad (\text{S43})$$

This result shows that the lowest order corrections to the FM Hamiltonian (S28) are spin independent. Spin dependent terms emerge in higher orders and are thus at least $\propto B^3/\Omega^2$.

The effective Hamiltonian can be fully diagonalized with additional rotation Eq. (S36)

$$U_1 = \text{diag}(\bar{u}, u), \quad (\text{S44})$$

the transformation in antibonding states is defined as

$$\bar{u} = e^{-i\bar{\psi}\sigma_z} e^{-i\bar{\phi}\sigma_y}, \quad \tan(2\bar{\psi}) = \bar{l}_y/\bar{l}_x, \quad \cos(2\bar{\phi}) = \bar{l}_z/|\bar{l}|$$

Applying the unitary transformation (S44) on block diagonalized effective Hamiltonian (S41), we get

$$\tilde{H}_{\text{eff}} = U_1^\dagger H_{\text{eff}} U_1 = \text{diag} \left(\frac{|\bar{l}| \sigma_z}{2} + \frac{\Omega}{2} \left(1 + \frac{\sin^2(2\theta) |\Delta \mathbf{l}|^2}{2\Omega^2} \right) \sigma_0, \frac{|\mathbf{l}| \sigma_z}{2} - \frac{\Omega}{2} \left(1 + \frac{\sin^2(2\theta) |\Delta \mathbf{l}|^2}{2\Omega^2} \right) \sigma_0 \right). \quad (\text{S45})$$

S17. Decoherence

To characterize decoherence in the FM qubit, we study the effect on the FM qubit of fluctuations of tunneling, detuning, and g -tensors, δ_t , δ_ϵ , and δg_i respectively. These are characterized by the Hamiltonian

$$H_N = \frac{\delta_\epsilon}{2} \tau_z + \delta_t \tau_x + \sum_{i=1,2} \frac{\boldsymbol{\delta}_i^i \cdot \boldsymbol{\sigma}}{2} \frac{\tau_0 + (-1)^i \tau_z}{2}, \quad (\text{S46})$$

where $\boldsymbol{\delta}_i^i = \mu_B \mathbf{B} \cdot \delta g_i$. In the orbital eigenstates (see Eq. (S25)) reads

$$\tilde{H}_N = \tilde{U}^\dagger H_N \tilde{U} = \frac{\delta_\epsilon \cos(2\theta) + 2\delta_t \sin(2\theta)}{2} \tau_z - \frac{\delta_\epsilon \sin(2\theta) - 2\delta_t \cos(2\theta)}{2} \tau_x + \frac{1}{2} \begin{pmatrix} \bar{\boldsymbol{\delta}}_l \cdot \boldsymbol{\sigma} & \sin(2\theta) \Delta \boldsymbol{\delta}_l \cdot \boldsymbol{\sigma} \\ \sin(2\theta) \Delta \boldsymbol{\delta}_l \cdot \boldsymbol{\sigma} & \boldsymbol{\delta}_l \cdot \boldsymbol{\sigma} \end{pmatrix}, \quad (\text{S47})$$

where in analogy to above we introduce the quantities

$$\boldsymbol{\delta}_l = (1 - \eta) \boldsymbol{\delta}_l^1 + \eta \boldsymbol{\delta}_l^2, \quad (\text{S48})$$

$$\bar{\boldsymbol{\delta}}_l = \eta \boldsymbol{\delta}_l^1 + (1 - \eta) \boldsymbol{\delta}_l^2, \quad (\text{S49})$$

$$\Delta \boldsymbol{\delta}_l = \frac{\boldsymbol{\delta}_l^1 - \boldsymbol{\delta}_l^2}{2}, \quad (\text{S50})$$

and, explicitly,

$$\sin(2\theta) = \frac{2t_c}{\hbar\Omega} = 2\sqrt{\eta(1-\eta)}, \quad \cos(2\theta) = \frac{\varepsilon_{12}}{\hbar\Omega} = 1 - 2\eta. \quad (\text{S51})$$

Including perturbatively the hybridization between antibonding and bonding states via the second order SW generator S in Eq. (S40), we find, keeping only the terms linear in B ,

$$H_{\text{eff},N} = \frac{\delta_\epsilon}{2} \cos(2\theta) \tau_z - \frac{\delta_\epsilon}{2} \sin(2\theta) \tau_x + \frac{\delta_\epsilon}{2} \sin(2\theta) \frac{\Delta \mathbf{l} \cdot \boldsymbol{\sigma}}{\Omega} \begin{pmatrix} -\sin(2\theta) & -\cos(2\theta) \\ -\cos(2\theta) & \sin(2\theta) \end{pmatrix} \quad (\text{S52a})$$

$$+ \delta_t \sin(2\theta) \tau_z + \delta_t \cos(2\theta) \tau_x + \delta_t \sin(2\theta) \frac{\Delta \mathbf{l} \cdot \boldsymbol{\sigma}}{\Omega} \begin{pmatrix} \cos(2\theta) & -\sin(2\theta) \\ -\sin(2\theta) & -\cos(2\theta) \end{pmatrix} \quad (\text{S52b})$$

$$+ \frac{1}{2} \begin{pmatrix} \bar{\boldsymbol{\delta}}_l \cdot \boldsymbol{\sigma} & \sin(2\theta) \Delta \boldsymbol{\delta}_l \cdot \boldsymbol{\sigma} \\ \sin(2\theta) \Delta \boldsymbol{\delta}_l \cdot \boldsymbol{\sigma} & \boldsymbol{\delta}_l \cdot \boldsymbol{\sigma} \end{pmatrix} + \mathcal{O}(B^2). \quad (\text{S52c})$$

1. First-order decoherence process

From the noise Hamiltonian in Eq. (S52), we can directly extract the matrix elements resulting in dephasing and relaxation for the FM qubit. By diagonalizing the qubit energy by the transformation in Eq. (S36), which rotates the vectors defined in lab into the qubit frame

$$\Delta \mathbf{l} \rightarrow \tilde{\Delta} \mathbf{l} = R_y(-2\phi)R_z(-2\psi)\Delta \mathbf{l}, \quad \boldsymbol{\delta}_l \rightarrow \tilde{\boldsymbol{\delta}}_l = R_y(-2\phi)R_z(-2\psi)\boldsymbol{\delta}_l \quad (\text{S53})$$

we find

$$H_{\text{FM,N}} = \frac{\tilde{\boldsymbol{\delta}}_N \cdot \boldsymbol{\sigma}}{2}, \quad \tilde{\boldsymbol{\delta}}_N = \tilde{\boldsymbol{\delta}}_l + \frac{\tilde{\Delta} \mathbf{l}}{\Omega} [\sin(2\theta)^2 \delta\epsilon - 2 \cos(2\theta) \sin(2\theta) \delta t], \quad (\text{S54})$$

including fluctuations of g -tensors that directly couple to the spin and of detuning that couple to the spin via spin-orbit interaction. Dephasing is characterized by the longitudinal part of $\tilde{\boldsymbol{\delta}}_N$ ($\tilde{\boldsymbol{\delta}}_N$)_z,

$$\begin{aligned} (\tilde{\boldsymbol{\delta}}_N)_z &= \boldsymbol{\delta}_N \cdot \frac{\mathbf{l}}{|\mathbf{l}|} = \frac{(1-\eta)^2 \boldsymbol{\delta}_l^1 \cdot \mathbf{l}_1 + \eta^2 \boldsymbol{\delta}_l^2 \cdot \mathbf{l}_2 + \eta(1-\eta)(\boldsymbol{\delta}_l^1 \cdot \mathbf{l}_2 + \boldsymbol{\delta}_l^2 \cdot \mathbf{l}_1)}{\Delta E} \\ &\quad + 2\sqrt{(1-\eta)\eta} \frac{[(1-\eta)E_{z,1} + \eta E_{z,2}](E_{z,1} - E_{z,2}) + 2E_{z,1}E_{z,2}(1-2\eta)\sin^2(\theta_{12}/2)}{\Omega \Delta E} \left[\sqrt{\eta(1-\eta)}\delta\epsilon + (1-2\eta)\delta t \right], \end{aligned} \quad (\text{S55})$$

and relaxation by its transversal part ($\tilde{\boldsymbol{\delta}}_N$)_⊥, that can be written as

$$(\tilde{\boldsymbol{\delta}}_N)_\perp^2 = \left| \boldsymbol{\delta}_N \times \frac{\mathbf{l}}{|\mathbf{l}|} \right|^2 = (\tilde{\boldsymbol{\delta}}_N)_x^2 + (\tilde{\boldsymbol{\delta}}_N)_y^2 = |\tilde{\boldsymbol{\delta}}_N|^2 - (\tilde{\boldsymbol{\delta}}_N)_z^2. \quad (\text{S56})$$

$$|\tilde{\boldsymbol{\delta}}_N|^2 \approx \sum_{n=(x,y,z)} (\boldsymbol{\delta}_l)_n^2 + \frac{4(1-\eta)\eta}{\Omega^2} \left[(E_{z,1} - E_{z,2})^2 + 4E_{z,1}E_{z,2}\sin^2\left(\frac{\theta_{12}}{2}\right) \right] [\eta(1-\eta)\delta\epsilon^2 + (1-2\eta)^2\delta t^2]. \quad (\text{S57})$$

We neglect cross terms coupling fluctuations of different parameters. At zero-detuning $\eta = 1/2$ and $\Omega = 2t_c/h$, the FM qubit decoherence is determined by

$$(\tilde{\boldsymbol{\delta}}_N)_z = \frac{(\boldsymbol{\delta}_l^1 + \boldsymbol{\delta}_l^2) \cdot (\mathbf{l}_1 + \mathbf{l}_2)}{4\Delta E_{\text{FM}}} + \frac{E_{z,1}^2 - E_{z,2}^2}{8t_c\Delta E_{\text{FM}}} \delta\epsilon, \quad (\text{S58})$$

$$(\tilde{\boldsymbol{\delta}}_N)_\perp = \frac{|(\boldsymbol{\delta}_l^1 + \boldsymbol{\delta}_l^2) \times (\mathbf{l}_1 + \mathbf{l}_2)|}{4\Delta E_{\text{FM}}} + \frac{E_{z,1}E_{z,2}\sin(\theta_{12})}{4t_c\Delta E_{\text{FM}}} \delta\epsilon. \quad (\text{S59})$$

2. Second-order relaxation process

The second-order process involves the off diagonal block matrix elements. In the qubit frame, the modulus of transition matrix off diagonal block elements are (with $\tilde{M}_\epsilon = U_1^\dagger M_\epsilon U_1$, $\tilde{M}_t = U_1^\dagger M_t U_1$ and $\tilde{M}_\delta$):

$$\begin{aligned} |\tilde{M}_{t,14}| &= \left| \frac{|\epsilon_{12}|}{\Omega} \sin\left(\frac{\gamma}{2}\right) - \frac{4t_c^2}{\Omega^3} (|\Delta \mathbf{l}_\perp| \cos\left(\frac{\gamma}{2}\right) - \Delta \mathbf{l}_\parallel \sin\left(\frac{\gamma}{2}\right)) \right|, \\ |\tilde{M}_{t,23}| &= \left| \frac{|\epsilon_{12}|}{\Omega} \sin\left(\frac{\gamma}{2}\right) + \frac{4t_c^2}{\Omega^3} (|\Delta \mathbf{l}_\perp| \cos\left(\frac{\gamma}{2}\right) - \Delta \mathbf{l}_\parallel \sin\left(\frac{\gamma}{2}\right)) \right|, \\ |\tilde{M}_{t,13}| &= \left| \left(\frac{|\epsilon_{12}|}{\Omega} + \frac{4t_c^2 \Delta \mathbf{l}_\parallel}{\Omega^3} \right) \cos\left(\frac{\gamma}{2}\right) + \frac{4t_c^2 |\Delta \mathbf{l}_\perp|}{\Omega^3} \sin\left(\frac{\gamma}{2}\right) \right|, \\ |\tilde{M}_{t,24}| &= \left| \left(\frac{|\epsilon_{12}|}{\Omega} - \frac{4t_c^2 \Delta \mathbf{l}_\parallel}{\Omega^3} \right) \cos\left(\frac{\gamma}{2}\right) - \frac{4t_c^2 |\Delta \mathbf{l}_\perp|}{\Omega^3} \sin\left(\frac{\gamma}{2}\right) \right| \end{aligned} \quad (\text{S60})$$

$$\begin{aligned}
|\tilde{M}_{\epsilon,14}| &= \left| \frac{t_c}{\Omega} \sin\left(\frac{\gamma}{2}\right) + \frac{t_c|\epsilon_{12}|}{\Omega^3} (|\Delta\mathbf{l}_\perp| \cos\left(\frac{\gamma}{2}\right) - \Delta\mathbf{l}_\parallel \sin\left(\frac{\gamma}{2}\right)) \right|, \\
|\tilde{M}_{\epsilon,23}| &= \left| \frac{t_c}{\Omega} \sin\left(\frac{\gamma}{2}\right) - \frac{t_c|\epsilon_{12}|}{\Omega^3} (|\Delta\mathbf{l}_\perp| \cos\left(\frac{\gamma}{2}\right) - \Delta\mathbf{l}_\parallel \sin\left(\frac{\gamma}{2}\right)) \right|, \\
|\tilde{M}_{\epsilon,13}| &= \left| \left(\frac{t_c}{\Omega} - \frac{t_c|\epsilon_{12}|\Delta\mathbf{l}_\parallel}{\Omega^3} \right) \cos\left(\frac{\gamma}{2}\right) - \frac{t_c|\epsilon_{12}||\Delta\mathbf{l}_\perp|}{\Omega^3} \sin\left(\frac{\gamma}{2}\right) \right|, \\
|\tilde{M}_{\epsilon,24}| &= \left| \left(\frac{t_c}{\Omega} + \frac{t_c|\epsilon_{12}|\Delta\mathbf{l}_\parallel}{\Omega^3} \right) \cos\left(\frac{\gamma}{2}\right) + \frac{t_c|\epsilon_{12}||\Delta\mathbf{l}_\perp|}{\Omega^3} \sin\left(\frac{\gamma}{2}\right) \right|,
\end{aligned} \tag{S61}$$

where we define the transverse B field gradient $\Delta\mathbf{l}_\perp = |\Delta\mathbf{l} \times \hat{l}| = \frac{E_{z,1}E_{z,2}\sin(\theta_{12})}{2\Delta E}$, longitudinal B-field gradient $\Delta\mathbf{l}_\parallel = \frac{E_{z,1}^2 - E_{z,2}^2}{4\Delta E} + \frac{\epsilon_{12}}{\Omega} \frac{|\Delta\mathbf{l}|^2}{\Delta E}$ and the angle γ between the FM Zeeman vector in the bonding and anti-bonding states $\cos(\gamma) = \hat{\mathbf{l}} \cdot \hat{\hat{\mathbf{l}}}$.

At zero detuning ($\hat{\mathbf{l}} = \hat{\hat{\mathbf{l}}}$, $\gamma = 0$), we observe that the second-order channel survives only in the tunneling channel, mediated by spin-preserving inter-orbital tunneling

$$|\tilde{M}_{t,13}| = |\tilde{M}_{t,24}| = \left| \frac{E_{z,1}^2 - E_{z,2}^2}{8t_c\Delta E_{FM}} \right|, \tag{S62}$$

and spin flip inter orbital tunneling

$$|\tilde{M}_{t,14}| = |\tilde{M}_{t,23}| \approx \frac{|\Delta\mathbf{l}_\perp|}{2t_c} = \frac{1}{4t_c} \frac{E_{z,1}E_{z,2}\sin(\theta_{12})}{\Delta E_{FM}}. \tag{S63}$$

S18. Relaxation

1. General model

The complete Hamiltonian of the FM qubit including an environment reads

$$\tilde{H} = \tilde{H}_{\text{eff}} + \tilde{H}_{\text{eff,N}} + H_{\text{bath}}, \tag{S64}$$

with $\tilde{H}_{\text{eff}}$ and $H_{\text{eff,N}}$ given in Eqs. (S45) and (S52) with an additional rotation $\tilde{H}_{\text{eff,N}} = \tilde{U}_1^\dagger H_{\text{eff,N}} \tilde{U}_1$ into the qubit frame, respectively, and H_{bath} representing a general environment Hamiltonian. The bath-system Hamiltonian in interaction picture is

$$\tilde{H}_{\text{eff,N}} = \sum_{\nu \in \epsilon, t} \tilde{M}_\nu \delta_\nu, \quad \tilde{M}_\nu = \sum_{n,m} \tilde{M}_{\nu,nm} e^{i\omega_{nm}t} |n\rangle \langle m|, \tag{S65}$$

$\tilde{M}_\nu$ is the bath-system coupling matrix for the ν^{th} channel and δ_ν is the fluctuation of noise of the ν^{th} channel caused by the bath in the unit of energy. The detailed matrix elements are shown in Eq. (S60) and (S61). Consider the general multilevel system, the equation of motion of a qubit system operator C under the framework of the quantum Langevin equation is

$$\dot{C} = -\frac{i}{\hbar} [C, \tilde{H}] - \sum_\nu \sum_{nm} \Gamma_\nu^+(\omega_{nm}) [C, \tilde{M}_{\nu,nm}] \tilde{M}_{\nu,nm}^\dagger - \Gamma_\nu^-(\omega_{nm}) \tilde{M}_{\nu,nm}^\dagger [C, \tilde{M}_{\nu,nm}], \tag{S66}$$

the asymmetric noise spectral density is

$$\Gamma_\nu^+(\omega_{nm}) = \frac{1}{\hbar^2} \int_0^\infty d\tau e^{i\omega_{nm}\tau} \langle \delta_\nu(\tau) \delta_\nu(0) \rangle, \tag{S67}$$

$$\Gamma_\nu^-(\omega_{n,m}) = \frac{1}{\hbar^2} \int_0^\infty d\tau e^{i\omega_{nm}\tau} \langle \delta_\nu(0) \delta_\nu(\tau) \rangle. \tag{S68}$$

Relaxation processes in our system occur not only because of direct spin-flip transitions in the bonding orbital ground state, but also occur because of higher-order processes involving fast orbital transitions between bonding and antibonding states. To describe this effect, we consider the rate equation for the populations $P_\mu = |\mu\rangle\langle\mu|$ of each flopping mode state μ . Since P_μ is a projection operator, the equation (S66) becomes

$$\begin{aligned}\dot{P}_\mu &= \sum_\nu \sum_{n,m} \Gamma_\nu^+(\omega_{n,m}) [P_\mu, \tilde{M}_{\nu,nm}] \tilde{M}_{\nu,nm}^\dagger - \Gamma_\nu^-(\omega_{n,m}) \tilde{M}_{\nu,nm}^\dagger [P_\mu, \tilde{M}_{\nu,nm}] \\ &= \sum_\nu \sum_n |\tilde{M}_{\nu,\mu n}|^2 \{ [\Gamma_\nu^+(\omega_{n,\mu}) + \Gamma_\nu^-(\omega_{\mu,n})] P_n - [\Gamma_\nu^+(\omega_{\mu,n}) + \Gamma_\nu^-(\omega_{n,\mu})] P_\mu \},\end{aligned}\quad (\text{S69})$$

where in the second line we have used $\tilde{M}_{\nu,nm}^\dagger [P_\mu, \tilde{M}_{\nu,nm}] = [\tilde{M}_{\nu,nm}, P_\mu]$, $\tilde{M}_{\nu,nm}^\dagger = |\tilde{M}_{\nu,nm}|^2 (\delta_{\mu n} P_m - \delta_{\mu m} P_\mu)$. Then we can write a set of rate equations describing the occupation rate of each energy level P_i

$$\dot{P}_\mu = \sum_j (W_{\mu j} P_j - W_{j\mu} P_\mu) = -W_\mu P_\mu + \sum_{j \neq \mu} W_{\mu j} P_j, \quad (\text{S70})$$

where $W_{\mu j} = W_{j \rightarrow \mu}$ are the transition rates from state j to state μ with energies E_j and E_μ , respectively, and we introduce the total rate $W_\mu = \sum_{j \neq \mu} W_{j\mu}$. The energy level transition rates are given by

$$W_{nm} = W_{m \rightarrow n} = \sum_\nu |\tilde{M}_{\nu,nm}|^2 [\Gamma_\nu^+(\omega_{mn}) + \Gamma_\nu^-(\omega_{nm})] = \sum_\nu |\tilde{M}_{\nu,nm}|^2 \frac{S_\nu(\omega_{nm})}{\hbar^2}, \quad (\text{S71})$$

which connects to the symmetric power spectral density defined as

$$S_\nu(\omega_{nm}) = \int_{-\infty}^{\infty} d\tau e^{-i\omega_{nm}\tau} \langle \delta_\nu(\tau) \delta_\nu(0) \rangle. \quad (\text{S72})$$

To estimate the relaxation rates of the FM qubit encoded in the spin state of the bonding state, we perform adiabatic elimination of the spin states in the antibonding states. This procedure assumes that the excited orbital states (antibonding) quickly decay to the bonding state and their population transition rate is constantly zero; we thus set $\dot{P}_{3,4} = 0$. We also note that the direct spin transition between the spin antibonding states is much slower than the orbital transition rate, thus for simplicity we neglect the corrections coming from W_{34} and $W_{4,3}$. Solving for P_3 and P_4 and inserting back into the equation S70, we find

$$\dot{P}_1 = -P_1 \left(W_1 - \frac{W_{13}W_{31}}{W_3} - \frac{W_{14}W_{41}}{W_4} \right) + P_2 \left(W_{12} + \frac{W_{13}W_{32}}{W_3} + \frac{W_{14}W_{42}}{W_4} \right), \quad (\text{S73})$$

$$\dot{P}_2 = P_1 \left(W_{21} + \frac{W_{23}W_{31}}{W_3} + \frac{W_{24}W_{41}}{W_4} \right) - P_2 \left(W_2 - \frac{W_{23}W_{32}}{W_3} - \frac{W_{24}W_{42}}{W_4} \right). \quad (\text{S74})$$

From this system of equations, we find the relaxation rate $1/T_1$ as the sum of its off-diagonal elements:

$$\frac{1}{T_1} = \frac{1}{T_1^{(1)}} + \frac{1}{T_1^{(2)}}, \quad (\text{S75})$$

where we distinguish between first and second order processes

$$\begin{aligned}\frac{1}{T_1^{(1)}} &= W_{12} + W_{21}, \\ \frac{1}{T_1^{(2)}} &= \frac{W_{13}W_{32} + W_{23}W_{31}}{W_3} + \frac{W_{14}W_{42} + W_{24}W_{41}}{W_4}.\end{aligned}\quad (\text{S76})$$

We now discuss explicitly different possible sources of relaxation.

2. Relaxation by a bosonic bath

We model the environment by considering a bosonic bath of n orthogonal modes with wavevector $\mathbf{q}$ and frequency $\omega_{n,\mathbf{q}}$, i.e., $H_{\text{bath}} = \sum_{n,\mathbf{q}} \hbar \omega_{n,\mathbf{q}} a_{\mathbf{q},n}^\dagger a_{\mathbf{q},n}$, that couples to tunneling, detuning, and g -tensor as

$$\delta_\epsilon = \sum_{n,\mathbf{q}} \delta\epsilon_{n,\mathbf{q}}^* a_{n,\mathbf{q}} + \delta\epsilon_{n,\mathbf{q}} a_{n,\mathbf{q}}^\dagger, \quad \delta_t = \sum_{n,\mathbf{q}} \delta t_{n,\mathbf{q}}^* a_{n,\mathbf{q}} + \delta t_{n,\mathbf{q}} a_{n,\mathbf{q}}^\dagger, \quad \delta_l = \sum_{n,\mathbf{q}} \delta l_{n,\mathbf{q}}^* a_{n,\mathbf{q}} + \delta l_{n,\mathbf{q}} a_{n,\mathbf{q}}^\dagger. \quad (\text{S77})$$

We postpone the discussion about the explicit form of the couplings $\delta\epsilon_{n,\mathbf{q}}$, $\delta t_{n,\mathbf{q}}$, $\delta l_{n,\mathbf{q}}$ to the next section and provide here general results for arbitrary bosonic baths.

For convenience, we group all the prefactors of $a_{n,\mathbf{q}}$ in $H_{\text{eff},N}$ (S52) in definition (S65). The system-environment coupling Hamiltonian is

$$\tilde{H}_{\text{eff},N} = \tilde{M}_\epsilon \delta_\epsilon + \tilde{M}_t \delta_t, \quad (\text{S78})$$

with $\tilde{M}_\epsilon = U_1^\dagger M_\epsilon U_1$ and $\tilde{M}_t = U_1^\dagger M_t U_1$ where

$$M_\epsilon = \frac{1}{2} \cos(2\theta) \tau_z - \frac{1}{2} \sin(2\theta) \tau_x + \frac{\Delta \mathbf{l} \cdot \boldsymbol{\sigma}}{2\Omega} \sin(2\theta) \begin{pmatrix} -\sin(2\theta) & -\cos(2\theta) \\ -\cos(2\theta) & \sin(2\theta) \end{pmatrix} \quad (\text{S79})$$

$$M_t = \sin(2\theta) \tau_z + \cos(2\theta) \tau_x + \frac{\Delta \mathbf{l} \cdot \boldsymbol{\sigma}}{\Omega} \sin(2\theta) \begin{pmatrix} \cos(2\theta) & -\sin(2\theta) \\ -\sin(2\theta) & -\cos(2\theta) \end{pmatrix} \quad (\text{S80})$$

Further, the g -tensor can be modulated by the gate voltage, as shown in Tab. S1. We connect the bosonic-bath-induced gate fluctuation to the local g -tensor fluctuation by

$$\delta_l^i = \mu_B \mathbf{B} \cdot \left(\frac{\partial \mathbf{g}_i}{\partial V_{P2}} \frac{\alpha_{21} + \alpha_{11}}{\alpha_{11} \alpha_{22}} - \frac{\partial \mathbf{g}_i}{\partial V_{P1}} \frac{1}{\alpha_{11}} \right) \delta_\epsilon,$$

resulting in additional transition components to the detuning channel $\tilde{M}_{\delta_l} = U_1^\dagger M_{\delta_l} U_1$:

$$M_{\delta_l} = \frac{1}{2} \begin{pmatrix} \tilde{\delta}_l \cdot \boldsymbol{\sigma} & \sin(2\theta) \Delta \delta_l \cdot \boldsymbol{\sigma} \\ \sin(2\theta) \Delta \delta_l \cdot \boldsymbol{\sigma} & \delta_l \cdot \boldsymbol{\sigma} \end{pmatrix}. \quad (\text{S81})$$

We will show that the gate modulated g -tensor noise has a small contribution compared to the detuning and tunneling noise channels in the FM and small-detuning regime. In the large-detuning regime, where we return to the LD qubit limit, relaxation is dominated by the gate-modulated g -tensor noise channel that couples to the Johnson-Nyquist noise source.

Assuming that the bath is in a thermal state with population $N_{n,\mathbf{q}} = (e^{\hbar \omega_{n,\mathbf{q}}/k_B T} - 1)^{-1}$ the relaxation rates in Eq. (S71) equation reduces to the Fermi Golden rule (using $\delta(\hbar \omega_i) \rightarrow \frac{1}{\hbar} \delta(\omega_i)$)

$$W_{ij} = W_{j \rightarrow i} = \frac{2\pi}{\hbar} \sum_{\nu, n, \mathbf{q}} |\delta_{n,\mathbf{q}}|^2 |\tilde{M}_{\nu,ij}|^2 \left[N_{n,\mathbf{q}} \delta(E_i - E_j - \hbar \omega_{n,\mathbf{q}}) + (1 + N_{n,\mathbf{q}}) \delta(E_i - E_j + \hbar \omega_{n,\mathbf{q}}) \right] = \frac{1}{\hbar^2} \sum_{\nu} S_{\nu}(\omega_{ij}) |\tilde{M}_{\nu,ij}|^2, \quad (\text{S82})$$

$$W_{ji} = W_{i \rightarrow j} = W_{j \rightarrow i} e^{(E_i - E_j)/k_B T}, \quad (\text{S83})$$

with the power spectral density (S72) under the rotating wave approximation (dropping all $\delta(2\omega_{ij} \pm \omega_{n,\mathbf{q}})$ terms) is

$$S_{\delta}(\omega_{i,j}) = 2\pi \sum_{n,\mathbf{q}} |\delta_{n,\mathbf{q}}|^2 [N_{n,\mathbf{q}} \delta(\omega_{ij} - \omega_{n,\mathbf{q}}) + (N_{n,\mathbf{q}} + 1) \delta(\omega_{ij} + \omega_{n,\mathbf{q}})] \quad (\text{S84})$$

leading to

$$\begin{aligned} \Gamma_1^{(1)} &= \frac{1}{T_1^{(1)}} = \frac{2\pi}{\hbar} \coth \left(\frac{\Delta E}{2k_B T} \right) \sum_{\nu, n, \mathbf{q}} |\delta_{n,\mathbf{q}}|^2 |M_{\nu,ij}|^2 \delta(\Delta E - \hbar \omega_{n,\mathbf{q}}), \\ \Gamma_1^{(2)} &= \frac{1}{T_1^{(2)}} \approx 2e^{-\frac{\Omega}{k_B T}} \left(\frac{1}{W_{13}^{-1} + W_{23}^{-1}} + \frac{1}{W_{14}^{-1} + W_{24}^{-1}} \right). \end{aligned} \quad (\text{S85})$$

In the last approximation we considered that in our experiment thermal energy $k_B T$ is higher than the Zeeman splitting, but not larger than the orbital energy Ω . This equation shows that second-order decay rates are exponentially suppressed as a function of the orbital energy.

3. Phonons

We now estimate explicitly the couplings $\delta\epsilon_{n,\mathbf{q}}$, $\delta t_{n,\mathbf{q}}$ caused by phonons. A phonon bath couples to the charge degree of freedom by

$$H_{\text{ph-charge}} = i \sum_{n,\mathbf{q}} C_{n,\mathbf{q}} \left(\Phi_n(\mathbf{q}) e^{i\mathbf{q}\cdot\mathbf{r}} a_{n,\mathbf{q}} - \Phi_n^*(\mathbf{q}) e^{-i\mathbf{q}\cdot\mathbf{r}} a_{n,\mathbf{q}}^\dagger \right), \quad (\text{S86})$$

where the potential for a 3D phonon with wavevector $\mathbf{q}$ and polarization $\mathbf{c}_n$ coupled to pure HH states (confined along z) is [66]

$$\Phi_n(\mathbf{q}) = \left(a + \frac{b}{2} \right) \mathbf{q} \cdot \mathbf{c}_n - \frac{3}{2} b q_z (\mathbf{c}_n)_z, \quad (\text{S87})$$

with normalization

$$C_{n,\mathbf{q}} = \sqrt{\frac{\hbar}{2\rho V \omega_{n,\mathbf{q}}}}. \quad (\text{S88})$$

Here, V is the volume of the sample, ρ is the density, and a, b are deformation potentials. We neglect anisotropies in the deformation potential.

There are 3 bulk phonon modes, one longitudinal with velocity $v_l = \sqrt{(\lambda + 2\mu)/\rho}$ and $\mathbf{c}_l = \mathbf{q}/q = (\sin \Theta \sin \Phi, \sin \Theta \cos \Phi, \cos \Theta)$ and two transversal with velocity $v_t = \sqrt{\mu/\rho}$ and $\mathbf{c}_{t1} \perp \mathbf{q}$ (e.g. $(-\cos \Phi, \sin \Phi, 0)$) and $\mathbf{c}_{t2} = \mathbf{c}_{t1} \times \mathbf{q} = (\cos \Theta \sin \Phi, \cos \Theta \cos \Phi, -\sin \Theta)$ [67].

Using spherical coordinates for $\mathbf{q}$

$$\Phi_l(\mathbf{q}) = q \left[\left(a + \frac{b}{2} \right) - \frac{3}{2} b \cos^2(\Theta) \right], \quad (\text{S89})$$

$$\Phi_{t1}(\mathbf{q}) = 0, \quad (\text{S90})$$

$$\Phi_{t2}(\mathbf{q}) = \Phi_t(\mathbf{q}) = \frac{3b}{4} q \sin(2\Theta), \quad (\text{S91})$$

such that the coupling is only from two phonon modes.

In a DQD the coupling to detuning and tunneling is

$$H_{\text{ph-DQD}} = \frac{1}{2} \tau_z \sum_{n,\mathbf{q}} (\delta\epsilon_{n,\mathbf{q}} a_{\mathbf{q},n} + \delta\epsilon_{n,\mathbf{q}}^* a_{\mathbf{q},n}^\dagger) + \tau_x \sum_{n,\mathbf{q}} (\delta t_{n,\mathbf{q}} a_{\mathbf{q},n} + \delta t_{n,\mathbf{q}}^* a_{\mathbf{q},n}^\dagger) \quad (\text{S92})$$

with

$$\delta\epsilon_{n,\mathbf{q}} = \Phi_n(\mathbf{q}) C_{n,\mathbf{q}} [\langle \psi_L | e^{i\mathbf{q}\cdot\mathbf{r}} | \psi_L \rangle - \langle \psi_R | e^{i\mathbf{q}\cdot\mathbf{r}} | \psi_R \rangle], \quad (\text{S93})$$

$$\delta t_{n,\mathbf{q}} = \frac{\Phi_n(\mathbf{q})}{2} C_{n,\mathbf{q}} [\langle \psi_L | e^{i\mathbf{q}\cdot\mathbf{r}} | \psi_R \rangle + \langle \psi_R | e^{i\mathbf{q}\cdot\mathbf{r}} | \psi_L \rangle]. \quad (\text{S94})$$

We neglect here the direct coupling of the phonons with the spin degree of freedom through the electric tunability of the g -tensor.

Assuming for simplicity that the two QDs are identical isotropic Gaussians with width l , shifted by $\pm d/2$, having an overlap $s = e^{-d^2/4l^2} \ll 1$, we find that

$$\delta\epsilon_{n,\mathbf{q}} \approx 2C_{n,\mathbf{q}} \Phi_n(\mathbf{q}) e^{-\frac{q^2 l^2}{4}} \sin \left(\frac{qd}{2} \sin \Theta \sin \Phi \right), \quad (\text{S95a})$$

$$\delta t_{n,\mathbf{q}} \approx C_{n,\mathbf{q}} \Phi_n(\mathbf{q}) s e^{-\frac{q^2 l^2}{4}}. \quad (\text{S95b})$$

Combining Eqs. (S95) and (S82), we find that the transition rates can be written as

$$W_{ij} = |\tilde{M}_{\epsilon,ij}|^2 \sum_n (\gamma_\epsilon)_{ij}^n + |\tilde{M}_{t,ij}|^2 \sum_n (\gamma_t)_{ij}^n, \quad (\text{S96})$$

with

$$(\gamma_\epsilon)_{ij}^n = \frac{2\pi}{\hbar} \sum_{\mathbf{q}} |\delta\epsilon_{n,\mathbf{q}}|^2 \left[N_{n,\mathbf{q}} \delta(E_i - E_j - \hbar\omega_{n,\mathbf{q}}) + (1 + N_{n,\mathbf{q}}) \delta(E_i - E_j + \hbar\omega_{n,\mathbf{q}}) \right] \quad (\text{S97})$$

$$(\gamma_t)_{ij}^n = \frac{2\pi}{\hbar} \sum_{\mathbf{q}} |\delta t_{n,\mathbf{q}}|^2 \left[N_{n,\mathbf{q}} \delta(E_i - E_j - \hbar\omega_{n,\mathbf{q}}) + (1 + N_{n,\mathbf{q}}) \delta(E_i - E_j + \hbar\omega_{n,\mathbf{q}}) \right]. \quad (\text{S98})$$

Using $\sum_{\mathbf{q}} \rightarrow \frac{V}{(2\pi)^3} \int q^2 \sin(\theta_q) dq d\theta_q d\phi_q$, $\omega_{n,\mathbf{q}} = v_n q$, $N(\omega_{n,\mathbf{q}}) = N_{n,\mathbf{q}}$, $\hbar\omega_{ij} = E_i - E_j$, and $q_{ij}^n = \omega_{ij}/v_n$ we find

$$\begin{aligned} (\gamma_\epsilon)_{ij}^n &= \frac{(q_{ij}^n)^2 e^{-\frac{(q_{ij}^n)^2 l^2}{2}}}{2\rho\pi^2 \hbar\omega_{ij} v_n} \left[N(\omega_{ij}) \Theta(\omega_{ij}) + [1 + N(\omega_{ji})] \Theta(\omega_{ji}) \right] \int d\theta_q d\phi_q |\Phi_n(q_{ij}^n, \theta_q)|^2 \sin \theta_q \sin^2 \left(\frac{q_{ij}^n d}{2} \sin \theta_q \sin \phi_q \right) \\ &\approx \frac{(q_{ij}^n)^2 e^{-\frac{(q_{ij}^n)^2 l^2}{2}}}{2\rho\pi^2 \hbar\omega_{ij} v_n} \left[N(\omega_{ij}) \Theta(\omega_{ij}) + [1 + N(\omega_{ji})] \Theta(\omega_{ji}) \right] I_\epsilon^n, n \in \{l, t2\} \end{aligned} \quad (\text{S99})$$

$$\begin{aligned} (\gamma_t)_{ij}^n &= s^2 \frac{(q_{ij}^n)^2 e^{-\frac{(q_{ij}^n)^2 l^2}{2}}}{8\rho\pi^2 \hbar\omega_{ij} v_n} \left[N(\omega_{ij}) \Theta(\omega_{ij}) + [1 + N(\omega_{ji})] \Theta(\omega_{ji}) \right] \int d\theta_q d\phi_q |\Phi_n(q_{ij}^n, \theta_q)|^2 \sin \theta_q \\ &\approx s^2 \frac{(q_{ij}^n)^2 e^{-\frac{(q_{ij}^n)^2 l^2}{2}}}{8\rho\pi^2 \hbar\omega_{ij} v_n} \left[N(\omega_{ij}) \Theta(\omega_{ij}) + [1 + N(\omega_{ji})] \Theta(\omega_{ji}) \right] I_t^n, n \in \{l, t2\}, \end{aligned} \quad (\text{S100})$$

where the step function $\Theta(\omega)$ is defined as

$$\Theta(x) = \begin{cases} 1, & x > 0 \\ 0, & x < 0 \end{cases}. \quad (\text{S101})$$

Here, each integration has an analytical expression:

$$\begin{aligned} I_\epsilon^l &= \frac{\pi x_l^2}{d^2} \left(a^2 (2 - 2J_0(x_l)) + ab \left(\frac{6J_1(x_l)}{x_l} - 2J_0(x_l) \right) + b^2 \left(-\frac{J_0(x_l)}{2} + \frac{3J_1(x_l)}{x_l} - \frac{27J_2(x_l)}{2x_l^2} + \frac{2}{5} \right) \right) \\ I_\epsilon^{t2} &= \frac{\pi x_t^2}{d^2} b^2 \left(\frac{3}{5} - \frac{9J_1(x_t)}{2x_t} + \frac{27J_2(x_t)}{2x_t^2} \right), \end{aligned} \quad (\text{S102})$$

and

$$\begin{aligned} I_t^l &= \frac{4\pi (5a^2 + b^2) x_l^2}{5d^2} \\ I_t^{t2} &= \frac{6\pi b^2 x_t^2}{5d^2}, \end{aligned} \quad (\text{S103})$$

where we have defined $x_l = \frac{d\omega_{ij}}{v_l}$ and $x_t = \frac{d\omega_{ij}}{v_t}$. J_n is the n -th Bessel function.

We introduce the parameter that we used for estimating the energy level transition rate induced by phonon bath in Eq. (S99) and (S100). The hole-phonon coupling is governed by the valence-band absolute and shear deformation potentials, with parameters $a_v = 2.1$ eV [68] and $b = -2.32$ eV [69], respectively. The Lamé constants, estimated from the elastic constants of germanium with the Voigt-Reuss-Hill Approximation [70–72] ($c_{11} = 126$ GPa, $c_{12} = 44$ GPa, and $c_{44} = 67.7$ GPa), yield $\lambda = 34.4$ GPa and $\mu = 55.4$ GPa. The longitudinal and transverse sound velocities are then calculated from the Lamé constants, $c_l = \sqrt{(\lambda + 2\mu)/\rho} = 5220$ m/s and $c_t = \sqrt{\mu/\rho} = 3230$ m/s, using a Ge mass density of $\rho = 5330$ kg/m³ [73]. The geometry of the double quantum dot is modeled with an inter-dot distance of $d = 100$ nm and a characteristic dot wavelength of $l_{dot} = 55$ nm. The effective temperature of the phonon thermal bath is assumed to be $T_{ph} = 20$ mK. Furthermore, the spin-orbit rotation and the effective magnetic field orientation are defined by the angles $\theta_{so} \approx 14.7^\circ$, $\theta_n \approx 78.3^\circ$, and $\phi_n \approx -33.7^\circ$.

The results of the energy level transition rates induced by phonon are plotted in Fig. S11. Around the FM qubit frequency ($\omega_{ij} < 1$ GHz), we observe that for both channels, the excitation and relaxation are comparable, indicating that the system is immersed in a phonon bath. At the higher frequency, the relaxation is larger than the excitation by several orders of magnitude, indicating the consequences of spontaneous emission and detailed balance.

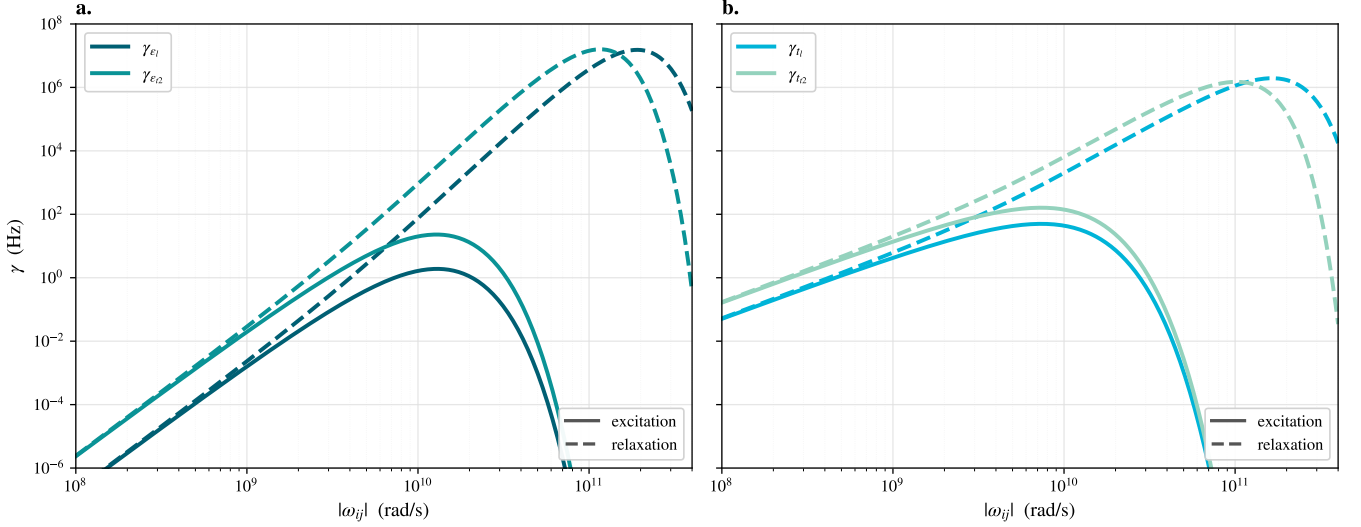


FIG. S11. The excitation and relaxation rate of energy levels for ϵ channel and t channel are presented in **a** using Eq. (S99) and in **b** using Eq. (S100) respectively.

4. Johnson noise

We consider the relaxation induced by thermal photon, the transition matrix $\tilde{M}$ in Eq. (S96) is identical to the phonon case but with different level transition rate given by

$$(\gamma_\nu)_{ij} = \frac{S_\nu(\omega_{ij})}{\hbar^2} = \frac{|\alpha_\nu|^2}{\hbar^2} 2\pi \left[|\delta(\omega_{ij})|^2 N(\omega_{ij}) \Theta(\omega_{ij}) + |\delta(\omega_{ji})|^2 [N(\omega_{ji}) + 1] \Theta(\omega_{ji}) \right]. \quad (\text{S104})$$

Here we consider the general bosonic bath coupled to the system via the fluctuation of the top gates, with the same voltage fluctuation amplitude δ for different channels $\nu \in \{t, \epsilon\}$. This describes the fluctuation of the gates induced shifting in detuning and tunneling (S77). The lever arms α_ν are extracted from the experiment.

We consider the Johnson-Nyquist noise, with asymmetric power spectral density

$$S_{jn,\nu}(\omega_{ij}) = 2|\alpha_\nu|^2 R_J \left[\hbar\omega_{ij} N(\omega_{ij}) \Theta(\omega_{ij}) + \hbar\omega_{ji} (N(\omega_{ji}) + 1) \Theta(\omega_{ji}) \right]. \quad (\text{S105})$$

This is equivalent to the standard symmetric Johnson-Nyquist PSD for $\omega_{ij} = -\omega_{ji} > 0$

$$S_{VV}(\omega_{ij}) = S_\nu(\omega_{ij}) + S_\nu(\omega_{ji}) = 2|\alpha_\nu|^2 R_J \left[\hbar\omega_{ij} N(\omega_{ij}) + \hbar\omega_{ji} (N(\omega_{ji}) + 1) \right] = 2|\alpha_\nu|^2 R \hbar\omega_{ij} \coth \frac{\hbar\omega_{ij}}{2k_B T}, \quad (\text{S106})$$

In the high temperature limit, this results in white noise with

$$S_{VV}(\omega_{ij}) = 4\alpha_\nu^2 R_J k_B T. \quad (\text{S107})$$

The Johnson-Nyquist noise-equivalent resistance is extracted from the experimental PSD by measuring thermal photons and converting the signal into resistance, which gives $R_J \approx 250\Omega$ at temperature $T = 300$ mK.

5. $1/f$ noise

The power spectral density of $1/f$ charge noise is

$$S_{1/f}(\omega_{nm}) = S_{1/f}(-\omega_{nm}) = \int_{-\infty}^{\infty} d\tau e^{-i\omega_{nm}\tau} \langle \delta(\tau) \delta(0) \rangle = \frac{A}{|\omega_{nm}/2\pi|^\alpha}, \quad (\text{S108})$$

where δ is defined as the classical fluctuation, and A is the amplitude of gate fluctuations at 1 Hz in units of V^2 .

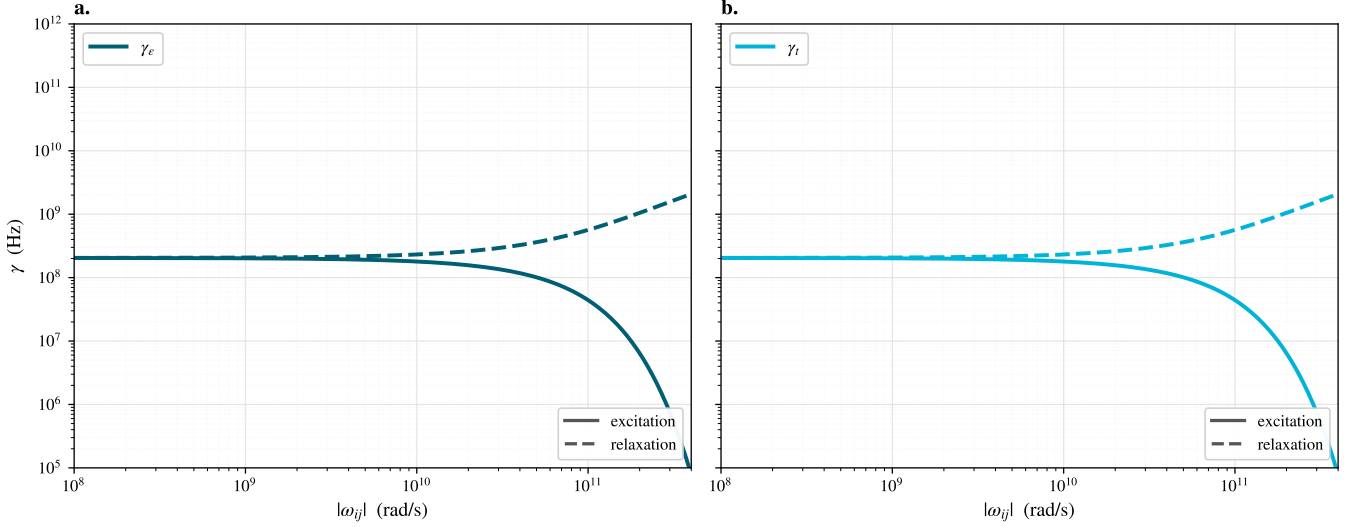


FIG. S12. The excitation and relaxation rate of energy levels for ϵ channel and t channel are presented in **a** and in **b** using Eq. (S104). We assumed a equal lever arm for detuning and tunnling channel ($\alpha_t = \alpha_\epsilon$).

The transition rate equation (S71) holds generally for any system-environment coupling; therefore, the transition rate associated with $1/f$ charge noise is

$$W_{nm} = |M_{\epsilon,nm}|^2 \gamma_\epsilon(\omega_{nm}) + |M_{t,nm}|^2 \gamma_t(\omega_{nm}), \quad \gamma_\nu(\omega_{nm}) = \begin{cases} \frac{|\alpha_\epsilon|^2 S_{1/f}(\omega_{nm})}{\hbar^2} \\ \frac{|\alpha_t|^2 S_{1/f}(\omega_{nm})}{\hbar^2} \end{cases} \quad (\text{S109})$$

The exponent extracted at zero detuning is $\alpha = 1$, and the PSD after conversion to energy is experimentally determined to be $\alpha_\epsilon^2 S_{1/f}^{1.3} = 2 \times 10^{-12} \text{ eV}^2/\text{Hz}$. We neglected the tunneling related $1/f$ noise channel as it was not measured in the experiment. We emphasize that the high-frequency PSD cannot be measured in this experimental setup. It is hence only possible to extrapolate the PSD to the qubit frequency using Eq. (S109). This extrapolation may overestimate the PSD and hence the $1/f$ charge-noise-induced relaxation rate.

6. Comparison of different relaxation mechanisms

We discuss the magnetic-field dependence of the first- and second-order relaxation processes for different noise sources at the FM operation point $\epsilon_{12} = 0$ at low qubit frequencies, and the results are summarized in Fig. S13.

First, for all noise sources, the second-order relaxation process is proportional to B^2 . This is because the second-order process involves only spin-flip transitions between bonding and antibonding states. According to Fermi's Golden Rule in Eq. (S82), the transition frequency is well approximated as $\hbar\omega_{ij} = 2t_c \pm \Delta E_{\text{FM}} \approx 2t_c$. The magnetic-field dependence arises because the B -field dependent Larmor-vector gradient between the two QDs mediates the spin-flip transition, providing the finite matrix elements in Eq. (S61) and Eq. (S60). Note that at the zero-detuning point, the detuning relaxation channel $\tilde{M}_\epsilon$ in Eq. (S61) only permits first-order processes, whereas the tunneling relaxation channel $\tilde{M}_t$ in Eq. (S60) only permits second-order relaxation processes.

For the first order relaxation process, different noise sources have different B field dependence. We first consider two relaxation channels, each consisting of two phonon modes. By combining Eq. (S99), (S100), (S102) and (S103), we find that the detuning phonon channel γ_ϵ shows a $\Gamma_{ph,\epsilon}^{(1)} \propto B^6$ dependence, while the tunneling phonon channel γ_t

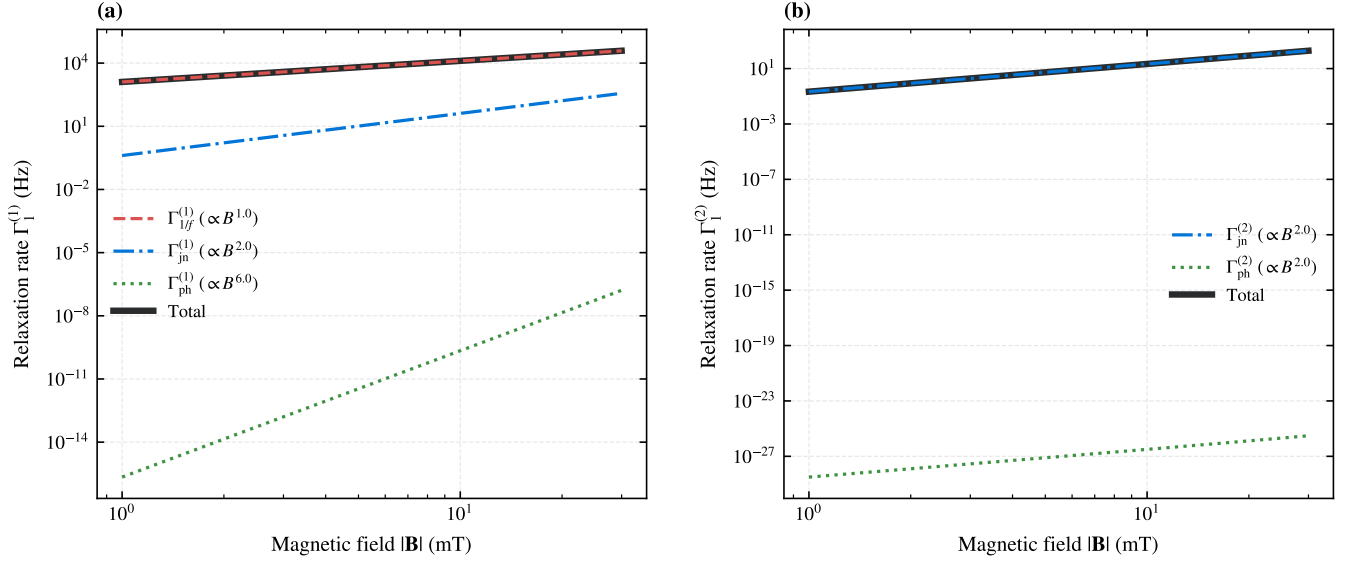


FIG. S13. The first order relaxation rate (a) and the second order relaxation (b) from all major relaxation source including $\Gamma_{1/f}^{(1)}$, $\Gamma_{jn}^{(1)}$, and $\Gamma_{ph}^{(1)}$ as function of magnitude of $\mathbf{B}$ field are calculated via Eq. (S85) with the level transition rate for each source obtained from Eq. (S109), (S104), (S99) and (S99) respectively.

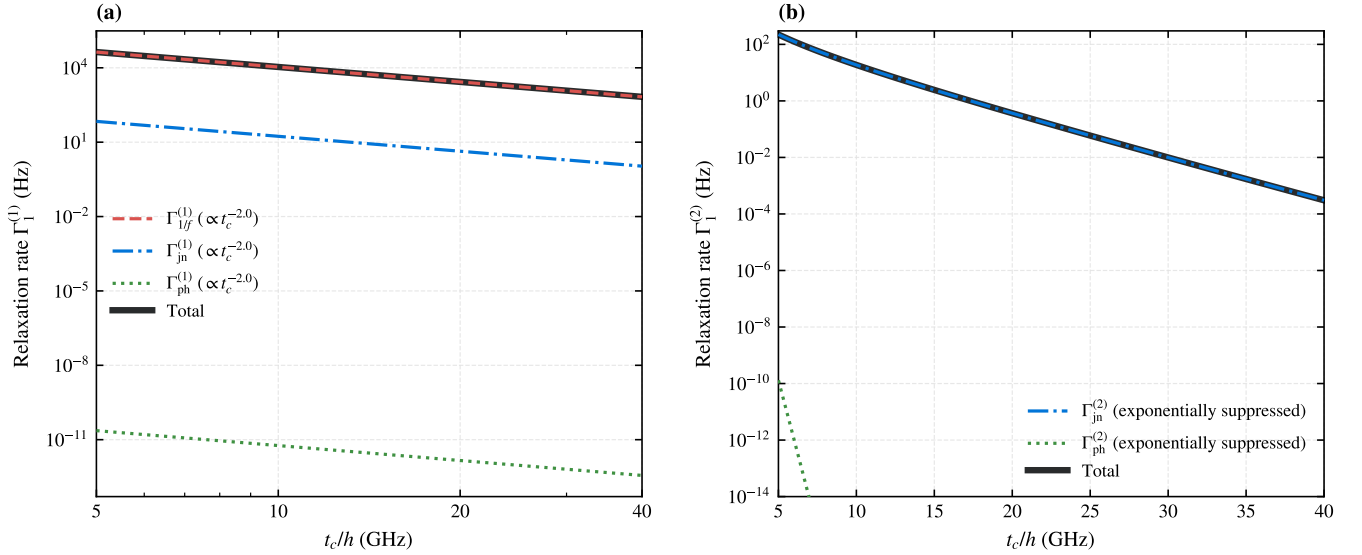


FIG. S14. The first order relaxation rate (a) and the second order relaxation (b), as function of t_c from all major relaxation source including $\Gamma_{1/f}$, Γ_{jn} , and Γ_{ph} are calculated via Eq. (S85) with the level transition rate for each source obtained from Eq. (S109), (S104) and (S99) and (S100) respectively.

shows a $\Gamma_{ph,t}^{(1)} \propto B^4$ relation, with a factor of B^2 contributed by the matrix elements $|\tilde{M}|^2$ from both channels.

$$(\gamma_\epsilon)_ij^l \approx \frac{d^2 k_B T_{ph} (35a^2 + 14ab + 5b^2) \omega_{ij}^4}{210\pi\rho\hbar^2 v_l^7} \quad (S110)$$

$$(\gamma_\epsilon)_ij^t \approx \frac{3b^2 d^2 k_B T_{ph} \omega_{ij}^4}{70\pi\rho\hbar^2 v_t^7} \quad (S111)$$

$$(\gamma_t)_ij^l \approx \frac{s^2 k_B T_{ph} (5a^2 + b^2) \omega_{ij}^2}{10\pi\rho\hbar^2 v_l^5} \quad (S112)$$

$$(\gamma_t)_ij^t \approx \frac{3b^2 s^2 k_B T_{ph} \omega_{ij}^2}{20\pi\rho\hbar^2 v_t^5} \quad (S113)$$

The Johnson-Nyquist noise is a white noise at high temperature limits ($E_z \ll k_B T$). The transition rate Eq. (S104) at high temperatures for detuning and tunneling channels is

$$\gamma_\epsilon = \frac{2\alpha_\epsilon^2 k_B R T}{\hbar^2}, \quad (S114)$$

$$\gamma_t = \frac{2\alpha_t^2 k_B R T}{\hbar^2}. \quad (S115)$$

This indicates that the energy transition rate at qubit subspace is frequency independent. For the orbital energy, the Johnson-Nyquist noise is not a white noise ($\Omega \gg k_B T$), but it is also not B-field dependent. Considering the B^2 contribution from the matrix elements, we find that both first- and second-order relaxation processes induced by Johnson-Nyquist noise show a $\Gamma_{jn}^{(1)} \propto B^2$ dependence.

The $1/f$ charge noise PSD in Eq. (S108) and Eq. (S109) shows $S_{1/f} \propto \frac{1}{B^\alpha}$, where α is fitted to be 1. Therefore, the first-order relaxation process shows a $\Gamma_{1/f}^{(1)} \propto B$ dependence.

In the experiment, a clear relation of $\Gamma_1 = \Gamma_1^{(1)} + \Gamma_1^{(2)} \propto B^2$ is shown in Fig. 4 (d). Based on this B-field dependence, we conclude that the likely dominant noise source is either second-order phonons or a combination of first- and second-order Johnson-Nyquist noise. However, as shown in Fig. S13, at low temperature ($T_{ph} = 20$ mK) and low magnetic field ($B = 5$ mT), the phonon contribution is many orders of magnitude lower than the Johnson-Nyquist noise for both first- and second-order processes. Therefore, Johnson-Nyquist noise is more likely the main relaxation process than phonons.

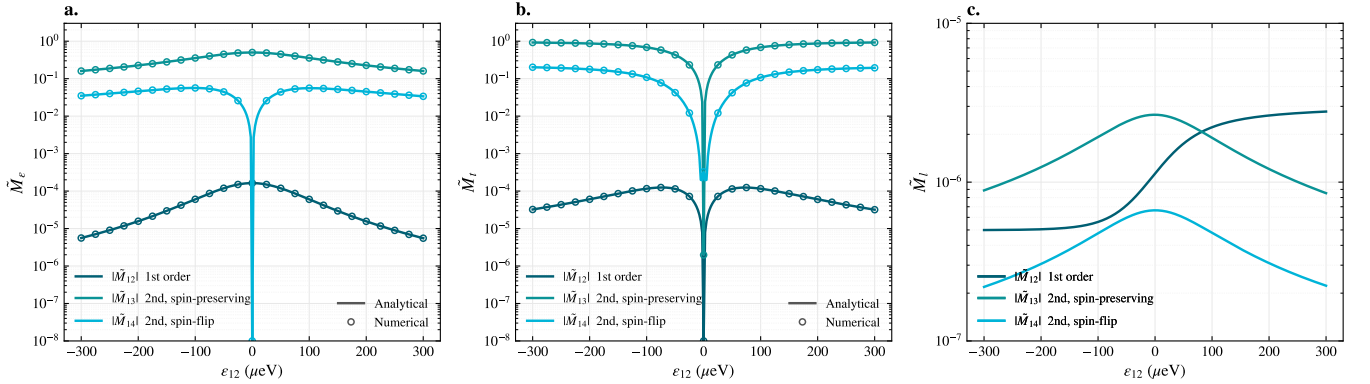


FIG. S15. The detuning channel **a**, tunneling channel **b**, and g-tensor modulation channel **c** are plotted. In **a** and **b**, we provide the comparison of analytical expression in Eq. (S60) and (S61) with the numerical expression $\tilde{M}_t = U_q^\dagger M_t U_q$ and $\tilde{M}_\epsilon = U_q^\dagger M_\epsilon U_q$. In plot **c**, we only numerically calculate the transition matrix element as their analytical expression is lengthy and the contribution from g-tensor modulation channel at FM regime compared to the other two channel is negligible.

To investigate whether the first-order or second-order relaxation process is dominant, we introduce additional evidence from Fig. 4 (e): as the tunnel coupling increases, the B-field dependence becomes $\Gamma_1 \propto B$, indicating that at high t_c , T_1 is limited by $1/f$ charge noise.

To understand this, we examine the relaxation rates from different noise sources as a function of the tunnel coupling t_c . For all first-order processes, the PSD is independent of t_c , and the matrix element is proportional to t_c^{-1} , as

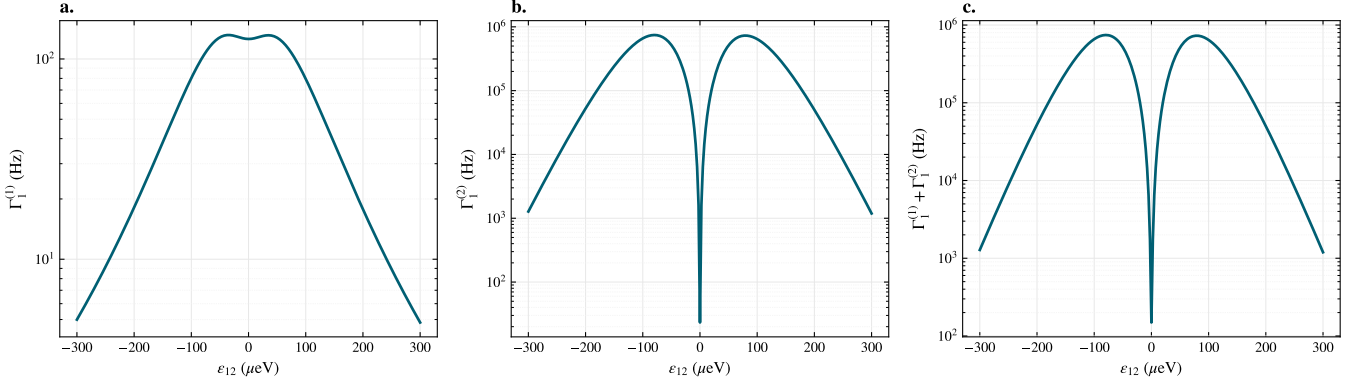


FIG. S16. The relaxation rate for the first order process in **a**, the second order process in **b** and total process in **c**.

indicated by Eq. (S80), Eq. (S79) and Eq. (S59). Therefore, all the first order relaxation processes give $\Gamma^{(1)} \propto t_c^{-2}$. For a bosonic bath, the bosonic PSD has a polynomial dependence on t_c , similar to its B -field dependence. However, due to detailed balance, the second-order relaxation process is exponentially suppressed by the energy gap to higher orbitals, $\Gamma_1 \propto e^{-2t_c/k_B T}$, as indicated by Eq. (S85). Classical $1/f$ charge noise is likely negligible at $t_c/h = 11$ GHz. Nonetheless, if we naively consider a $\frac{1}{f^\alpha}$ PSD that is non-negligible at high frequency, we observe a $\Gamma_1 \propto t_c^{-2-\alpha}$ relation.

Therefore, if the system T_1 is only limited by first-order Johnson–Nyquist noise at low t_c , we should not observe a transition in B -field dependence from $\propto B^2$ into $\propto B$ as t_c increases because both $1/f$ charge noise and Johnson–Nyquist noise are suppressed as t_c^2 . Therefore, a plausible explanation for the trends in both Fig. 4 (d) and Fig. 4 (e) is that a significant second-order relaxation process is involved.

Further evidence supporting the second-order relaxation process is that T_1 exhibits a peak-and-dip structure as a function of detuning, as shown in Fig. S17 (b.), Fig. 4 (f.) and Fig. S15. In the range $\epsilon_{12} < 2t_c$, from Eq. (S60) and Eq. (S61), we find that the two qubit subspace transition matrix elements

$$|\tilde{M}_{\epsilon,12}| = \frac{2t_c^2 |\Delta \mathbf{l}_\perp|}{\Omega^3} \quad |\tilde{M}_{t,12}| = \frac{2t_c |\epsilon_{12}| |\Delta \mathbf{l}_\perp|}{\Omega^3}$$

vary by less than an order of magnitude over this range due to the suppression given by the factor $1/\Omega^3$. Therefore, if the first-order process dominates at the zero-detuning point, we do not expect $\Gamma_1^{(1)} \propto \frac{(2t_c^2 + t_c \epsilon_{12}) |\Delta \mathbf{l}_\perp|}{\Omega^3}$ to drop by an order of magnitude at a slightly detuned point. For the second order matrix element,

$$|\tilde{M}_{\epsilon,14}| = \frac{t_c |\epsilon_{12}|}{2\Omega^2} \frac{E_{z,1} E_{z,2} \sin \theta_{12}}{\Delta E \Delta \bar{E}} \quad |\tilde{M}_{t,14}| = \frac{\epsilon_{12}^2}{2\Omega^2} \frac{E_{z,1} E_{z,2} \sin \theta_{12}}{\Delta E \Delta \bar{E}} + \frac{4t_c^2 |\Delta \mathbf{l}_\perp|}{\Omega^3}.$$

We first recall that only the tunneling channel allows the second order relaxation process at $\epsilon_{12} = 0$. Further, we observe that in the denominator, the second-order matrix element is only suppressed as $\propto 1/\Omega^2$, which means that slight detuning will activate the second-order relaxation channel, leading to the double dips observed in Fig. 4, Fig. S17 (b) and Fig. S16.

We show T_1 calculated by taking into account both the first-order and second-order relaxation processes described earlier, in Fig. S17a. The T_1 and T_2^* sweet lines are represented by black and white dashed lines in Fig. S17a. T_1 sweet line corresponds to the magnetic field directions where the tilt θ_{12} between the Larmor vectors of the two LD qubits in each QD ($\mathbf{l}_i$) becomes minimized. We emphasize that this is the global sweet-line for both first-order and second-order relaxation processes. On the other hand, T_2^* sweet line denotes the collection of magnetic field directions where the size of the two Larmor vectors, i.e. the Zeeman energy, becomes identical. For the studied system in this work, there is no magnetic field direction where T_1 and T_2^* sweet lines overlap. Nevertheless, for the low-magnetic field operation regime, we find that T_1 is not a main limiting factor for T_2^* and T_2^{Hahn} when operated on the T_2^* sweet line.

Furthermore, we present the calculated T_1 as a function of ϵ_{12} at the magnetic field direction $\hat{b}_3$ (see Fig. 2a) in Fig. S17b. The solid (dashed) curve corresponds to T_1 calculated taking into account the bosonic bath noise ($1/f$ charge noise). Here, we use the charge noise amplitude extracted from the PSD measurements shown in Supplementary section S10. While the calculation suggests that $1/f$ charge noise dominates the relaxation at $\epsilon_{12} = 0$, the observed $1/B^2$ scaling of T_1 in Fig. 4d indicates that the relaxation is governed by second-order process rather than

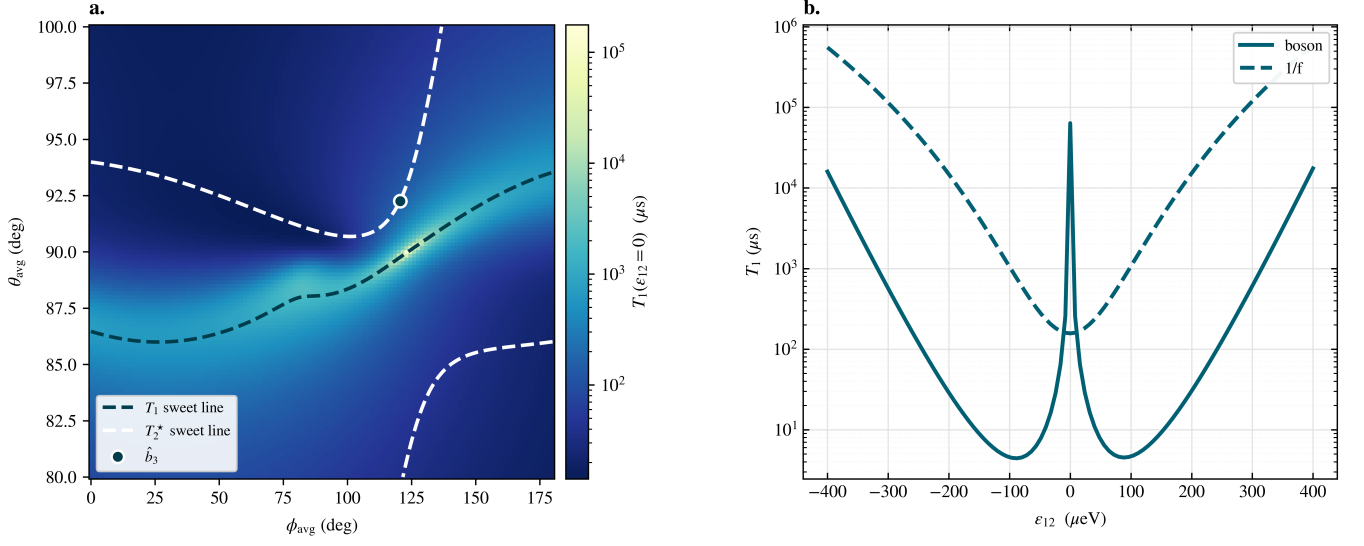


FIG. S17. **a.** The FM qubit relaxation time T_1 at $\epsilon_{12} = 0$ as a function of ϕ_{avg} and θ_{avg} . T_1 sweet line (black dashed line) for both first order Eq. (S59) and second order relaxation process (S63) is calculated by finding the minimization angle of $\sin(\theta_{12})$ at a given ϕ_{avg} . The T_2^* sweet lines (white dashed lines) are calculated using Eq. (S58). **b.** T_1 as a function of ϵ_{12} at the magnetic field corresponding to $\hat{\mathbf{b}}_3$ (green point in Fig. 2a), calculated using Eq. (S76) with energy level transition rate estimated from Eq. (S109) (dashed curve, 1/f noise case), Eq. (S99), (S100) and (S104) respectively (solid curve, bosonic bath case).

the 1/f charge noise. We attribute this discrepancy to the uncertainty in the charge-noise amplitude extracted in Supplementary section S10, which calls for detailed investigation.